



Evaluating Personalised Phishing Awareness Training: A
Mixed Methods Study on Emotional Response, Engagement,
and User Preference

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Abstract

Phishing campaigns continue to provide the single largest entry point for successful cyberattacks. Simulated phishing campaigns are currently used to improve employee awareness, but evidence suggests that generic communications and impersonal simulation content can impact the success of training campaigns. The aim of this research was to investigate whether users' emotional responses and user-stated preferences can influence engagement and behavioural outcomes of phishing awareness training. This study comprised two main components: a review of relevant literature, and the collection and analysis of empirical data. A mixed methods approach was used to conduct the empirical research with quantitative and qualitative data collected via two self-administered surveys and a semi-structured expert-led interview. The findings from this research provide evidence that whilst the professional focus on training engagement lies almost exclusively in simplistic outcome metrics, almost half (44.8%) of end-users indicate a greater inclination to engage when provided more holistic training features, particularly supportive communication and flexible delivery mechanisms. The main conclusions drawn from this study were that click-through-rates in isolation should not be relied upon to provide accurate measures of long-term behavioural change, and that incorporating non-demographic user factors (including user-driven personalisation) into training programme design may deliver improved training outcomes. This dissertation recommends such an approach, based on supportive and collaborative organisational communications, with a call for further research to be undertaken into the real-world effects of user-stated personalisation on long-term phishing awareness training outcomes.

Keywords: *phishing awareness, human factors, personalised training, cybersecurity training*

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Abbreviations

AI	Artificial Intelligence
CREST	Centre for Research and Evidence on Security Threats
CSS	Cascading Style Sheets
CTR	Click-Through-Rate
DSIT	Department for Science, Innovation and Technology
EDA	Exploratory Data Analysis
FP	False Positive
GDPR	General Data Protection Regulation
GUI	Graphical User Interface
HTML	Hyper Text Markup Language
ID	Identifier
IMAP	Internet Message Access Protocol
LSE	London School of Economics
M	Mean
NCSC	National Cyber Security Centre

NIST	National Institute for Standards and Technology
OHE	One-Hot Encoding
SD	Standard Deviation
SEPA	Scottish Environment Protection Agency
SMTP	Simple Mail Transport Protocol
SQL	Structured Query Language
USD	United States Dollars
VM	Virtual Machine
VS Code	Visual Studio Code

Glossary

Term	Definition
3rd Party Perspective Framing	When responding to survey questions, participants frame their answers through the imagined views of others, rather than directly expressing their own point of view.
Click-Through Rate (CTR)	A percentage of users who actively click on links provided to them.
Deductive Coding	A thematic coding process driven by predefined concepts, theory, or research questions.
Dichotomous Variable	A variable with only two resulting values (e.g., yes/no responses).
Exploratory Data Analysis (EDA)	The process of exploring data prior to formal analysis, often using descriptive statistics (e.g., mean and standard deviation).
Inductive Coding	A thematic coding process driven by patterns identified in raw data.
Nominal Data	Categorical data with no inherent order (e.g., job role, gender).
One-Hot Encoding (OHE)	A method of representing categorical variables in a binary format, used for statistical analysis.
Ordinal Data	Data that represents ordered categories (e.g., Likert scale responses) - the intervals between values are not necessarily equal.
Thematic Analysis	A method for identifying, analysing, and reporting patterns (themes) within qualitative data.

1 Introduction

1.1 Background

The National Cyber Security Centre (NCSC) defines phishing as a type of cybersecurity attack that attempts to deceive users into disclosing sensitive information or performing unsafe actions via email or messaging platforms (NCSC, 2024). According to the Department for Science, Innovation and Technology (DSIT), 85% of successful cyber-attacks were attributable to phishing emails, making it the most common attack vector across the organisations surveyed (DSIT, 2025). Figure 1.1 shows the top five attack methods identified, demonstrating the dominance of phishing as an entry point for cyber-attacks.

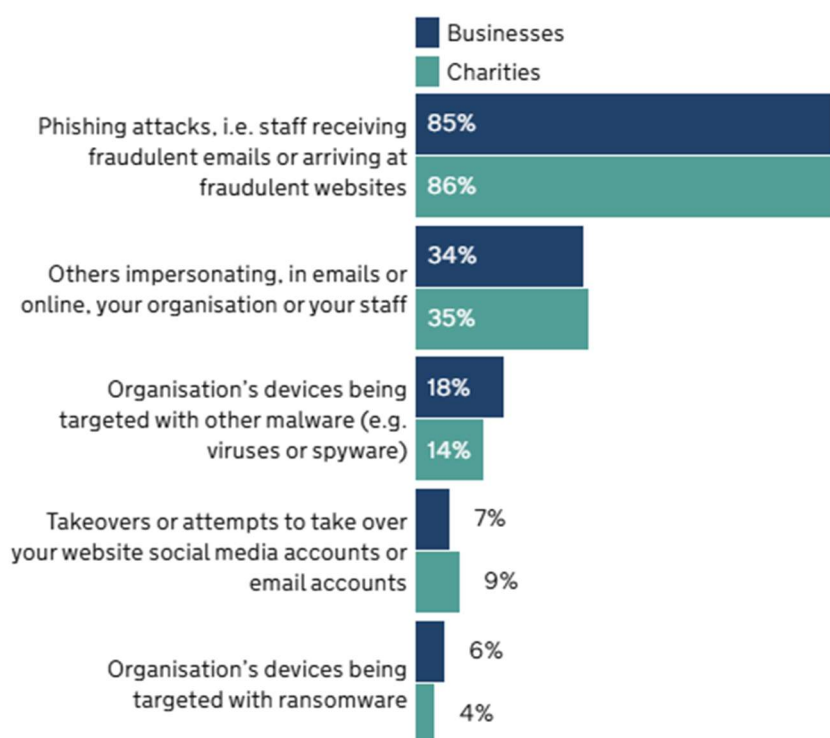


Figure 1.1: Percentage of types of breaches or attacks among the organisations that have identified any breaches or attacks (DSIT, 2025)

The effects of phishing attacks can be multifaceted and long-lasting, as demonstrated by the 2020 ransomware incident at the Scottish Environment Protection Agency (SEPA), which continued to cause organisational damage for over a year (Audit Scotland, 2022). Phishing was also identified as the most disruptive attack vector by 65% of businesses in DSIT's report (2025), with the cost of the most

serious incidents averaging £12,560 across medium/large organisations (DSIT, 2025). Figure 1.2 shows the comparative figures for costs across other organisation categories.

Only across organisations identifying a breach or attack with an outcome	All businesses	Micro/small Businesses	Medium/large businesses	All charities
Mean cost	£8,260	£7,960	£12,560	£21,540
Median cost	£370	£370	£1,150	£200
Base	181	101	80	63

Figure 1.2: Average total cost of the most disruptive breach or attack (DSIT, 2025)

According to Proofpoint's State of the Phish report (2024), 71% of users who received a suspicious email undertook risky behaviour (such as clicking on a link from an unknown sender); of these, 96% were aware that their behaviour was risky, and 69% of the surveyed organisations suffered ransomware infections as a result. These figures highlight the importance of ensuring that users not only recognise phishing emails but also understand the organisational consequences of engaging with them. Proofpoint (2024) suggests that security awareness training should be part of a broader programme to address user knowledge gaps and reduce risk arising from human factors - an approach already widely endorsed within the cybersecurity industry (Whitman and Mattord, 2019). Given the continued dominance of phishing as an attack vector, awareness programmes should emphasise appropriate user behaviours when handling suspicious emails (Tschakert and Ngamsuriyaroj, 2019). Additionally, awareness programmes should remain agile to retain their relevance and effectiveness over time (Proofpoint, 2024).

Simulations are the most common method used to improve staff awareness of phishing emails (Althobaiti and Alsufyani, 2024). These exercises send realistic phishing emails to observe employee responses and determine whether further training is necessary. Campaigns can be targeted at teams, departments, groups of individuals, or an entire organisation. Typically, the contents of fake emails and the workflows associated with user actions are the same across all participants. Despite

their popularity, these approaches have not prevented ongoing disruptions caused by phishing (NCSC, 2021), raising questions about their long-term effectiveness.

Schneier (2000) posits that people are the “weakest link” in the security chain and are “chronically responsible for the failure of security systems” (Schneier, 2000, p. 255). This mindset of blame, focussed on the end-user, has permeated into organisations, despite evidence suggesting that it undermines the effectiveness of cybersecurity training (Renaud, Musarurwa and Zimmermann, 2021). This same study highlights that a punitive culture, where individuals are punished for failing to meet awareness expectations, can damage overall security posture by fostering distrust, guilt, fear, and disengagement from training.

1.2 Research Focus

Educating users not to interact with suspicious messages presents a complex challenge, as the underlying vulnerability lies not in a technical component but a psychological one (Jari, 2022). Phishing email contents are specifically crafted to exploit emotional triggers, rather than technical flaws, making them difficult to defend against with traditional security measures. While simulations are a widely adopted response, allowing an administrator to test user responses with realistic fake emails, this approach raises questions about how best to assess user behaviour and measure improvements. In most instances, users who “fail” the test are offered targeted training, though reports suggest that some organisations may employ more punitive measures, including formal disciplinary procedures (Segal, 2022). This variation in follow-up responses suggests a lack of consensus about how improved outcomes are best achieved, resulting in inconsistent views on the most appropriate metrics for training success.

This uncertainty extends to how simulations are received by end-users. There are conflicting findings in the literature regarding employee perceptions of phishing simulations. Rizzoni *et al.* (2022) found that these campaigns can instil feelings of distrust, leading some staff members to believe that they were being scrutinised. Feedback indicated that participants would have been more receptive to the programme if individualised content had been used and if they had been informed of the exercise. Conversely, Schiller *et al.* (2024) found no evidence of negative emotional impact and reported that the programme successfully encouraged more

cautious email behaviours. These contrasting findings, and the reported controversy surrounding simulation-based training metrics, suggest that the overall effectiveness of such interventions may be significantly affected by user perceptions and delivery mechanisms.

Jari (2022) highlighted the lack of customisable content in phishing simulations as a contributor to poor training outcomes. Targeted users were less likely to engage with campaigns when email content felt irrelevant, undermining the intended learning objectives. A comparative literature review conducted by Jampen *et al.* (2020) echoed this argument, though their recommendations centred around demographic factors, rather than user emotional response. More recently, Schiller *et al.* (2024) conducted an empirical study exploring the effect of personalised phishing training and found that allowing users to indicate their preferred training delivery mode significantly improved engagement and knowledge gain. Despite these promising findings, the evaluation of personalised delivery methods, particularly those shaped by user-stated preferences, is under-represented in the broader research field.

1.3 Research Aim and Objectives

The overall aim of this research was to investigate whether emotional responses and user-stated preferences can influence engagement and behavioural outcomes of phishing awareness training. The specific objectives of this study were as follows:

1. *Clarify* what outcomes are expected of phishing training campaigns and how they are measured.
2. *Explore* the user-driven factors (e.g., participant satisfaction, age) that are believed to influence the effectiveness of delivered training programmes.
3. *Critically assess* current methods used to deliver training campaigns.
4. *Evaluate* the implementation of personalisation (through participant preferences) into a training campaign, including the impact this could have on desired outcomes.
5. *Formulate* recommendations to the cybersecurity community on how training approaches could be adjusted to achieve better outcomes.

Delivering a phishing training programme that satisfies both organisational and end-user needs presents a complex and nuanced challenge, with a need to incorporate both statistical and more holistic measures of success. The first objective was

necessary to ensure that training outcomes were framed in alignment with both professional practice and academic literature. Building upon this foundation, objectives 2 and 3 examined the effects of user factors and personalisation on training success, driving an understanding of how training success metrics could be updated to reflect research trends. Objective 4, involving the collection and analysis of empirical data, provided meaningful insights into the effects of phishing simulation training on end-user emotional response and engagement. The overarching goal of this dissertation was to make recommendations based on the findings of the literature review and empirical research (objective 5).

1.4 Overview of Research Approach

This research comprised two main components: a review of relevant literature (objectives 1-3), and the collection and analysis of empirical data (objectives 2-4). The secondary data (i.e., literature review) were drawn largely from academic journals (e.g., Information Management & Computer Security), conference proceedings (e.g., IEEE Symposium on Security and Privacy) and reports from well-reputed organisational bodies (e.g., Proofpoint). Literature searches were supported through established library database services, particularly ACM Digital Library, Elsevier, and IEEE Xplore.

Primary data (i.e., empirical research) were obtained using surveys and an interview. A preliminary survey focussing on emotional responses to phishing simulation training was first deployed, with participants further asked to comment on communication tone and their preferences around training delivery modes. A follow-up survey gathered user feedback on a proof-of-concept phishing awareness platform, which had been developed using insights from the preliminary survey.

Participants were recruited through a range of social and professional networks, with no eligibility restrictions applied. Several subject-area experts were identified via a workplace network and approached to provide professionally grounded feedback on the developed web platform during a semi-structured interview.

Survey responses were analysed using a mixed methods approach: Likert scale data were used to derive averaged sentiment scores for use as quantitative data (Bryman, 2012), while thematic analysis was applied to all qualitative data, including open-text survey responses and interview transcripts.

This research sought an in-depth understanding of how user-driven factors influence phishing awareness training outcomes. The combination of surveys and interviews to support a literature review enabled an integration of both theoretical and practical insights. Full details of the research methodology, including design justifications and information about the web application development, are provided in Chapter 3 (Methodology).

1.5 Value of This Research

This research adds timely and practical insights into one of the most persistent and prominent challenges in cybersecurity: the human factor. Technical solutions remain central to organisational defences and are strongly emphasised in official guidance (NCSC, 2024). They are, however, inherently limited in their ability to address the emotional and behavioural factors that social engineering attacks exploit. As Schneier (2000, p. 391) comments, “there are no technical solutions to social problems”, indicating that efforts to mitigate phishing risk should consider human responses more directly. This research aims to realise that principle by combining a review of current academic and industry literature with new empirical data, exploring more user-centric approaches to training design and delivery. In particular, it responds to calls in existing literature for the investigation of customisable phishing training (Jampen *et al.*, 2020; Sarker *et al.*, 2024; Schöni *et al.*, 2025) and considers how user preferences and emotional responses might be harnessed to improve engagement, knowledge gain, and behavioural outcomes.

A key strength of this research is its dual relevance to both academic and practitioner communities. The findings contribute to academic discourse by exploring the often-overlooked emotional response to training and by assessing the value of tailoring training delivery mechanisms to individual user needs. Concurrently, a developed proof-of-concept web application demonstrates how the theory of user-stated personalisation could be translated into practice. Stakeholders in information security management, training design, service provision, and policy development may benefit from the study’s conclusions, which offer a more holistic approach to designing effective awareness campaigns. End-users may also see tangible benefits, as supportive training approaches could help reduce their exposure to phishing threats both personally and professionally. More broadly, this research

promotes a shift towards an organisational culture that values user trust and provides psychologically safe spaces for all – conditions that are essential for meaningful engagement and long-term behavioural change (NIST, 2023).

1.6 Dissertation Structure

Following this chapter, the remainder of the dissertation is structured as follows:

Chapter 2: Literature Review

This chapter defines the term “phishing awareness training” and discusses the metrics used to measure training success, examines user factors (behavioural and demographic) thought to influence training outcomes, and assesses current training delivery mechanisms.

Chapter 3: Methodology

This chapter discusses and justifies the research strategy and collection techniques adopted for the empirical data. Additionally, details of data analysis methods (quantitative and qualitative), web application development, and study limitations are discussed.

Chapter 4: Results

This chapter presents the quantitative and qualitative results from the user surveys and interview transcript data.

Chapter 5: Discussion

This chapter interprets the findings presented in the Results chapter, specifically with regards to their significance to emotional responses, the role of user-stated personalisation, and the impact of demographic variables, all within the context of phishing training delivery. Trends and reflections are presented, with further discussion of the limitations of the study.

Chapter 6: Conclusion

This chapter revisits the specific objectives of this research, with findings summaries and conclusions linked to each one in turn. Recommendations derived from the conclusions are presented, and suggestions made for future work opportunities.

The next chapter (Literature Review) examines literature relevant to this study’s objectives, starting with an exploration into what is meant by the term “phishing awareness training”.

2 Literature Review

2.1 Introduction

The introduction to this dissertation provided background and context for this research, identifying five research objectives. This literature review will focus on objectives 1, 2, and 3 (reiterated below).

1. ***Clarify what outcomes are expected of phishing training campaigns and how they are measured.***
2. ***Explore the user-driven factors (e.g., participant satisfaction, age) that are believed to influence the effectiveness of delivered training programmes.***
3. ***Critically assess current methods used to deliver training campaigns.***

By grounding the study in established theory and current practice, the literature review forms a strong foundation for the subsequent investigation of objective 4. Given the broad range of interpretations of key terms across academia and industry, this review will begin by exploring how “phishing awareness training” is defined.

2.2 Defining Phishing Awareness Training

As discussed in the introduction, phishing training is a key strategy for reducing user vulnerability to socially engineered cyber threats (Darem, 2021). The cybersecurity training industry has seen significant growth in recent years, with global revenue rising from USD 2.6 billion to USD 5.2 billion in the period 2018-2024 (Grand View Research, 2024). This growth is forecast to continue, reaching an estimated US 13.7 billion by 2030, as illustrated in Figure 2.1 (Grand View Research, 2024).

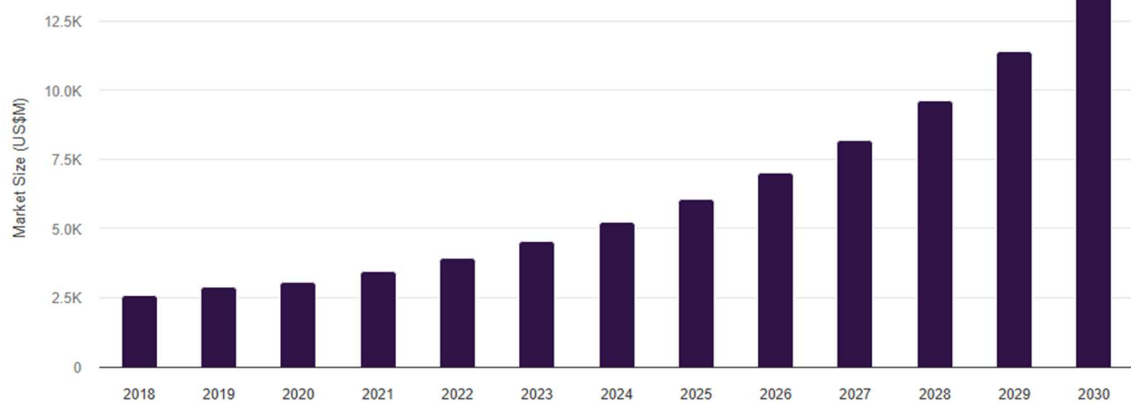


Figure 2.1: Global cybersecurity training market, 2018-2030 (US\$M) (Grand View Research, 2024)

During this period of growth, phishing awareness training (originally considered a core cybersecurity awareness training function but not a subject in its own right) has developed into a lucrative sub-sector. Its global revenue was estimated to contribute USD 1.2 billion to the broader cybersecurity training industry in 2024 (approximately 20%) and is predicted to see further expansion of 16.5% in the period up to 2033 (Verified Market Reports, 2025), visualised in Figure 2.2 below.

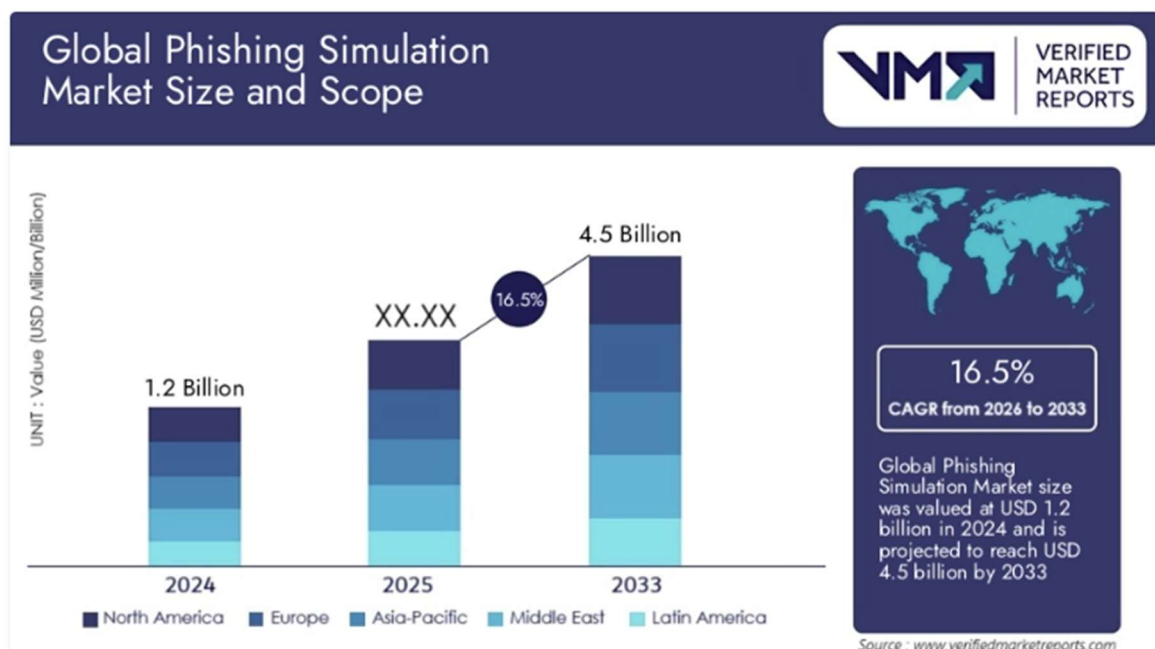


Figure 2.2: Global phishing simulation market size and scope (Verified Market Reports, 2025)

The training for this cybersecurity topic has been further categorised as being almost exclusively delivered in the form of simulation exercises. The U.S. National Institute of Standards and Technology (NIST) specifically defines phishing awareness training programmes as those that deliver “simulated phishing emails” (NIST, 2023) to employees for educational purposes. Despite the report’s definitive declaration of what phishing awareness training is, NIST’s Phish Report primarily focusses on phishing email content, rather than the use or analysis of the results of a campaign. This potentially limiting view of phishing awareness training has been largely adopted by the industry as a whole, as evidenced in the market research report referenced above. By contrast, the definition assumed in academia tends to be broader. Jampen *et al.* (2020) describe phishing awareness training more vaguely as a process that can involve multiple delivery mechanisms, not constrained to simulations alone. Sarker *et al.* (2024) further partition this process into three separate intervention types (training, education, and awareness). Thus, it appears there is a disconnect between the industry’s somewhat technical definition of what phishing awareness consists of and academia’s more holistic and human-centred approach.

Despite differing views on mindset and delivery, industry and academia do agree on the overarching goal for phishing awareness campaigns (Ho *et al.*, 2024): to develop the skills and knowledge of training participants, enabling them to respond to phishing emails more appropriately (Chen *et al.*, 2024). There are, however, further disassociations between industry and academia regarding exactly how these skills are measured, with industry favouring statistics and academia focussing on more behaviour-centric outcomes (Hillman, Harel and Toch, 2023). These competing perspectives are further complicated by the varying levels of success reported when exploring academic research around training effectiveness. Current metrics being used to measure the success (or otherwise) of training campaigns will be explored in section 2.3.

These inconsistent views around what exactly constitutes “phishing awareness training” appear to centre around scope. The prevailing definition in industry is narrow, pertaining largely to method of delivery (Lain *et al.*, 2024). In contrast, the academic community explores a wider definition, perhaps less focussed on the subject area itself and more on the concept of outcomes of training and education

(Jampen *et al.*, 2020). With these considerations in mind, this research will consider the definition of “phishing awareness training” as ***encompassing any programme designed to improve a user’s ability to deal with phishing emails in a manner consistent with established cybersecurity practices***. This includes, but is not limited to, those that incorporate phishing simulation as a delivery mechanism. Prior to discussing the current methods in use for conducting phishing awareness training, it is pertinent to discuss how exactly the outcomes of training campaigns are measured.

2.3 Measuring the Effectiveness of Phishing Awareness Training

The Kirkpatrick Model, developed by Donald Kirkpatrick in 1959 (Kirkpatrick, 1959), offers a framework for evaluating the effectiveness of training programmes. It identifies four levels of evaluation, demonstrated in Figure 2.3.

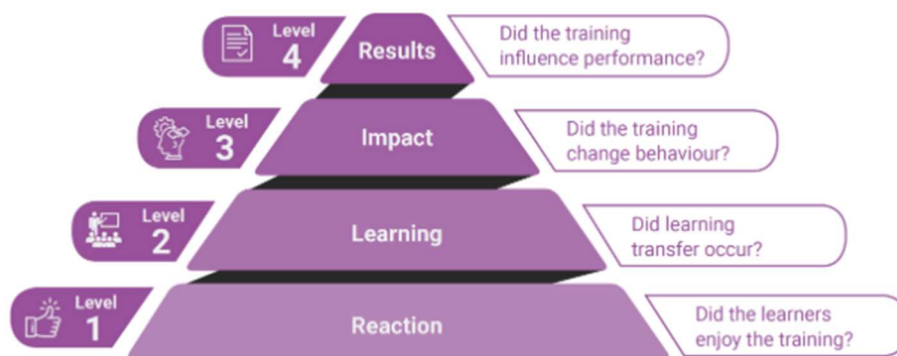


Figure 2.3: The four levels of training evaluation (Brentz, 2018)

According to the Kirkpatrick Model, administrators of training programmes should measure the effectiveness of their programmes in a multifaceted manner, looking at short-, medium- and long-term outcomes. It is therefore notable that phishing simulation is often evaluated by a single metric - that of the Click-Through Rate (CTR). It is commonly accepted that a low CTR for a training campaign is representative of an effective phishing training programme in action. This statistic, which measures the percentage of targeted users who click on a link or attachment sent in a simulated phishing email (NIST, 2023), is the most popular metric for measuring phishing training effectiveness (Hillman, Harel and Toch, 2023). Despite its popularity in both industry and academia, there are conflicting views about how

accurate this measurement is in interpreting the long-term goals of phishing simulation training.

In their extensive systematic review, Sarker *et al.* (2024) argue that basing conclusions on CTRs alone can lead to misleading or erroneous results. They hypothesise that links or attachments contained within simulated phishing emails could be opened by automated processes (such as email scanners or link preview features), leading to false positive (FP) results. This may result in a user being incorrectly informed that they have “failed” a test, which could lead to an undermining of trust in the training, possibly even eliciting frustration or resentment at the organisation. In this instance, levels one (reaction), three (impact), and four (results) of the Kirkpatrick Model are sacrificed in prioritising the one remaining level (learning), severely narrowing the scope of training effectiveness evaluation.

Sarker *et al.* (2024) highlight a further complicating matter for using CTRs as the sole measure of effectiveness – that of “prairie dogging”. This term refers to the scenario in which a user informs other users of an ongoing phishing simulation, having either received or failing it themselves. Such actions could influence other users not to click the link when they otherwise would have done, resulting in a falsely low CTR for the campaign.

Whilst the review conducted by Sarker *et al.* (2024) constitutes a significant contribution to the literature, it contains little in the way of qualitative research on this topic. A case study conducted by Schöps *et al.* (2024) frames the contradiction in using CTRs as a metric with feedback from Chief Information Security Officers (CISOs). Their qualitative research suggests that CTR metrics are favoured by CISOs because they enable the reporting of numerical information, feeding into the assumption that certain figures automatically represent good or bad outcomes definitively. The authors of this study also discuss the potential consequences of running phishing simulation campaigns in the workplace, hypothesising that simulations can increase stress levels in participants, in turn lowering perceived self-efficacy rates. They further argue that stress and perceived self-efficacy levels should form part of a set of metrics measured when assessing the effectiveness of training programmes, thus embodying levels one, three and four of the Kirkpatrick Model, allowing a more comprehensive evaluation to take place. Their

recommendations, based on both the quantitative and qualitative results obtained, advocate for critical thinking when using CTRs, agreeing not with the CISOs they interviewed, but with the wider research that this metric should be used with caution. This reinforces the need for user-centred, diversified training programmes – an aspect further explored through the personalised training approach proposed in this study.

The use of CTRs as a metric is argued for by Marshall, Sturman and Auton (2023), who compare it against measuring learning transfer with tests of knowledge (taking the form of quizzes or e-learning modules). They reason that between the two metrics, CTRs represent a more accurate measure of user susceptibility, enabling the identification of an organisation's riskiest users. They contend that using tests of knowledge as the deciding metric for evaluating effectiveness presents issues around the legitimacy of the results, as they are obtained under non-realistic circumstances.

Notably, the findings from the scoping review from Marshall, Sturman and Auton (2023) are not enforced with practical implementation. A proposal by Al Maquosi (2023) demonstrates this same limitation yet posits several alternative metrics that could be used. Despite the report's broad subject area and short length, the proposed alternatives would contribute greatly to addressing all four levels of the Kirkpatrick Model and have been categorised in a similar fashion to its own. The recommendations can be seen in Figure 2.4.

TABLE III. EVALUATION METRICS

Metric Category	Example Metrics
Engagement	Training enrollment, course completions
Comprehension	Knowledge tests, simulation performance
Skill Application	Phishing detection rates, password strength
Behavior Change	Policy compliance, malware infections

Figure 2.4: Sample alternative metrics (Al Maquosi, 2023)

Whilst Al Maquosi's proposal is written with cybersecurity training in general as its intended target, these alternative metrics could easily be applied to specific phishing awareness training. They also align with recommendations from Sarker *et al.* (2024) and Schöps *et al.* (2024) to adopt a broader set of metrics for measuring the effectiveness of training campaigns. Such a combination of metrics would allow for a broader assessment of a user's knowledge and skills that are not evaluated with a simple CTR. For example, CTRs do not measure the number of users who take the recommended action against a suspected phishing email. This metric is utilised by some training platforms (usually referred to as "report rate" or "reporting rate"). This statistic is considered to be of great importance when assessing an organisation's cybersecurity posture (NIST, 2023) and was utilised as the joint primary metric (along with CTR) to measure in a case study by Oner, Cetin and Savas (2025). Their study acknowledges that reporting rates amongst participants are not as high as would be desired, and that positive results in this area are often skewed by small groups of particularly vigilant users. This skew suggests a possible overestimation of programme effectiveness if reporting rates are considered in isolation. Their recommendations further add to the case for a set of metrics to be used in combination to model both individual risk levels and an organisation's resilience. They also propose that further studies should be done in this area, as their research was conducted on historical campaigns which they had no control over and was further limited to the finance sector, limiting the generalisability of their findings.

Metrics in use by phishing training platforms that measure actions that users take are widely discussed in the literature. By contrast, there is little to be found about statistics that measure apathetic behaviour displayed by users. Whilst there is consensus that dangerous behaviour towards phishing emails is represented by clicking a link or opening an attachment (NIST, 2023) and that desired behaviour is represented by reporting a suspicious email (NIST, 2023), measurement of actions of those users who simply delete or ignore the email remains a challenge. Likewise, little can be done to track users who forward suspicious emails to other users and the issue of users making FP identifications is scarcely discussed. Darem (2021) argues that the actions of users who report phishing emails that are in fact genuine or harmless spam should be considered when evaluating the effectiveness of a training programme. Unfortunately, this article, like those of Al Maquosi (2023) and

Schöps *et al.* (2024) presents a conceptual hypothesis with no practical implementation or empirical validation, and as such does not meet the criteria to be considered grounded theory.

A further area of scant research is that of long-term behavioural change, with the majority of studies focussing on outcomes of no more than eight weeks (Marshall, Sturman and Auton, 2023). Levels three (impact) and four (results) of the Kirkpatrick Model are aimed at evaluating the long-term effectiveness of a training programme, which CTRs do little to address. Furthermore, the scoping review conducted by Marshall, Sturman and Auton (2023) found that the level of engagement users have with their training, which would address level one (reaction) of the Kirkpatrick Model, is understudied, suggesting that further study is needed in that area. This scarcity of evidence makes it difficult to determine whether current training approaches lead to long-term behavioural change or meaningful user engagement.

The recent literature collectively signals a shift away from a singular metric (CTRs) to a more holistic set of metrics that would better encompass the medium- and long-term effectiveness of training programmes. These metrics would combine a greater level of qualitative measurements (e.g., user satisfaction and engagement) with recognition of positive behaviours (such as reporting rates) and apathetic practices (such as deleting suspicious emails) to deliver a more personalised approach to assessing user risk. The perceived inconsistencies in how training effectiveness is measured also raise important questions about whether a one-size-fits-all approach is appropriate. As later sections will explore, personalised training may offer a means to tailor interventions to user needs and thereby improve long-term outcomes.

Having established a foundation of effectiveness metrics, consideration will now be given to user factors, demographic and behavioural, that are perceived to affect training outcomes.

2.4 User-Driven Factors Affecting Training Outcomes

While a comprehensive exploration of all demographic factors affecting training outcomes is beyond the scope of this review, this section focusses on the two most frequently reported on in the literature: age and gender (Lain, Kostianen and Čapkun, 2022).

2.4.1 Age as an Affecting Factor

In a large meta-analysis conducted by Kubeck *et al.* (1996), the authors argue that older adults perform worse than younger adults in job-related training, showing less learning knowledge transfer and slower completion times. Their findings, whilst theoretical in nature, are supported by a large body of related work and offer a significant contribution to the research around the age-performance relationship. It is therefore reasonable to consider age as a potential affecting factor when measuring the effectiveness of phishing awareness training.

Although few recent studies have specifically examined the proposed link between age and training effectiveness, the empirical work conducted by Oner, Cetin and Savas (2025) partly addresses this relationship within a wider analysis of demographic factors. Their findings agree with those of Kubeck *et al.* (1996), adding that their findings suggest older adults are less likely to report suspicious emails. Despite aligning with cross-disciplinary research, their work, though robust in terms of sample size ($n=8102$) and study duration (two years), examines the effects of these demographics on user susceptibility, rather than training outcomes specifically.

The impact of age on phishing training effectiveness is often not considered when designing a study, with researchers focussing on whether a particular method of delivery is more effective than another. This demographic, along with others, can sometimes only be seen as an impacting factor once the empirical research has been conducted and results analysed. In a case study conducted by Canham, Posey and Constantino (2022), the authors ran a month-long competition for users at a large U.S.-based university. The focus of their research lies not in the effect of age but of a participant's department and job role, combined with the effect of the wording used in phishing emails. They also found, consequentially, that older participants were less successful at identifying simulated phishing emails. The validity of their consequential findings is notably limited by the artificial test conditions, with voluntary participants and financial incentives for better performance.

There is uncertainty in the connection between older adults and poorer performance, with some studies revealing the opposite. In fact, the study by Oner, Cetin and Savas (2025), despite proposing that older adults are less likely to report suspicious

emails, concludes that younger adults demonstrate less cautious behaviour. This is supported by work undertaken by Sumner *et al.* (2021) who argue that the wider body of research suggests that younger people are more likely to “fail” a phishing simulation exercise.

By contrast, there is emerging research that purports that age (and other demographics) has such little effect on training outcomes that it is not worth considering. In a case study highly focussed on the personalisation of training material, Schöni *et al.* (2025) removed all demographics from their regression models after determining their inclusion resulted in no improvement to the model’s fit. This finding is echoed in reviews undertaken by Tschakert and Ngamsuriyaroj (2019), Jampen *et al.* (2020), and Buckley *et al.* (2023), all of which conclude that age cannot be said to influence training outcomes. With such debate seen in the literature, it is reasonable to conclude that age alone should not be considered a deciding factor in training effectiveness, further strengthening the need for diverse and personalised training approaches (Sarker *et al.*, 2024; Sutter *et al.*, 2022).

2.4.2 Gender as an Affecting Factor

The contradictory findings into how age may affect training outcomes are mirrored in the studies that explore the impact of gender, with Lain, Kostinen and Čapkun (2022) going so far as to describe this demographic as “divisive”. Returning to the study by Sarker *et al.* (2024), they conclude, both in the review of literature and in their empirical research, that gender has a significant impact on training outcomes, with females more likely to “fail” simulated phishing tests on multiple occasions. It is their recommendation, based on their findings, that gender should be considered when designing and implementing phishing awareness training programmes. It’s important to highlight that their case study was restricted to historical data from a single organisation, limiting their ability to assess whether factors like email content influenced gender-specific responses.

Similar to the studies exploring age as an influencing factor, there exists strong evidence of inconsistencies around the effects of gender on training outcomes, even within single studies. The comparative literature review conducted by Jampen *et al.* (2020) discusses some instances in the wider body of research that support the findings of Sarker *et al.* (2024), but also directly contradict those same findings

(Flores *et al.*, 2014). Their review identifies several papers that report inconsistent or inconclusive findings with specific regard to the impact of gender, thereby complicating interpretation of the field's wider discoveries. Whilst the work of Jampen *et al.* (2020) is an excellent point of reference for research conducted by others into factors influencing phishing training outcomes, the paper is highly descriptive in nature and offers little insight into why certain correlations might have been found – only that they have been.

The inconsistencies reported by Jampen *et al.* (2020) are replicated in other reviews exploring the factors impacting phishing training effectiveness (Buckley *et al.*, 2023; Hillman, Harel and Toch, 2023; Sumner *et al.*, 2021). Whilst each of these studies makes valuable contributions to the field, each of them has limitations, further complicating definitive interpretation: the metric used by Buckley *et al.* (2023) was highly subjective (participants were asked to measure their own emotional state) and no control group was used. By contrast, the variables measured by Hillman, Harel and Toch (2023) in their phishing simulations are arguably too numerous, confusing the effects of any one individual factor. Furthermore, they discuss the possibility that training outcomes could also be impacted by behavioural factors, but this is not explored. The final review mentioned here, a case study conducted by Sumner *et al.*, (2021), was severely hampered by the COVID19 pandemic, making reliable results near impossible to achieve. As these studies show, the contradictory and inconclusive results relating to gender emphasise further the need for a more human-centred and holistic approach to the design and implementation of phishing awareness programmes – a direction this study intends to explore in broad terms, though not specifically relating to gender.

2.4.3 Behavioural Factors

Beyond demographic factors, a substantial body of research has explored participants' emotional responses to phishing simulations, with the results showing little of the inconsistencies suffered by the work into age and gender. There is overwhelming support for two complementary perspectives: positive emotional responses improve training outcomes, whilst negative user experiences have a detrimental effect. In a large-scale case study ($n=4554$), qualitative and quantitative research conducted by Lain *et al.* (2024) showed that positive attitudes towards a

phishing training programme increase engagement and learning rates. Participants reported a feeling of responsibility that was credited to a positive experience with their training, making them more likely to understand the risks involved with phishing emails and less likely to click on links or attachments in suspicious emails.

In a novel study conducted by Hull, Schuetz and Lowry (2023), phishing training was administered by way of narrative storytelling, enriching participant experience to stimulate learning transfer and knowledge retention. Their findings offer strong conclusions to favour this approach, seeing improvements in both behavioural (e.g., satisfaction and enjoyment) and objective (e.g., CTRs) metrics. This positive outlook is further strengthened by the findings of Schöps *et al.* (2024) who posit that not only do simulated phishing campaigns have the potential to cause higher levels of stress in those users who “fail”, but that those heightened levels of stress have an adverse effect on training outcomes.

The results of these and other studies exploring subtler measurements add to the argument that achieving better training outcomes requires a broader understanding of the user-driven factors influencing those outcomes. The concept of catering more heavily to these human-centric nuances (in combination with user demographics and broader metrics) is gaining traction within academia with results suggesting that offering employees a range of training options may dramatically improve training effectiveness (Schöni *et al.*, 2025; Schiller *et al.*, 2024). This area of knowledge is still being developed and warrants further investigation.

Building on the earlier analysis of metrics and user-driven factors that influence training outcomes, the following section explores the current approaches used to deliver phishing awareness training.

2.5 Current Approaches to Delivering Phishing Awareness Training

Although a full examination of specific training delivery methods (e.g., gamification, e-Learning, videos) lies beyond the capabilities of this review, this section concentrates on the most extensively documented approach in the literature: embedded phishing training.

Whilst there is no industry-wide approved definition of “embedded phishing training”, the description provided by Sutter *et al.* (2022) is an accurate representation of the mechanism as seen across industry and academia:

“Embedded training is an anti-phishing training methodology where participants are confronted with education material whenever they fail a phishing attack simulation. The basic idea of embedded training is to directly engage the participant with educational material whenever a dangerous activity such as clicking on a link or submitting credentials is performed. Often educational material in the form of videos, illustrations, games, or text is shown to participants when they fail one of the anti-phishing simulations.”

(Sutter *et al.*, 2022)

As can be seen in this definition, embedded phishing training is not restricted to one delivery method (e.g., e-Learning) but encompasses multiple methods. The key characteristic of embedded phishing training is that it is an instant response to a user clicking on a link or attachment in a simulated phishing email. This approach is grounded in the assumption that immediate feedback reduces participant confusion. The use of real-time responses aims to facilitate meaningful learning transfer (Sutter *et al.*, 2022; Jampen *et al.*, 2020; Schiller *et al.*, 2024), a principle supported by established learning science theory (Ho *et al.*, 2024). A case study by Singh *et al.* (2023) argues for this use of immediacy when delivering phishing awareness training. In it, the authors tested the vigour of immediate training when compared with a delayed training mechanism (classroom delivery) and found that users participating in immediate training spent significantly longer periods of time examining the training material presented to them than those in the delayed training group. The results of the study further showed that the group receiving immediate training went on to perform better in post-tests. Although the study presents its findings with methodological rigour, it is important to acknowledge that participants were aware of each phishing simulation as it occurred.

Schiller *et al.* (2024) argue that integrating training into a user’s routine supports the need for immediate, embedded phishing education. However, qualitative evidence from their own study suggests that many targeted users voiced a preference for

scheduled training sessions, allowing them to manage their time more effectively. The case study conducted by Lain *et al.* (2024) provides further evidence to contradict the argument for immediate training delivery. Their findings found that delaying training for users who have “failed” the test was at least as effective as immediate training. Furthermore, the feedback received from their empirical research suggests that participants receiving a delayed training course had a more positive outlook with regards to phishing awareness programmes, further improving training outcomes. The participants of this study were not subjected to mandatory training if they “failed” the test, which is not reflective of standard industry practice (Centre for Research and Evidence on Security Threats (CREST), 2020). This limitation, and the short study duration (two months), make generalisation of the results challenging.

There is growing evidence within this field of research to suggest that the effectiveness of embedded phishing training is, controversially, not as impressive as historical data might suggest (Lain *et al.*, 2024; Sarker *et al.*, 2024), with some work even suggesting that it is less effective than other delivery mechanisms (Lain, Kostinen and Čapkun, 2022). This perspective is further explored in a scoping review conducted by Marshall, Sturman and Auton (2023), which examines how the type of training employed (e.g., gamification) can influence the overall impact of embedded phishing training. Their findings acknowledge the benefits that can be delivered using immediacy but also recognise that this delivery method faces limitations in its effectiveness as users request detailed feedback in response to their mistakes. Participants also expressed a desire for increased input into the intensity of the training they receive, suggesting that material personalised to individual users would improve training outcomes.

Whilst Marshall, Sturman and Auton (2023) did not explore the concept of integrating customisation of training content, Schöni *et al.* (2025) address this gap by incorporating personalisation (based on user proficiency) into existing embedded phishing training. Their findings suggest that not only does content personalisation result in a significant improvement on training outcomes, but that users reported a more positive experience and expressed greater satisfaction with the personalised content they received. The arguments presented in the conclusion of this new research are directly at odds with the “one-size-fits-all” approach exemplified in standard embedded phishing training (Sarker *et al.*, 2024) and advocates for a more

granular approach to delivery of phishing awareness training. They acknowledge the limited attention this area of research has received and that their own findings lack long-term insights. The potential role of personalisation by way of user-selected training in shaping outcomes remains underexplored in the literature and will be further investigated as part of Research Objective 4.

2.6 Conclusion

This chapter makes a significant contribution to the research by reviewing the literature relating to the effectiveness of phishing training, emphasising methods of delivery, and how user-driven factors are perceived to affect training outcomes. This exploration provides the basis for meaningful discussion and critical analysis of phishing awareness training, particularly in relation to the limitations of current training methods and lack of personalisation in campaign delivery.

In summary, the academic community offers complex and often contradictory insights into the effectiveness of phishing awareness training in its current form. Training outcomes are particularly influenced by emotional responses to training delivery and content, underscoring the importance of designing training programmes that foster positive engagement with participants. While embedded phishing training remains the industry standard approach, emerging evidence suggests that greater benefits may be achieved by offering users increased flexibility and choice in training modes. There is a lack of research conducted into the potential benefits that offering user-selected training modes could generate. Further study should focus on integrating this into delivering more nuanced, adaptive training interventions that have the capability to reflect the diverse needs of users.

3 Methodology

The Literature Review highlighted the need for further research into the impact of incorporating end-user preferences into phishing training programmes on training outcomes. This chapter will describe the main research strategy adopted to explore this issue, including participant selection, artefact development, and data analysis methods. Furthermore, limitations and ethical implications of the research process will be identified and considered.

This chapter focusses on Research Objectives 2-4, collecting empirical data on user attitudes, evaluating current training features, and assessing the impact of personalisation within phishing awareness training. To recap, Research Objectives 2-4 were as follows:

- 2. *Explore the user-driven factors (e.g., participant satisfaction, age) that are believed to influence the effectiveness of delivered training programmes.***
- 3. *Critically assess current methods used to deliver training campaigns.***
- 4. *Evaluate the implementation of personalisation (through participant preferences) into a training campaign, including the impact this could have on desired outcomes.***

Providing a detailed, systematic, and transparent explanation of the research process will enable other researchers to critically assess this study's reliability and validity and support future replication or extension of the work.

3.1 Research Strategy

A mixed methods approach was adopted to carry out this empirical research. The mixed methods approach allows a researcher to collect and integrate both quantitative and qualitative data into a single study. Combining the strengths of these two types of data allows for greater understanding of a research problem than the use of either one in isolation (Bryman, 2012; Creswell, 2009).

This research focusses on attitudes and opinions of participants with regards to the ability to incorporate their training mode preferences into a mandatory phishing training programme. Whilst emotional and behavioural responses do not lend themselves easily to objective quantitative measurement, these responses can be

subjectively assessed using Likert-type ordinal scales (Harpe, 2015). Using such a scale enables interpretation of responses in a quantitative manner, the results of which can then be corroborated with corresponding qualitative data. This process, named triangulation, enables a researcher to form conclusions not just about individual participants, but also to identify patterns and broader trends within the sample (Bryman, 2012; Creswell, 2009). This triangulation feature of the mixed methods approach fits the needs of this research, enabling a deeper understanding of user responses than a purely qualitative study would have allowed.

A convergent parallel model was used to administer the mixed methods approach. Such a model is defined by the following characteristics (Creswell, and Plano Clark, 2015):

- Quantitative and qualitative data are collected concurrently.
- Both data types are weighted equally.
- Data types are treated separately during the analysis phase.
- Data are integrated when the results are interpreted.

This definition is further demonstrated in Figure 3.1 below.

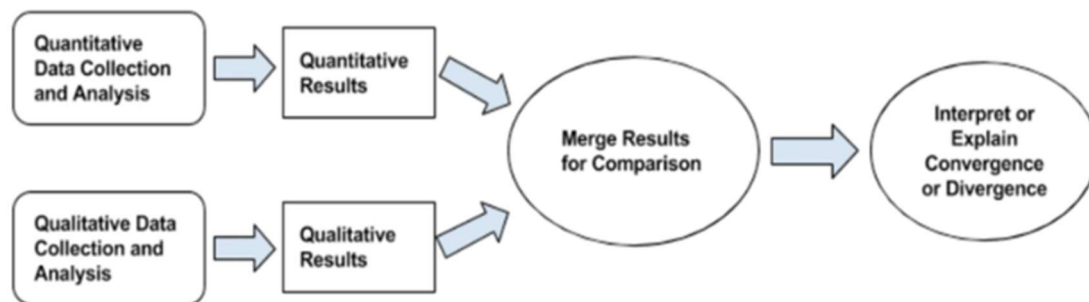


Figure 3.1: Convergent parallel mixed methods design (Creswell and Plano Clark, 2015, p.56)

In addition to collecting different types of data, this study also employed multiple collection techniques, consistent with the principles of mixed methods design. A combination of surveys and interviews was used. The former were distributed for collection of preliminary participant responses and later for feedback contributing to the evaluation of the artefact (an activity aligned with the principles of developmental research, discussed in section 3.3). The latter were employed solely to obtain expert views further evaluating the artefact. The specifics of each of these techniques will

be explored in sections 3.2-3.4. A diagram outlining the stages of the research strategy can be seen in Figure 3.2 below:

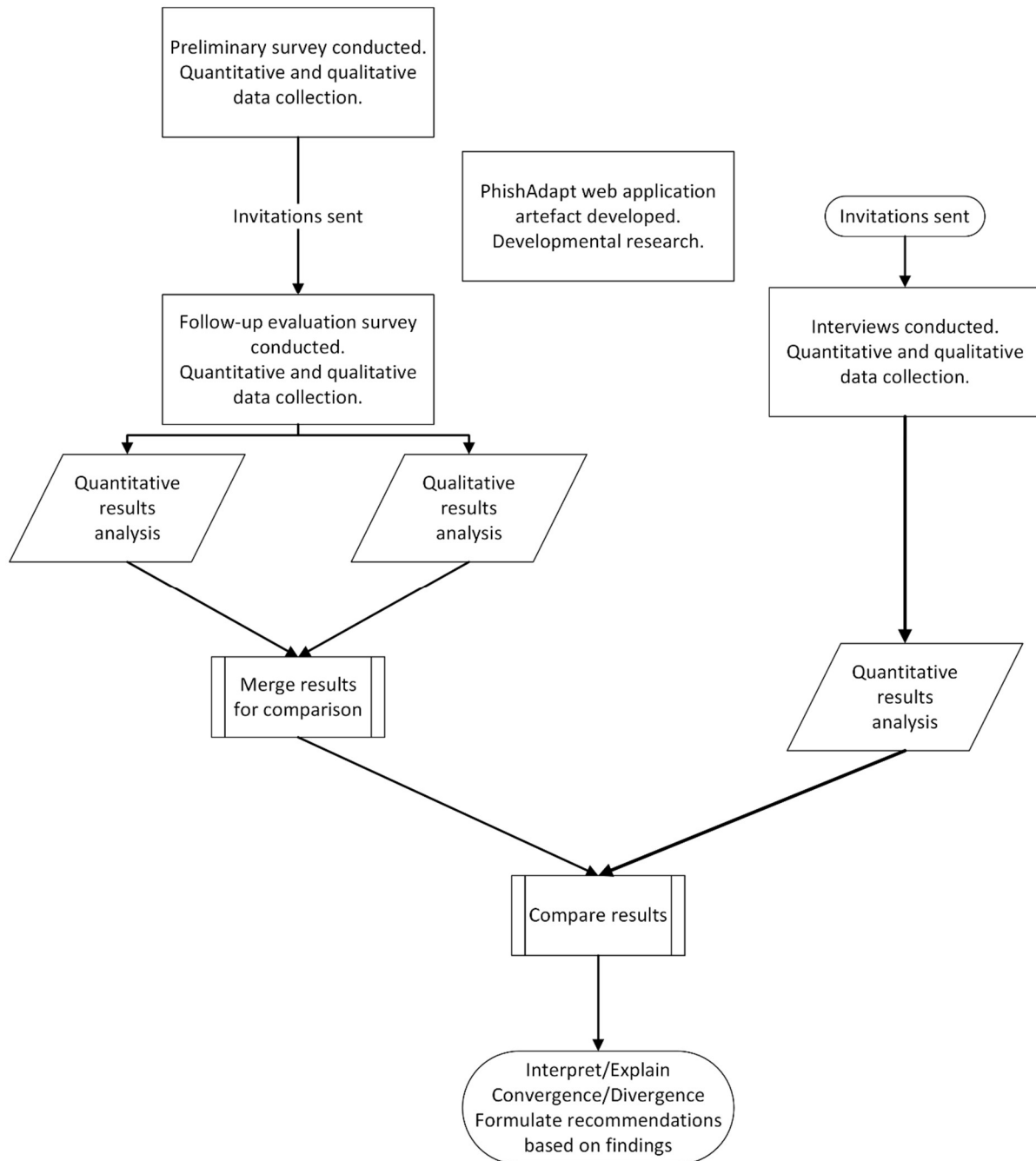


Figure 3.2: Research strategy process

Mixed methods research offers both breadth and depth of data (Burke Johnson, Onwuegbuzie and Turner, 2007), making it a well-suited choice for a study seeking to evaluate the impact of user-led personalisation on training outcomes. The

approach capabilities align directly with Research Objective 4 and allow the empirical findings to be meaningfully synthesised with the theoretical insights from the Literature Review.

The following sections will provide a more detailed account of the collection techniques, along with justification for their selection.

3.2 Survey Instrument

It is important to note that using surveys as part of a mixed methods research strategy is distinct from adopting a survey research method (Glasow, 2005). Whilst the latter depends almost exclusively on the former as a tool to collect data, surveys can be used as tools for research strategies other than the specific survey research method. The term “survey” is used in this paper to describe the specific data collection technique used as part of a mixed methods approach, not as a descriptor for a research strategy itself.

As outlined earlier, a preliminary survey was administered to gather participant feedback on phishing training programme approaches. Surveys are broadly defined as a method of asking a set of individuals a pre-defined set of questions, with the aim of generalising the results to a wider population (Goodfellow, 2023). They are further recognised as being an appropriate data collection technique for research that includes specific thoughts and opinions from individuals that cannot be obtained from any other source (Phillips, 2016). Given these characteristics, surveys are well-suited for collecting data from participants about their individual emotional responses to phishing simulation training programmes.

This preliminary survey contributed to the development of the artefact by capturing end-user perceptions relevant to Research Objectives 2, 3, and 4, particularly in relation to training preferences and emotional engagement. This survey was divided into four main sections:

1. Introductory questions.
2. Scenario-based questions (these made up the main body of the survey and consisted of four different scenarios, decreasing in tone negativity and severity with each iteration).
3. Concluding questions.
4. Collection of demographics.

The survey was designed using best practice principles formulated by Gehlbach and Artino (2018), with particular emphasis on those outlined in Table 3.1 below:

Table 3.1: Survey ethical considerations and their implementations

Guideline	Implementation
Avoid using negative language	Positive language used
Choose question formats that yield data formats appropriate to the question	Variety of question types used dependent on the question – further detail provided below
Use an appropriate number of response options	Likert scales use a 5-point scale as per best practice (Sullivan and Artino, 2013)
Ensure each question applies to every participant	Branching used where necessary
Use scales in preference to single-choice response options	Likert scales used for questions that have a range of possible responses
Ensure consistent visual layout throughout the survey	Template used for main body of survey
Place sensitive questions later in the survey	Demographic collection items placed on the final page of the survey

When designing the survey, the desired response type was taken into consideration. It was deemed necessary to use a variety of question types to ensure the response yielded was appropriate to the purpose of the question. Two main question types were used (Sauro, 2018):

- Open-ended – these enabled participants to provide free-text responses, e.g., “What do you understand ‘phishing’ to be?”. Acquisition of qualitative data pertinent to participant opinion was the purpose of these items.
- Closed-ended – these consisted of either multiple choice (single or multiple response), e.g., “Have you heard of ‘phishing’ before now?”, or rating scales, e.g., “Please read the following statements and select how strongly you agree or disagree with each one”. Acquisition of quantitative data pertinent to participant emotional response was the purpose of these questions.

A full copy of the preliminary survey, including all questions, is provided in Appendix A for reference.

Several ethical considerations were addressed during this stage of the research strategy. Fisher (2020) provides a summarised list of specific ethical considerations for all types of survey research. Table 3.2 below outlines how these were addressed in the implementation of the preliminary survey:

Table 3.2: Ethical considerations and their mitigation measures for the preliminary survey

Consideration	Mitigation
Confidentiality	Survey conducted using Microsoft Forms and OneDrive storage (hosted by research institution) – provides GDPR compliance (Microsoft, 2025)
Informed consent	All participants provided with information on survey content and purpose, as well as voluntary nature of research, in the information sheet
Anonymity	All identifying data (names and email addresses) removed from data prior to analysis
Persuasion/pressure	Participants informed of their right to withdraw without penalty in the debrief statement
Disclosure of interest	No conflicting interests to disclose

The information sheet was displayed before the commencement of the survey and included information about the voluntary nature of participation. Participants were required to confirm their understanding of the information provided to them with the use of declaration statements. Appendix B contains a copy of the information sheet, a copy of the debrief statement (provided to participants upon the completion of the survey), and a copy of the approval granted by the research institution's Ethics Committee for transparency. With these safeguards in place, the ethical concerns were considered appropriately addressed.

3.3 Artefact Design and Development

According to Ellis and Levy (2009), the core purpose of developmental research is to answer the question of how building a “thing” (artefact) can address a defined problem. Within this definition, the building of a web application for the purposes of deploying personalised training content fits the scope of developmental research. As identified in the Literature Review, there is a notable gap in existing studies exploring the effect of user-selected training modes on training outcomes. Furthermore, Research Objective 4 seeks to evaluate that effect, justifying the development of an artefact to explore the relationship between training personalisation and participant emotional response.

Ellis and Levy (2009) further posit that the developmental research process consists of three major elements:

1. Stipulation of criteria for the artefact.
2. Use of a formalised development process.
3. Evaluation of the artefact's ability to meet its criteria.

The evaluation component was addressed through the use of follow-up surveys and a semi-structured interview with experts – these will be discussed in section 3.4. The core functionality that the artefact needed to exhibit was the automated delivery of personalised training content in response to user interaction with simulated phishing emails. This process is illustrated in Figure 3.3 below:

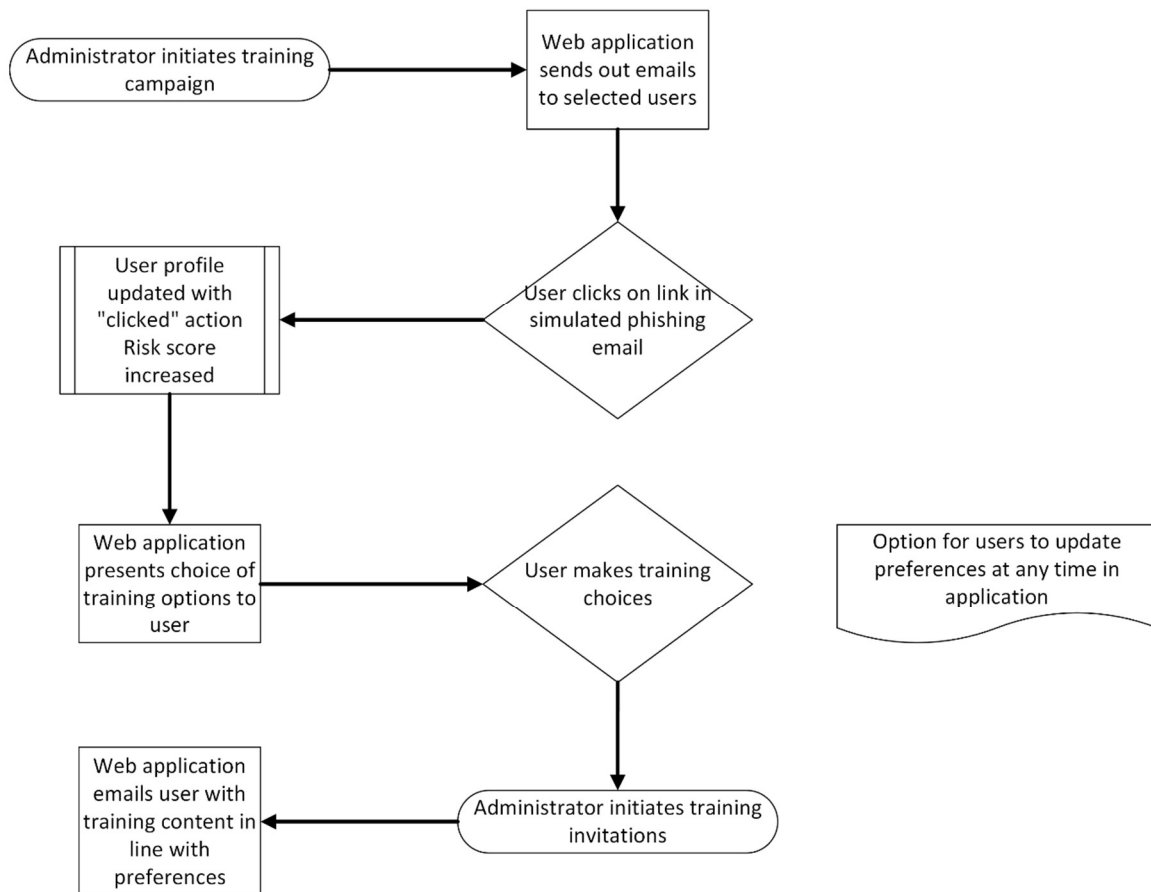


Figure 3.3: Web application core functionality process

With the criteria established, and a process for evaluation identified, the remainder of this section will now focus on the development process for the artefact. Insights obtained from the preliminary survey directly informed the design of the artefact by identifying the potential for emotional responses to phishing training content. As development was conducted in an isolated environment with no external participants¹, there were no ethical concerns during this stage of the research process.

3.3.1 Web Application Coding and Implementation

“PhishAdapt” (the name chosen to reflect both the central functionality of the application and its ability to offer adaptive content) was written using Microsoft Visual Studio Code (VS Code), a free open-source platform for code development. VS Code was selected owing to its versatility: support and extensions are provided for

¹ Profile information for internal web application users was generated using a fake profile generator

hundreds of coding languages. The platform is powerful yet has a user-friendly graphical interface (GUI) (Tan, Chen and Shuyin, 2023) and extensive support documentation is readily available².

To support the artefact's core features, the system was hosted on a virtual machine configured with SMTP and IMAP support for email handling. A graphical browser and email client were also installed. Full setup instructions and software specifications are provided in Appendix C.

The web application was developed using Python as the server-side coding language, Flask as the front-end web framework, and SQLite as the database language. Table 3.3 outlines the reasons for these choices.

With the system components specified, work began on identifying the functional requirements needed to support the intended functionality: automated delivery of personalised training via email. A directory tree of the underlying structure of the web application and a list of the features developed (along with their contributing files and functions) is provided in Appendix D. Furthermore, the full code for the developed application can be found in Appendix E or downloaded from the public GitHub repository³. Content for phishing simulation template emails was obtained from "Phish Bowl", a publicly available repository owned by the London School of Economics and Political Science displaying real-life phishing emails received within their organisation (LSE, n.d.a).

² <https://code.visualstudio.com/docs>

³ <https://github.com/janekpstudentuad/PhishAdapt>

Table 3.3: Web application components, their purpose, and reasoning for their selection

Component	Purpose	Reasoning
Python	Provided server-side logic for web application functionality	Simplicity and readability of language (Madeira, 2023) Easy integration with HTML, CSS, and SQL languages (Instaily Academy, 2023) Streamlined integration with back-end web frameworks (BrowserStack, 2025) Wealth of available libraries available for extra functionality, e.g., data analysis (Educative, 2025)
Flask	Front-end framework for building web pages and routing capabilities	Coding language closely follows Python conventions (Fox, 2023) Ability to implement “boilerplate code” to create dynamic yet reusable HTML templates (da Fonseca, 2024)
SQLite	Database for storing user, profile, and training campaign data	No requirement for a separate server or server process (Shinde, 2023) Embeds directly into web application logic with no configuration required (GeeksforGeeks, 2024)

In recognition of the role that effective visual branding can play in eliciting emotional responses (Xiaoyan and Xiaoling, 2020), a concept which aligns with the aims of Research Objective 4, a logo was created. To compensate for a lack of available graphic design experience, ChatGPT was used to generate a logo using the following prompt:

“Colourwise - the predominant colours in my research institution's crest are blue, yellow and a little red so any of these or a combination would work. For an icon, I like the idea of a fish morphing into some other marine animal, to represent the adaptive element of my application, but not too serious! I want it to look approachable without looking childish. Could I have a version with and without text? I don't have a preference for the shape, just not anything too angular.”

The resulting logo output by ChatGPT, and the landing page of the finished web application can be seen in Figure 3.4.

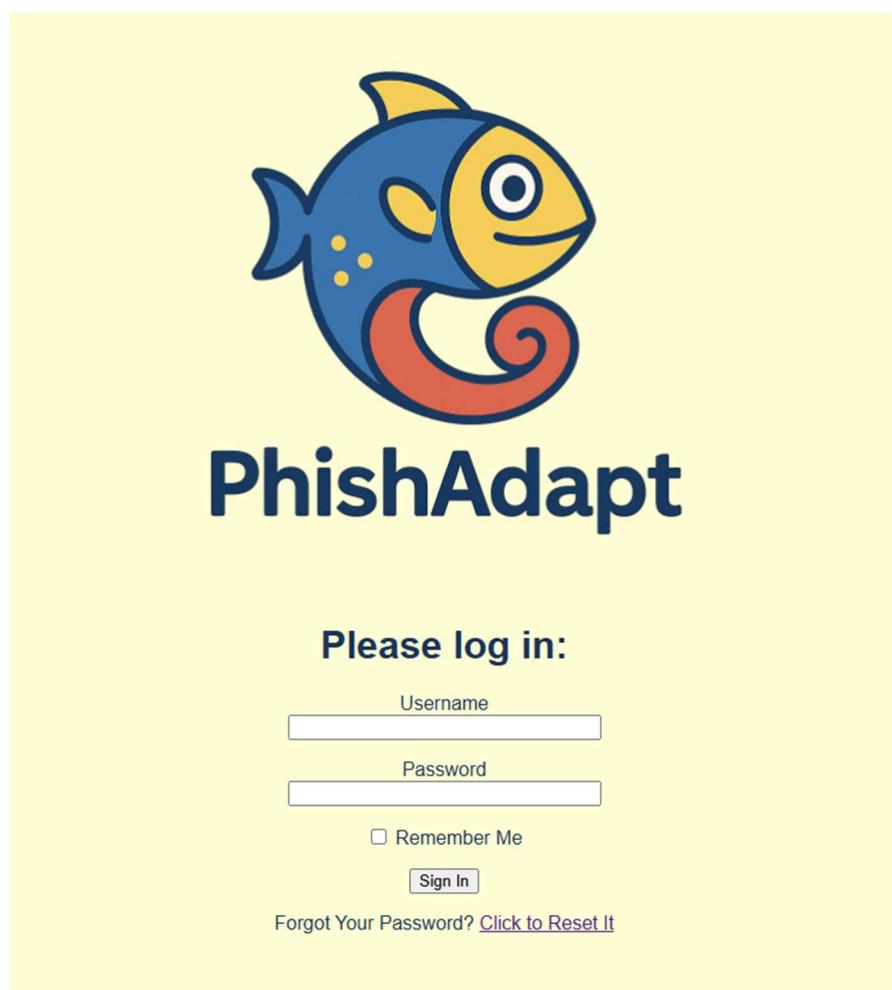


Figure 3.4: PhishAdapt landing page with AI generated logo

3.4 Evaluation Methods

In line with the final stage of the developmental research process outlined by Ellis and Levy (2009), it was necessary to evaluate PhishAdapt's features once the core functionality had been implemented. In order to satisfy this requirement, a follow-up survey and expert interview were conducted, selected for their suitability in conducting ex post (i.e., after implementation) evaluation of an artefact in instances where new functionality is representative of a research gap (Sonnenberg and vom Brocke, 2012). The responses obtained from this evaluation stage of the process, particularly those focussing on user engagement with a personalisation feature, would build upon those from earlier stages, providing enrichment and contextualisation to further inform and support Research Objectives 4 and 5.

3.4.1 Follow-up Surveys

The design of the follow-up survey followed the same principles as discussed in section 3.2 and will not be reiterated here. Furthermore, any ethical concerns encountered during the development of the preliminary survey remained when developing the follow-up survey and were addressed in the same manner. A further ethical concern (that of maintaining anonymity of participants across two separately collected datasets) was addressed with the use of randomised participant IDs, which were generated by ChatGPT with the following prompt:

"Can you generate a list of 67⁴ random 3-digit numbers? They must all be unique."

The ID numbers were sent to participants in an invitation email, a copy of which can be found in Appendix F, which also contains the contents of a reminder invitation that was sent to participants three days before the survey closed. The questions used for this second survey mirrored those in the first exactly to ensure fair comparisons could be made during the analysis phase. A full copy of the follow-up survey is provided in Appendix A for reference.

⁴ The number of responses received for the preliminary survey

3.4.2 Interviews

An initial approach was sent to identified participants, with a formal invite to interview issued upon confirmation of a willingness to take part in the research. The content for these communications can be found in Appendix F.

The interview questions were written with the specific purpose of obtaining expert insight into the core functionality of the PhishAdapt web application: the ability to deliver personalised training content in response to user preferences. A list of the questions to be used in the semi-structured interview, all of which were open-ended, is included in Appendix G.

Interviews, when used as a data collection method, share similar ethical concerns to surveys (Fisher, 2020). These were addressed in a slightly different manner given the difference in delivery method: considerations and mitigations are outlined in Table 3.4.

The information sheet was provided to the participants in the invitation to interview. Participants were required to confirm their understanding of the information provided to them by returning a signed copy prior to the interview taking place. A copy of the information sheet and the debrief statement (provided to participants upon the completion of the interview) is included in Appendix B for reference and transparency. With these safeguards in place, the ethical concerns were considered appropriately addressed.

Table 3.4: Ethical considerations and their mitigation measures for the interview

Consideration	Mitigation
Confidentiality	Interview conducted, recorded, and transcribed using Microsoft Teams and OneDrive storage (hosted by research institution) – provides GDPR compliance (Microsoft, 2025) No corporately sensitive information requested of participants
Informed consent	All participants provided with information on interview content and purpose, as well as voluntary nature of research, in the information sheet
Anonymity	All identifying data (names) removed from transcription data prior to analysis
Persuasion/pressure	Participants informed of their right to withdraw without penalty in the debrief statement
Disclosure of interest	No conflicting interests to disclose

3.5 Data Collection and Analysis

3.5.1 Collection

Whilst it is challenging to identify the number of phishing simulation training exercises in use globally, data from over 12.5 million users were analysed in a report published by KnowBe4, providing insight into the size, depth, and breadth of the industry (Grimes and Kraemer, 2023). With this in mind, participants for the preliminary survey were recruited using convenience sampling, a technique considered appropriate when the population is very large (Etikan, Musa and Alkassim, 2015). A range of personal, academic, and professional networks were utilised for survey distribution to ensure the collection of sufficient data for meaningful analysis. No exclusionary criteria were applied for participation.

The follow-up survey was distributed two weeks after deployment of the preliminary survey. Invitations were sent to all participants who included their email address in their original response. Provision of this information was optional, and participants were advised of the possibility that they might be contacted with follow-up questions at the time of preliminary survey completion.

Participants for the interviews were identified using purposive sampling (specifically expert sampling) through a workplace network, which is particularly appropriate in areas where there is a current lack of observational evidence available (Etikan, Musa and Alkassim, 2015). The criteria for their participation were determined by their administrative involvement with phishing simulation training campaigns within the workplace.

Response rates to survey and interview recruitment will be detailed in Chapter 4 (Results). Delivery methods and ethical concerns for each instrument are discussed in their respective subsections of this chapter.

3.5.2 Analysis

As outlined in section 3.1, this research involved both quantitative and qualitative data collection. Names and email addresses were removed from survey data (preliminary and follow-up) and anonymised labels given to interview participants prior to data analysis. The following subsections describe how data was categorised, the analytical methods applied for each type, and the tools used during the analysis process.

3.5.2.1 Quantitative Data

All closed-ended responses from the preliminary and follow-up surveys were categorised as quantitative. Table 3.5 details the types of data received from these responses with the data types allocated to each question outlined in Appendix H. This appendix also contains details of the labels and codes applied to the data pertaining to these responses, which allowed for accurate analysis.

Table 3.5: Types of quantitative data in survey responses

Type of data	Example of data
Dichotomous	Yes/No
Ordinal (categorical data with a clear order of rank)	Very unlikely/Very likely (and intermediary responses)
Nominal (categorical data with no ranking)	Full-time employed/Retired (and other values in this category)
Multi-response nominal (multi-value responses of categorical data with no ranking)	Videos/Quizzes (and other values in this category)

Sentiment scoring was used to compare participant responses across the four phishing training scenarios presented in the preliminary survey. This technique, averaging the sum of the scores of multiple responses, is known to produce a more reliable measure of overall participant emotional responses than considering the responses in isolation (Koo and Yang, 2025). To minimise potential bias in an overall sentiment score (per scenario), Likert scale response questions were grouped together by theme to produce an interim sentiment score. These interim scores were then averaged to provide a single composite overall score for the scenario. Any aggregated scores were treated as scalar data. Table 3.6 lists which variables were grouped together for this purpose, along with the names of the resulting output variables.

Table 3.6: Interim and composite sentiment score components

Scenario-based questions - same set for each scenario, distinguished with use of "sc ⁿ " prefix (e.g., sc1_positivity for this set of questions in Scenario 1)	
Summed variables	Output variable
- likely_respond - likely_engage	engagement_propensity
- cybersecurity_mindset_response - organisation_mindset_response	mindset
- angry_response - bullied_response	negativity
- empowered_response - motivated_response - invested_response - trust_response	positivity
- learn_lessons_response - skill_set_response	training_outcome
Theme-specific aggregates merged into overall sentiment score for each scenario, distinguished with use of "sc ⁿ " prefix (e.g., sc1_overall_sentiment_score for this set of sentiment scores in Scenario 1)	
- engagement_propensity - mindset - negativity⁵ - positivity - training_outcome	overall_sentiment_score

Exploratory data analysis (EDA) of the survey data (both preliminary and follow-up) was conducted on a per-question or per-scenario basis (the latter was used when dealing with sentiment scores). Table 3.7 outlines the types of statistical analysis applied to each type of data.

⁵ Variable was negatively scored to allow for consistent averaging to be applied

Table 3.7: Analytical methods applied per quantitative data type

Type of data	Analytical methods
Dichotomous	Frequency
Ordinal	Frequency distribution, descriptive statistics (median, mode)
Nominal	Frequency distribution, descriptive statistics (median, mode)
Multi-response nominal	Multi-value frequency distribution, descriptive statistics (modes)
Scalar	Frequency distribution, descriptive statistics (mean, standard deviation)

A Friedman test was used to analyse the differences between participant emotional responses to the five scenarios presented to them. This test was used as it is a recognised alternative to parametric tests when analysing repeated measures within one group in non-normally distributed data (Kim, 2014), as is often the case with survey-based data (Bono *et al.*, 2017). Normality of data was checked using the Shapiro-Wilk test, specifically recommended when analysing sample sizes of less than 50 (Ghasemi and Zahediasl, 2012), and visually inspected with Q-Q plots. Post-hoc tests on any statistically significant differences were conducted using individual Wilcoxon signed-rank tests with Bonferroni correction applied, in accordance with standard practice (Kim, 2014). Effect sizes between pairwise groups were calculated with the formula $r = |Z| / \sqrt{n}$ (n =sample size and Z =standardised test statistic produced by Wilcoxon signed-rank test) and categorised in accordance using standard thresholds for effect sizes, shown in Table 3.8 (Datanovia, 2020).

Table 3.8: Effect sizes in statistical correlation

<i>r</i> value	Effect size
0.1 - < 0.3	Small
0.3 - < 0.5	Moderate
>= 0.5	Large

A Kruskal-Wallis H test was used to check for statistically significant differences in scenario emotional response medians between demographically different groups. This test was used as it is a recognised robust, non-parametric alternative to the One-way ANOVA in cases of data that are not normally distributed (Lantz, 2012). Post-hoc tests on any statistically significant differences were conducted using pairwise comparison with Bonferroni correction applied, in accordance with standard practice (Kim, H., 2014).

Lastly, a Chi-Square test was used to investigate correlations between demographic groups and scenario preferences. This test was used as it is appropriate for exploring relationships between categorical variables, with a mixture of ordinal and nominal data, and is considered a robust test to use that can produce detailed information from non-normally distributed data (McHugh, 2013).

For all statistical tests conducted in this study (including normality assessments) a significance level of $p < 0.05$ was used to determine statistical significance as this is the most commonly used standard in applied statistics (Field, 2018).

3.5.2.2 Qualitative Data

All open-ended responses from the surveys (preliminary and follow-up) and all responses obtained from the interview were classed as qualitative data. These were analysed following the well-established framework set out by Braun and Clarke (2008). A hybrid deductive/inductive approach was used: a practice commonly used when research objectives drive the data analysis process (Fereday and Muir-Cochrane, 2006) as was the case with this research. This process is outlined in Figure 3.5.

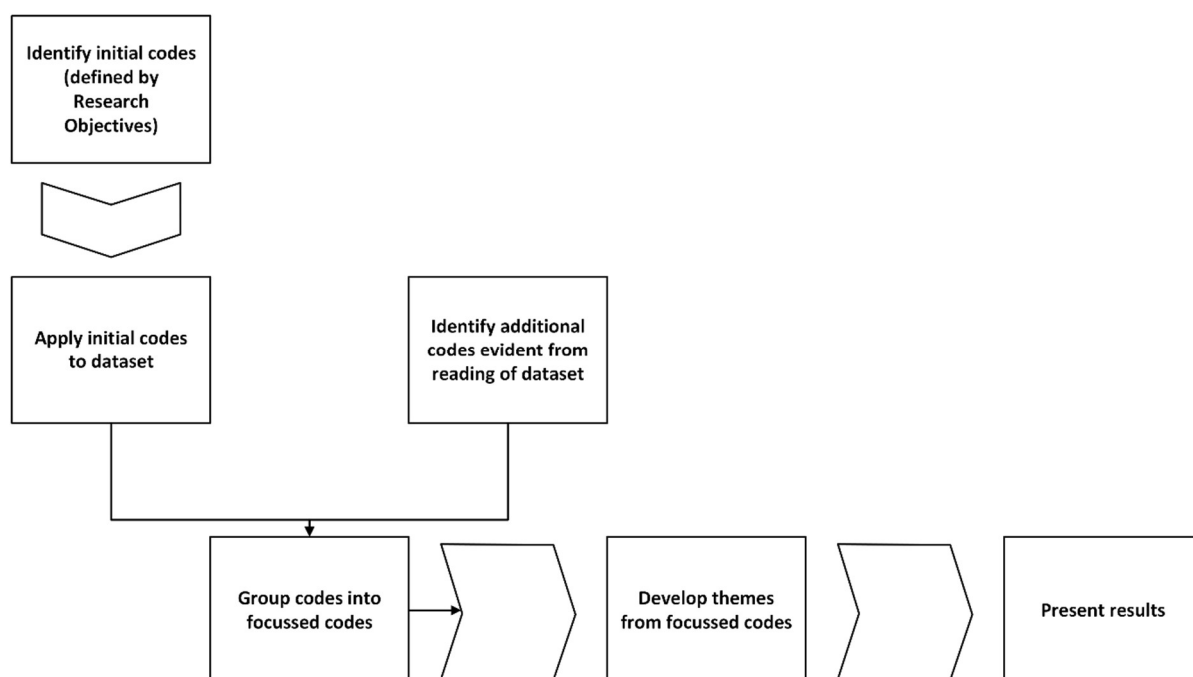


Figure 3.5: Hybrid deductive/inductive qualitative data analysis process

The list of initial codes developed from the Literature Review and Research Objectives (particularly 2, 3, and 4) can be found in Appendix H. For the scenario-based section of the survey, responses to the two open-ended questions associated with each scenario were grouped by scenario. This took place after the coding and theming process, enabling analysis of participants' qualitative feedback in the specific context of each training scenario. Results were presented in a theme-led interpretation of responses.

3.5.2.3 Tools

With the preliminary and follow-up surveys conducted using Microsoft Forms, data from these instruments were exported in .xlsx (Microsoft Excel) format. Interview response data was captured in a .docx (Microsoft Word) document. Microsoft Teams was used to record and transcribe the interview proceedings, with the transcription checked manually. The tools used to conduct the analysis of the quantitative and qualitative data, respectively, are detailed in Table 3.9 below.

Table 3.9: Tools used for quantitative and qualitative data analysis

Tool	Role	Reasoning
Python	OHE for quantitative data ⁶ Qualitative data organisation (export responses into per-participant ⁷ and per-question ⁸ documents)	Simplicity and readability of language (Madeira, 2023) Powerful and efficient when performing data structuring and organisational tasks (Scalosoft, 2023)
IBM SPSS Statistics	Quantitative data analysis	Easy to use Powerful Performs a wide range of statistical procedures (LSE, n.d.b)
Pandas and matplotlib libraries (for Python)	Visualisation of multiple bar charts in one plot for Likert scales ⁹ Creation of word cloud ¹⁰	More flexibility and versatility than SPSS (Tancred, 2021)
Microsoft Excel	Calculation of r scores for Wilcoxon Signed-Rank tests (effect sizes)	Functionality to calculate r not included in SPSS Quickly and easily applies functions dynamically to data (Microsoft, 2024)
NVIVO	Qualitative data analysis	Ease of use Allows clear organisation of codes (Kent State University, 2025)

⁶ Example code provided in Appendix H

⁷ Example code provided in Appendix H

⁸ Example code provided in Appendix H

⁹ Example code provided in Appendix H

¹⁰ Example code provided in Appendix H

3.6 Limitations

Having outlined the research methods and analytical processes, critical reflection on their limitations should be discussed, with a focus on those applicable to the strategy and data collection techniques used. These are summarised in the numbered list below for ease of reference:

1. A case study is likely to have provided a more robust evaluation of the effects of user-defined training modes, as opposed to the purely theoretical approach implemented.
2. Use of convenience sampling can introduce bias, with the sample failing to represent the wider population accurately (Etikan, Musa and Alkassim, 2015).
3. Purposive sampling often results in small sample sizes, making generalisation of the findings from the data obtained challenging (Etikan, Musa and Alkassim, 2015).
4. Relying on self-reported data, as was the case with the surveys and interview, may result in response bias or tendency to respond in a socially desirable way (Arnold and Feldman, 1981).
5. The extent of emotional response to PhishAdapt's features may not have been fully captured due to technology constraints (e.g., participants were asked to evaluate static images of the artefact).

The absence of a wider skillset (to include graphic and game design experience) and the limited time available for this research were the main causes of each of the points detailed but have particular relevance to points five and (to a lesser extent) one. Measuring long-term behavioural change requires a minimum of five months (Jampen *et al.*, 2020), which exceeded the scope of this project. Despite these constraints, the research design was carefully constructed to ensure methodological rigour, and the data collection and analysis were conducted in a transparent and systematic manner. As such, this study provides a valid and reliable foundation for future work. It is positioned as a preliminary exploration that can inform and support more extensive, case study-based investigations in the future.

3.7 Summary

This chapter has outlined the research strategy adopted for this research, including justification for the mixed methods approach. Data collection methods, analysis techniques, and artefact development have also been detailed. Limitations of the research design have been discussed, particularly with respect to sampling, potential bias, and working within the time constraints of a limiting deadline. Despite these limitations, the methodological decisions taken were grounded in relevant literature and chosen to best address Research Objectives 2, 3, and 4. The following chapter will present the results obtained through these methods, structured around the key components of the data collection process.

4 Results

Having described the strategies, methods, and techniques used to conduct this study in the previous chapter, this chapter presents the empirical findings, beginning with the survey data, which is structured chronologically according to the sequence of the research instrument implementation. Quantitative and qualitative results are included, with particular focus given to key themes, participant sentiment, and statistical comparisons across survey stages. This chapter ends with a presentation of the results from the interview. Interpretative analysis is reserved for Chapter 5 (Discussion).

4.1 Preliminary Survey Results

The preliminary survey was advertised online to personal and professional contacts through social media platforms such as LinkedIn and Facebook. All participants, of which there were 67, completed the survey online using Microsoft Forms. The full anonymised dataset from the survey is included as a supplemental file submitted alongside this dissertation (Survey Responses Appendix.xlsx).

4.1.1 Quantitative Data

4.1.1.1 Demographics

All age groups were represented, with just over a third (34.3%) of respondents aged between 45-54, and a further 25.4% aged between 34-44. Gender representation skewed slightly towards female participants who comprised two thirds (66.7%) of the total sample. The distributions for age and gender are visualised in Figure 4.1 and Figure 4.2, respectively, with the full demographic breakdown provided in Table 4.1 for completeness and transparency.

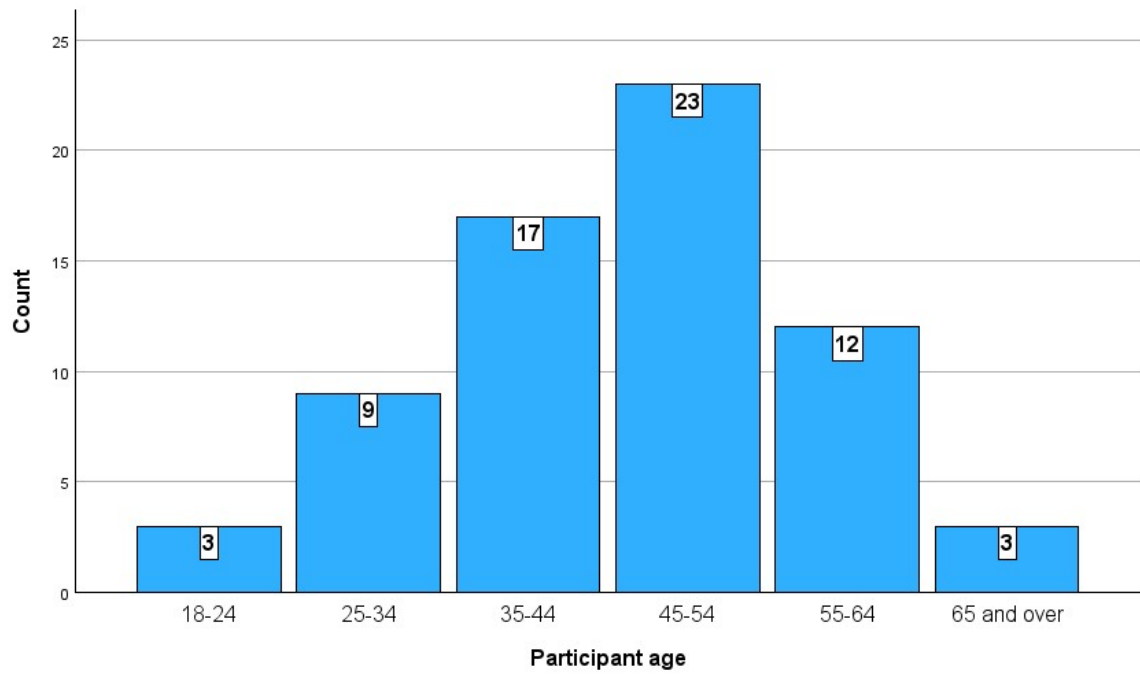


Figure 4.1: Bar chart of participant age frequencies

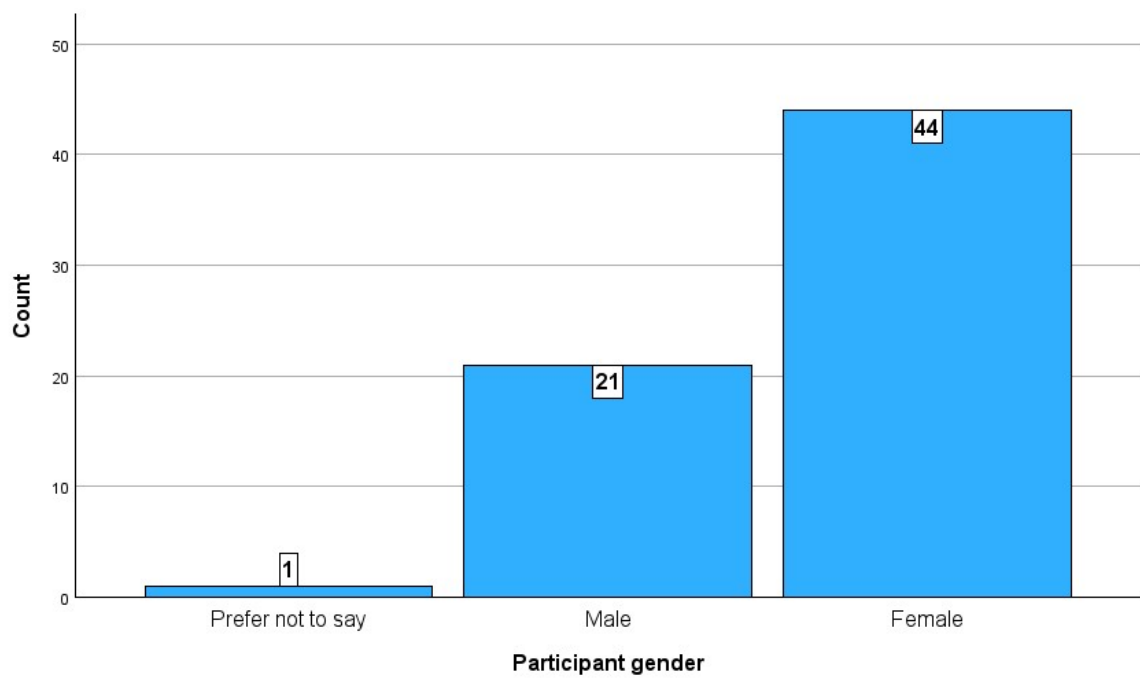


Figure 4.2: Bar chart of participant gender frequencies

Table 4.1: Demographic group breakdown of participants

		Count	Column N %
Participant age	18-24	3	4.5%
	25-34	9	13.4%
	35-44	17	25.4%
	45-54	23	34.3%
	55-64	12	17.9%
	65 and over	3	4.5%
Participant gender	Prefer not to say	1	1.5%
	Male	21	31.8%
	Female	44	66.7%
Participant employment status		1	1.5%
	Other	1	1.5%
	Full-time employed	53	79.1%
	Part-time employed	5	7.5%
	Self-employed	3	4.5%
	Unemployed	0	0.0%
	Student	0	0.0%
	Retired	3	4.5%
	Homemaker/carer	0	0.0%
	Unable to work due to disability or illness	1	1.5%
Highest level of education completed by participant		1	1.5%
	No formal education	0	0.0%
	Primary education	1	1.5%
	Secondary education	3	4.5%
	Vocational/technical qualification	4	6.0%
	College/further education	10	14.9%
	Bachelor's degree or equivalent	33	49.3%
	Postgraduate degree (Master's, PhD)	15	22.4%
Participant country	UK	65	97.0%
	NL	1	1.5%
	US	1	1.5%
Participant first language		2	3.0%
	English	63	94.0%
	Dutch	1	1.5%
	German	1	1.5%

4.1.1.2 Introductory Questions

When asked about existing knowledge, 66 (82.09%) participants reported awareness of phishing, 59 of whom felt that their understanding did not change when presented with the NCSC definition. These findings are presented in Table 4.2.

Table 4.2: Table of participant knowledge of phishing

		Count	Column N %
Participant has heard of phishing before	Not sure/don't know	0	0.0%
	No	1	1.5%
	Yes	66	98.5%
	Not applicable	0	0.0%
Participant understanding of phishing has changed after NCSC definition	Not sure/don't know	2	3.0%
	No	59	88.1%
	Yes	6	9.0%
	Not applicable	0	0.0%

Over four fifths of participants were both aware of phishing simulations being used as a training tool and had participated in such simulations in the past (82.1% and 79.1%, respectively). Of those that had participated, over a third (35.9%) stated either that they had not learned anything from them, or that they were unsure if they had. These findings are visualised in Figure 4.3.

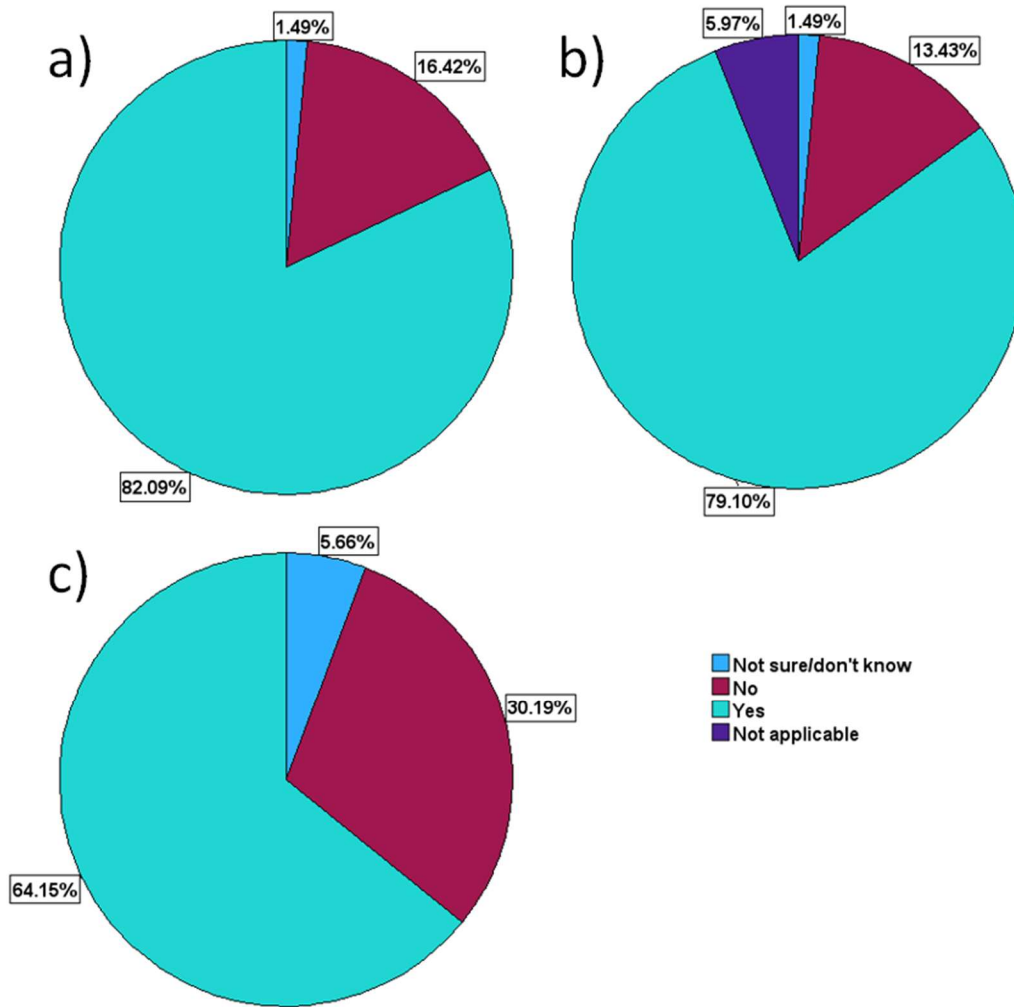


Figure 4.3: Pie charts of experience with phishing simulation training: a) Participant aware b) Participant has participated c) Participant has learned

4.1.1.3 Scenario-based Questions

For each scenario, an overall sentiment score (ranging from 0 to 5) was calculated as detailed in the Methodology chapter. Visualisations for interim sentiment scores are presented in Appendix J. Only the overall sentiment scores are analysed statistically in this chapter.

Exploratory analysis of the overall sentiment scores for Scenarios 1–4 is summarised in Table 4.3. The boxplots presented below (Figures 4.4–4.7) illustrate the distribution of scores for each. Scenario 1 showed the lowest mean score ($M=2.62$, $SD=0.90$), with scores largely between 2 and 3.25. Scenario 4 received the

highest sentiment scores ($M=4.04$, $SD=0.95$), with most responses falling between 3.5 and 4.75.

Table 4.3: EDA of overall sentiment scores

			Statistic	Std. Error
Overall sentiment score (Scenario 1)	Mean		2.6216	.11085
	95% Confidence Interval for Mean	Lower Bound	2.4003	
		Upper Bound	2.8430	
	5% Trimmed Mean		2.5977	
	Median		2.4000	
	Variance		.823	
	Std. Deviation		.90734	
	Minimum		1.00	
	Maximum		4.65	
	Range		3.65	
	Interquartile Range		1.25	
	Skewness		.549	.293
	Kurtosis		-.660	.578
Overall sentiment score (Scenario 2)	Mean		3.5933	.10690
	95% Confidence Interval for Mean	Lower Bound	3.3799	
		Upper Bound	3.8067	
	5% Trimmed Mean		3.6253	
	Median		3.7000	
	Variance		.766	
	Std. Deviation		.87498	
	Minimum		1.50	
	Maximum		5.00	
	Range		3.50	
	Interquartile Range		1.30	
	Skewness		-.585	.293
	Kurtosis		-.270	.578

Overall sentiment score (Scenario 3)	Mean		3.6791	.09285
	95% Confidence Interval for Mean	Lower Bound	3.4937	
		Upper Bound	3.8645	
	5% Trimmed Mean		3.6931	
	Median		3.8000	
	Variance		.578	
	Std. Deviation		.75999	
	Minimum		2.00	
	Maximum		5.00	
	Range		3.00	
	Interquartile Range		1.30	
	Skewness		-.193	.293
	Kurtosis		-.680	.578
Overall sentiment score (Scenario 4)	Mean		4.0410	.11616
	95% Confidence Interval for Mean	Lower Bound	3.8091	
		Upper Bound	4.2730	
	5% Trimmed Mean		4.1110	
	Median		4.3000	
	Variance		.904	
	Std. Deviation		.95079	
	Minimum		1.00	
	Maximum		5.00	
	Range		4.00	
	Interquartile Range		1.60	
	Skewness		-1.037	.293
	Kurtosis		.394	.578

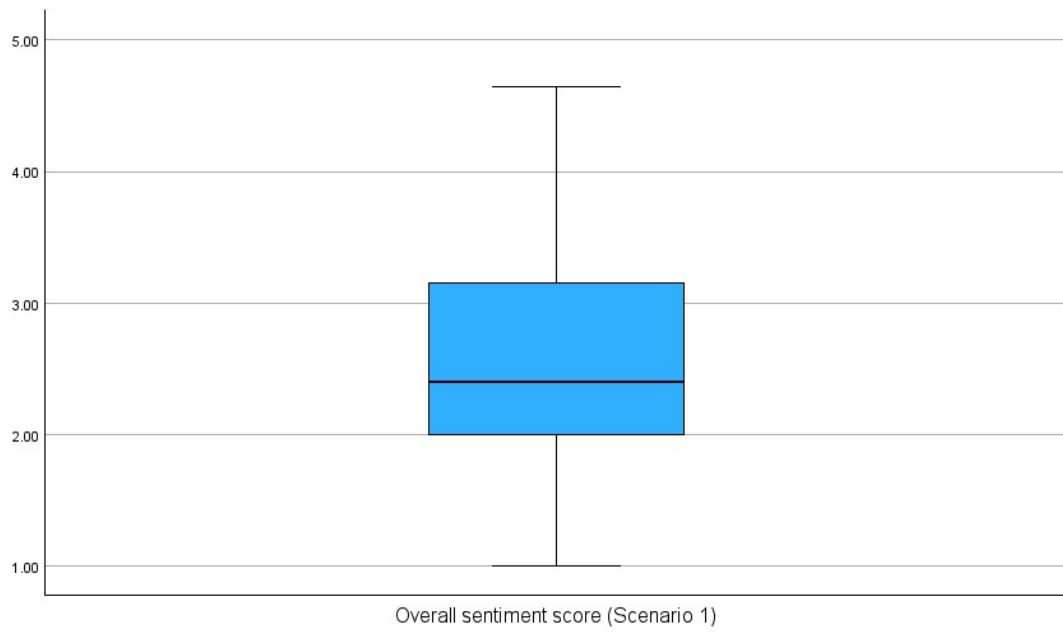


Figure 4.4: Boxplot for Scenario 1 overall sentiment scores

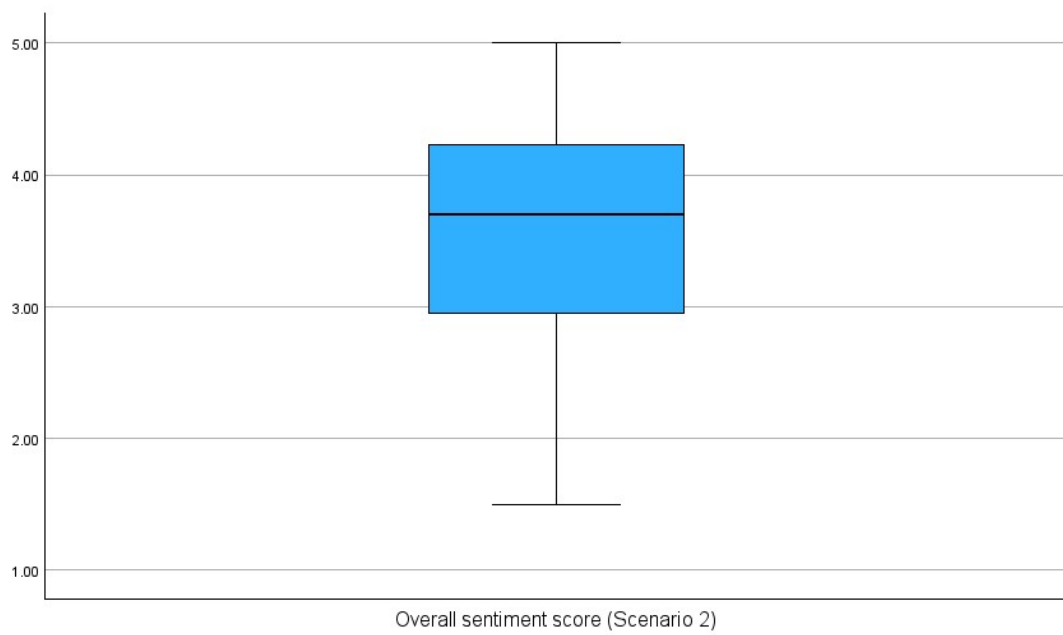


Figure 4.5: Boxplot for Scenario 2 overall sentiment scores

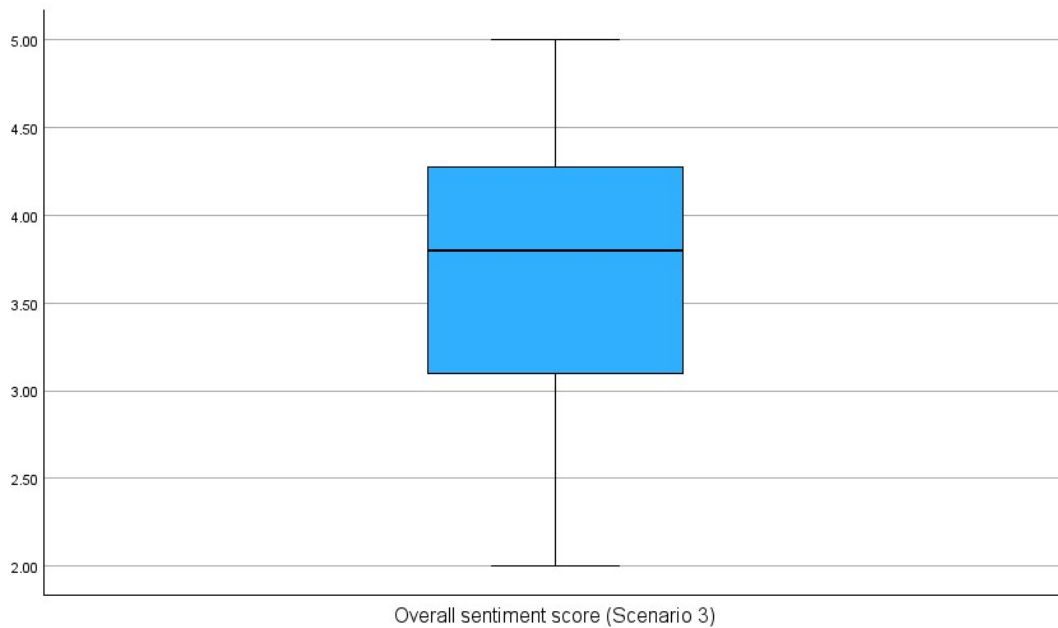


Figure 4.6: Boxplot for Scenario 3 overall sentiment scores

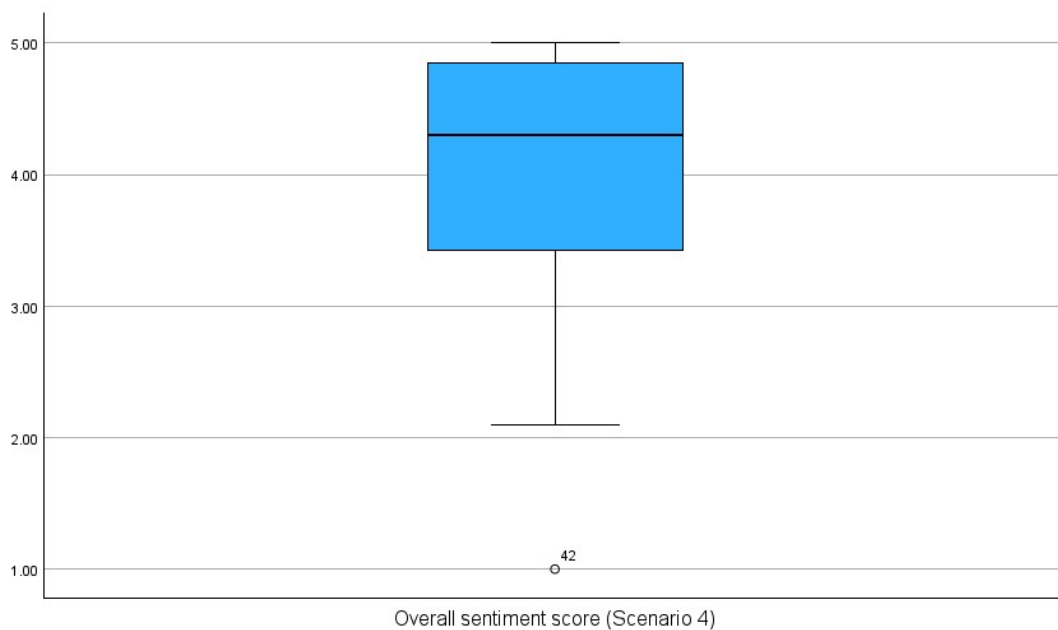


Figure 4.7: Boxplot for Scenario 4 overall sentiment scores

4.1.1.4 Concluding Questions

When asked to specify their preferred scenario, almost half (44.8%) of participants identified Scenario 4 (which used a supportive tone and offered training personalisation). This was also the most commonly specified scenario that respondents deemed most likely to engage with (49.2%) and most likely to improve their skill set (56.7%). For reference, Scenario 1 featured an authoritative, punitive tone; Scenario 2 used a formal but less punitive tone with a short deadline for

mandatory training completion; Scenario 3 adopted a supportive tone with no formal deadlines. Figure 4.8 shows frequency distributions for these three questions, with full breakdowns provided in Appendix J.

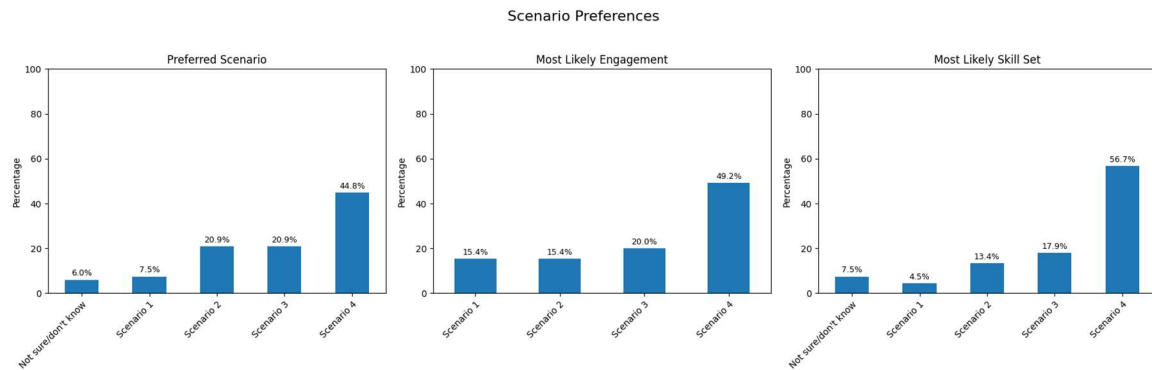


Figure 4.8: Bar charts for frequency distributions of: a) Preferred scenario, b) Scenario most likely to engage with c) Scenario most likely to improve skill set

Participants signified quizzes and e-learning as the most popular desired modes of training. Reading was identified as the least popular. Frequency distribution for all choices offered is visualised in Figure 4.9.

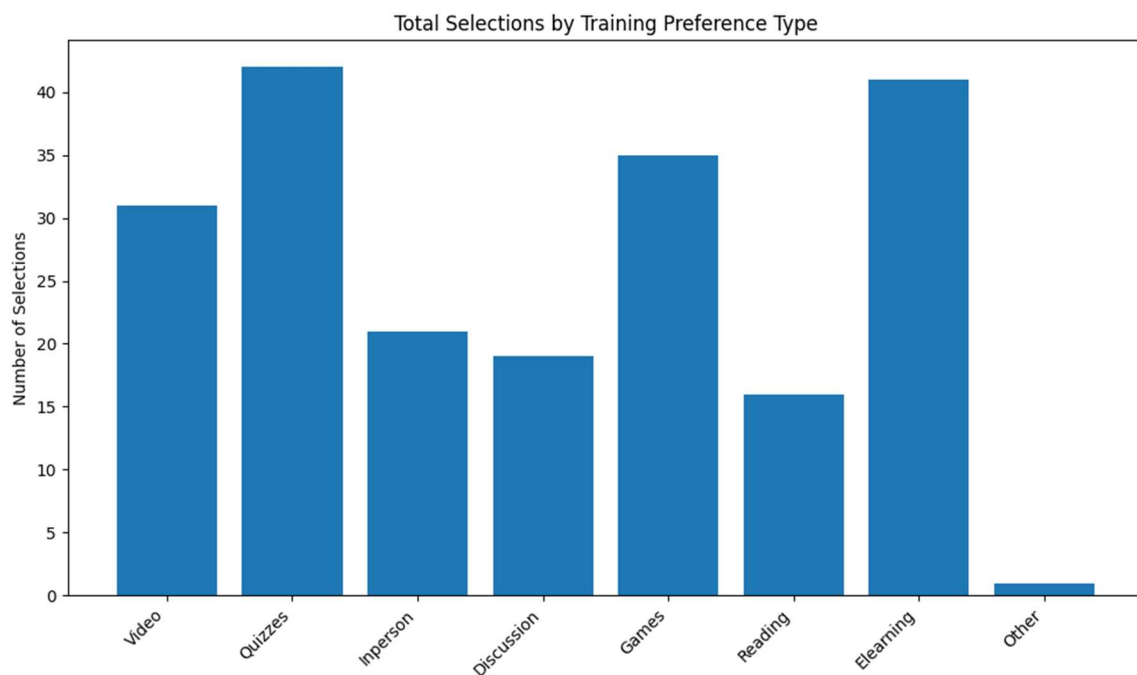


Figure 4.9: Frequency distribution of desired training modes

4.1.2.2 Scenario-based Questions

4.1.2.2.1 Thoughts on Scenario 1 Messaging

Scenario 1's tone was authoritative and stern, with an emphasis on the recipient's "failure". It stipulated consequences for non-compliance with completing urgent mandatory training. Nine focussed codes were identified amongst participant responses to this scenario's open-ended questions. The most common themes were *Emotional and Psychological Responses to Phishing Training* and *Outcomes and Organisational Impacts of Phishing Training*, appearing in 54.7% and 57.9% of responses, respectively. Participants consistently emphasised the stress and worry that they perceived this particular messaging tone would cause them:

- P439 – "I would feel terrible and stressed"
- P366 – "Scared and worried about to lose my job"

Respondents also frequently commented on the tone severity being used, perceiving it to be an unreasonable reaction to the initiating circumstances:

- P140 – "Aggressive and disproportionate"

4.1.2.2.2 Thoughts on Scenario 2 Messaging

The tone used for Scenario 2 was more polite and less severe than for Scenario 1. It removed some of the personal accountability from the recipient but retained a moderate sense of urgency with the use of a deadline for mandatory training to be completed. The dominant theme across responses was *Evaluation of Training Design and Delivery*, seen in over half of responses (58.5%) pertaining to this theme. Several comments centred on the deadline aspect of the message, with many participants feeling that it was too demanding:

- P473 – "Not enough time to engage"
- P892 – "Unrealistic due to other business needs"

4.1.2.2.3 Thoughts on Scenario 3 Messaging

The messaging for Scenario 3 adopted a friendlier and more supportive tone without accusation of "failure". It emphasised the training on offer and was more suggestive than directive, with no specific deadline for completion of mandated training. Eight of the nine developed focussed codes were identified within the questions pertaining to this scenario. The most prominent themes were *Evaluation of Training Design and Delivery* and *Outcomes and Organisational Impacts of Phishing Simulation Training*,

contributing to 69.5% and 63.4% of responses, respectively. Many participants perceived the tone of this scenario would prove to be ineffective in motivating them to complete the mandatory training:

- P472 – “[...] would probably just delete the response and not follow up on the training.”
- P342 – “Sounds a bit like you could ignore it though if they don't give a deadline or check that you have completed the training.”

Respondents also repeatedly focussed on the supportive tone of the message:

- P529 – “This response makes me feel better about myself”
- P239 – “Much more positive motivator”

4.1.2.2.4 Thoughts on Scenario 4 Messaging

The final scenario presented in the preliminary survey was empathetic, inclusive, and collaborative, offering customisable learning for individuals. It refocussed the responsibility of “failure” away from the individual, sharing this with the organisation involved. A total of seven focussed codes were observed across participant responses to this scenario’s open-ended questions. The most commonly observed theme was *Evaluation of Training Design and Delivery* (70.9%), which manifested itself repeatedly in positive comments regarding personalisation of training options:

- P464 – “I feel that it is a good offer to provide different learning styles”
- P241 – “the options would most likely encourage me to complete some training.”

As with Scenario 3, there were no occurrences of the focussed code *Impact on user behaviour and reporting* in the responses for this scenario although participants often noted the supportive tone as contributing to a positive reaction to the message:

- P239 – “I would respond really well to this feedback being put so clearly with support”

4.1.2.3 Concluding Questions

In a final set of questions, participants were asked to indicate which of the four scenarios they preferred and which they felt most likely to engage with and learn from. They were also invited to comment on the reasons behind their choices, with the aim of gaining insight into the factors that influence end-user satisfaction with

training programmes. A total of seven focussed codes were observed across participant responses to this section's open-ended questions. The most prominent themes were *Evaluation of Training Design and Delivery* and *Outcomes and Organisational Impacts of Phishing Simulation Training*, appearing in 58.1% and 55.4% of responses, respectively. Many comments centred on the use of a supportive tone, indicating that it promoted encouragement and instilled a sense of collaboration:

- P432 – “You feel supported and encouraged to learn.”
- P817 – “It makes it more collaborative not you have been naughty!”

When discussing the training design, participants frequently commented on the appeal of content personalisation. This particular initial code (*like the personalisation of content*) was seen in over three quarters (76.9%) of all responses in which the respondent identified Scenario 4 as their preferred scenario (the only scenario which included this characteristic).

- P187 – “I like them asking how they can make training engaging in a multitude of ways”
- P725 – “It allows me to pick the most suited method of training”

Notably, there were no occurrences of the focussed codes *Impact on user behaviour and reporting* or *Perceived effectiveness of simulations* in the responses for this set of questions.

4.2 Follow-up Survey Results

Follow-up survey invitations were sent to all preliminary survey respondents who had provided an email address (59), with almost 60% (35) responding. The follow-up survey was conducted online using Microsoft Forms. The full anonymised dataset from the survey is included as a supplemental file submitted alongside this dissertation (Survey Responses Appendix.xlsx).

4.2.1 Quantitative Data

As with the preliminary survey scenarios, an overall sentiment score (ranging from 0 to 5) was calculated as detailed in the Methodology chapter. Visualisations for interim sentiment scores are presented in Appendix J. Only the overall sentiment score is analysed statistically in this chapter.

Exploratory analysis (presented in Table 4.4) on the overall sentiment score for PhishAdapt shows the mean score is 4.25 with a standard deviation of 0.77. The boxplot (Figure 4.11) shows that the majority of the scores fall between 3.75 and 4.75.

Table 4.4: EDA for PhishAdapt overall sentiment scores

			Statistic	Std. Error
Overall sentiment score (PhishAdapt scenario)	Mean		4.2529	.13091
	95% Confidence Interval for Mean	Lower Bound	3.9868	
		Upper Bound	4.5189	
	5% Trimmed Mean		4.3389	
	Median		4.5500	
	Variance		.600	
	Std. Deviation		.77450	
	Minimum		1.40	
	Maximum		5.00	
	Range		3.60	
	Interquartile Range		.90	
	Skewness		-1.791	.398
	Kurtosis		4.303	.778

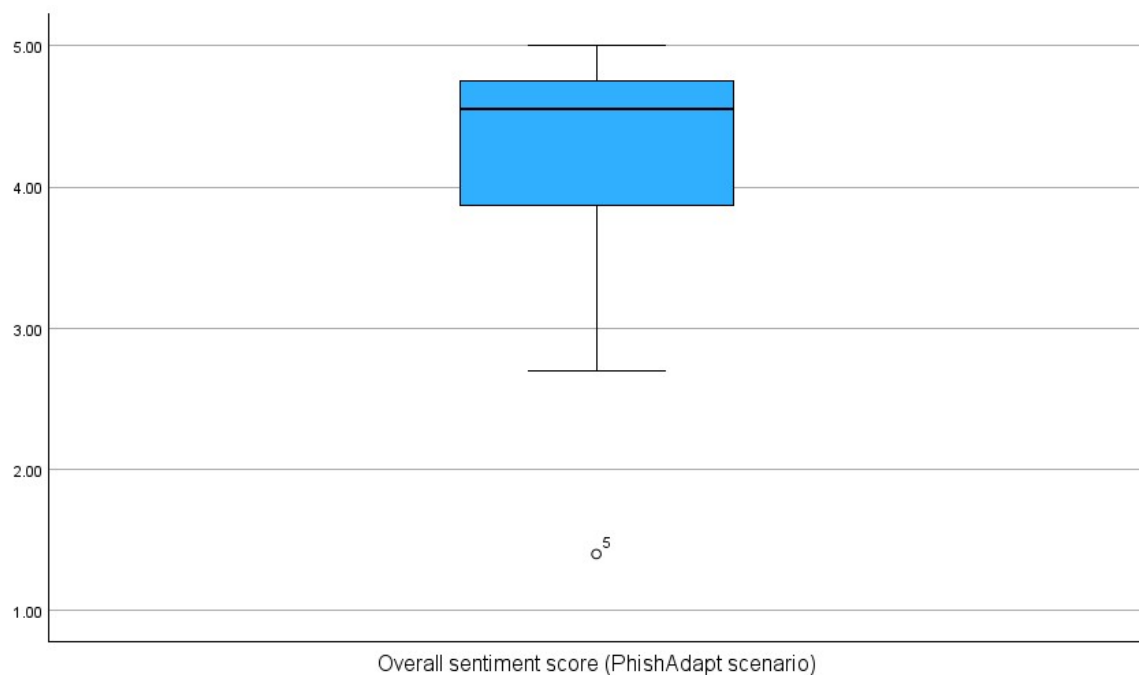


Figure 4.11: Boxplot for PhishAdapt overall sentiment scores

4.2.2 Qualitative Data

Similar to the scenario-based questions in the preliminary survey, participants were asked to provide an evaluation of messaging offered by PhishAdapt. The tone was similar to that of Scenario 4, but with the added flexibility that users would be able to actively choose their preferred mode of training delivery. A total of eight focussed codes were observed from participant evaluation responses to PhishAdapt.

Appearing in almost three quarters of all responses (70.9%), *Outcomes and Organisational Impacts of Phishing Simulation Training* was the most prominent theme. Participants often remarked on the supportive tone of the communication:

- P702 – “I would feel supported by my employer to learn and improve my skills”
- P312 – “The wording is empathetic and approachable.”

Notably, there were no instances of the focussed code *Impact on user behaviour and reporting* in the responses for this scenario.

4.3 Survey Results Comparisons and Correlations

4.3.1 Quantitative data

The results of the Shapiro-Wilk test of normality showed that two of the five sentiment score distributions were not normally distributed (Table 4.5): Scenario 4 ($W(35)=0.967$, $p=<0.001$) and PhishAdapt ($W(35)=0.825$, $p=<0.001$). The Friedman’s two-way analysis of variance test identified statistically significant differences between seven pairwise groupings of overall sentiment scores. The results of this test are presented in Table 4.6.

The size of the differences between pairs of sentiment scores are shown in Table 4.7, with the full results of the Wilcoxon signed-ranks tests included in Appendix J. Among the seven pairwise comparisons, Scenario 1 consistently scored lowest, while PhishAdapt scored highest ($r = 0.832$). Most effect sizes were large, with only two falling into the medium range (Table 4.7).

Table 4.5: Normality tests for overall sentiment scores (non-normal distributions highlighted)

	Kolmogorov-Smirnov ¹¹			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Overall sentiment score (Scenario 1)	.124	35	.192	.944	35	.075
Overall sentiment score (Scenario 2)	.118	35	.200 ¹²	.959	35	.220
Overall sentiment score (Scenario 3)	.091	35	.200 ¹³	.973	35	.542
Overall sentiment score (Scenario 4)	.167	35	.015	.867	35	<.001
Overall sentiment score (PhishAdapt scenario)	.175	35	.008	.825	35	<.001

¹¹ Lilliefors Significance Correction

¹² This is a lower bound of the true significance.

¹³ This is a lower bound of the true significance.

Table 4.6: Friedman's test results (statistically significant differences highlighted)

Pairwise Comparisons ¹⁴					
Sample 1-Sample 2	Test Statistic	Std. Error	Std. Test Statistic	Sig.	Adj. Sig. ¹⁵
Overall sentiment score (Scenario 1)-Overall sentiment score (Scenario 2)	-1.143	.378	-3.024	.002	.025
Overall sentiment score (Scenario 1)-Overall sentiment score (Scenario 3)	-1.286	.378	-3.402	<.001	.007
Overall sentiment score (Scenario 1)-Overall sentiment score (Scenario 4)	-2.243	.378	-5.934	<.001	.000
Overall sentiment score (Scenario 1)-Overall sentiment score (PhishAdapt scenario)	-2.471	.378	-6.539	<.001	.000
Overall sentiment score (Scenario 2)-Overall sentiment score (Scenario 3)	-.143	.378	-.378	.705	1.000
Overall sentiment score (Scenario 2)-Overall sentiment score (Scenario 4)	-1.100	.378	-2.910	.004	.036
Overall sentiment score (Scenario 2)-Overall sentiment score (PhishAdapt scenario)	-1.329	.378	-3.515	<.001	.004
Overall sentiment score (Scenario 3)-Overall sentiment score (Scenario 4)	-.957	.378	-2.532	.011	.113
Overall sentiment score (Scenario 3)-Overall sentiment score (PhishAdapt scenario)	-1.186	.378	-3.137	.002	.017
Overall sentiment score (Scenario 4)-Overall sentiment score (PhishAdapt scenario)	-.229	.378	-.605	.545	1.000

¹⁴ Each row tests the null hypothesis that the Sample 1 and Sample 2 distributions are the same. Asymptotic significances (2-sided tests) are displayed. The significance level is .005.

¹⁵ Significance values have been adjusted by the Bonferroni correction for multiple tests.

Table 4.7: Effect sizes (r) between pairwise groups

Scenario 1	Scenario 2	n	z	r	Higher performing scenario	Effect size
1	2	67	5.942	0.726	2	Large
1	3	67	5.274	0.644	3	Large
1	4	67	6.251	0.764	4	Large
1	PhishAdapt	35	4.923	0.832	PhishAdapt	Large
2	4	67	3.412	0.417	4	Medium
2	PhishAdapt	35	3.555	0.601	PhishAdapt	Large
3	PhishAdapt	35	2.933	0.496	PhishAdapt	Medium

When analysing age and gender groups with their associated sentiment scores, the Kruskal-Wallis tests identified no statistically significant differences between pairwise groupings. These findings are presented in Tables 4.8 and 4.9.

Table 4.8: Results of Kruskal-Wallis test with age as a grouping factor

	Null Hypothesis	Test	Sig. ¹⁶	Decision
1	The distribution of Overall sentiment score (Scenario 1) is the same across categories of Participant age.	Independent-Samples Kruskal-Wallis Test	.166	Retain the null hypothesis.
2	The distribution of Overall sentiment score (Scenario 2) is the same across categories of Participant age.	Independent-Samples Kruskal-Wallis Test	.420	Retain the null hypothesis.
3	The distribution of Overall sentiment score (Scenario 3) is the same across categories of Participant age.	Independent-Samples Kruskal-Wallis Test	.236	Retain the null hypothesis.
4	The distribution of Overall sentiment score (Scenario 4) is the same across categories of Participant age.	Independent-Samples Kruskal-Wallis Test	.480	Retain the null hypothesis.
5	The distribution of Overall sentiment score (PhishAdapt scenario) is the same across categories of Participant age.	Independent-Samples Kruskal-Wallis Test	.070	Retain the null hypothesis.

¹⁶ The significance level is .005. Asymptotic significance is displayed.

Table 4.9: Results of Kruskal-Wallis test with gender as a grouping factor

	Null Hypothesis	Test	Sig. ¹⁷	Decision
1	The distribution of Overall sentiment score (Scenario 1) is the same across categories of Participant gender.	Independent-Samples Kruskal-Wallis Test	.286	Retain the null hypothesis.
2	The distribution of Overall sentiment score (Scenario 2) is the same across categories of Participant gender.	Independent-Samples Kruskal-Wallis Test	.440	Retain the null hypothesis.
3	The distribution of Overall sentiment score (Scenario 3) is the same across categories of Participant gender.	Independent-Samples Kruskal-Wallis Test	.861	Retain the null hypothesis.
4	The distribution of Overall sentiment score (Scenario 4) is the same across categories of Participant gender.	Independent-Samples Kruskal-Wallis Test	.811	Retain the null hypothesis.
5	The distribution of Overall sentiment score (PhishAdapt scenario) is the same across categories of Participant gender.	Independent-Samples Kruskal-Wallis Test	.452	Retain the null hypothesis.

The results of the Shapiro-Wilk test showed that neither age nor gender data are normally distributed (Tables 4.10 and 4.11, respectively). Chi-Square tests found no significant associations between age or gender and participants' preferred scenario, perceived likely engagement, or perceived likely training value. As no meaningful correlations were found, these results are not reported in detail.

¹⁷ The significance level is .005. Asymptotic significance is displayed.

Table 4.10: Normality test for age group data distribution

	Kolmogorov-Smirnov ¹⁸			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Participant age	.195	67	<.001	.934	67	.001

Table 4.11: Normality test for gender group data distribution

	Kolmogorov-Smirnov ¹⁹			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
Participant gender	.419	66	<.001	.630	66	<.001

Figures 4.12 and 4.13 show training mode preferences by gender and age, respectively. While similarities are observable (e.g., consistent popularity of quizzes and e-learning), these findings are interpreted further in Chapter 5.

¹⁸ Lilliefors Significance Correction

¹⁹ Lilliefors Significance Correction

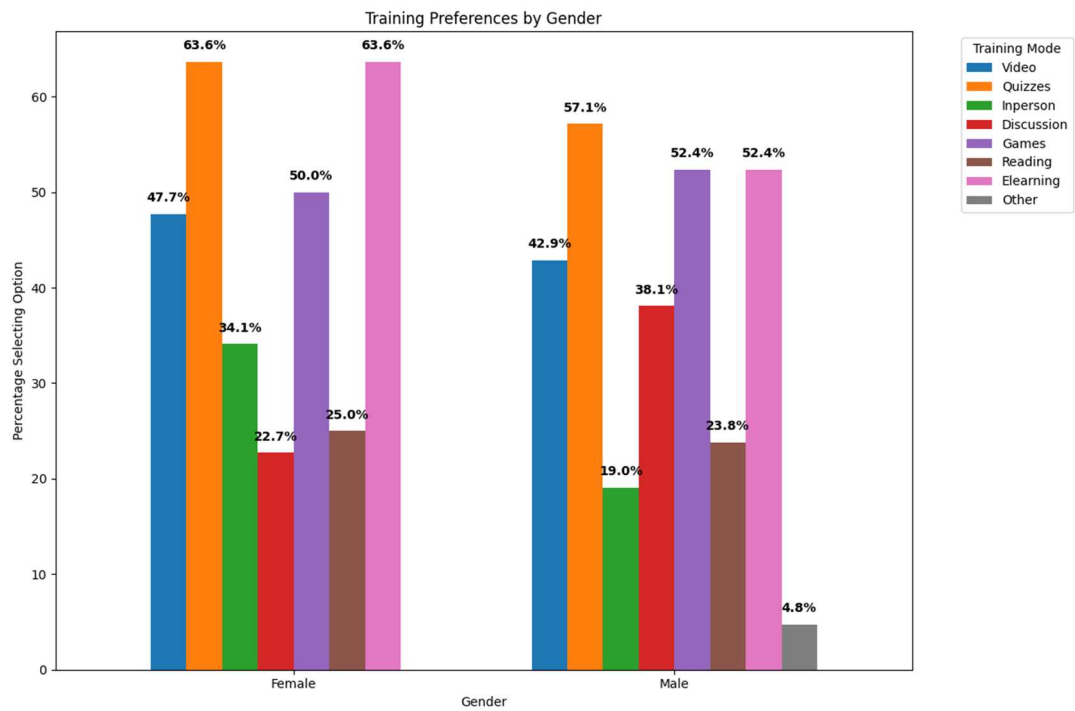


Figure 4.12: Frequencies of training mode preference, grouped by gender group

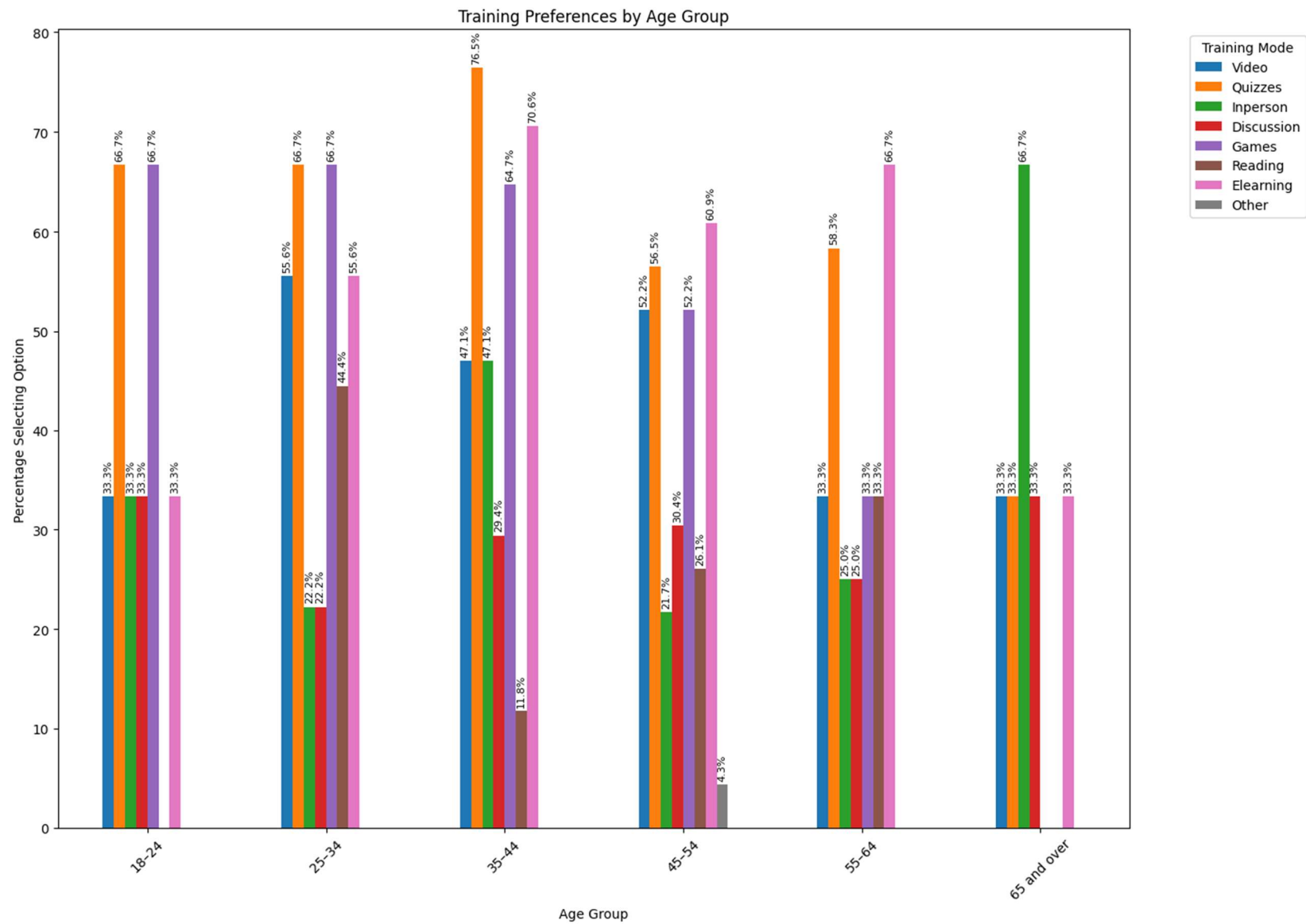


Figure 4.13: Frequencies of training mode preference, grouped by age group

Finally, Figure 4.14 demonstrates the frequencies with which participants perceived that they had witnessed the use of training material in line with each scenario's tone in their current or previous workplace with Scenario 2 being the most common.

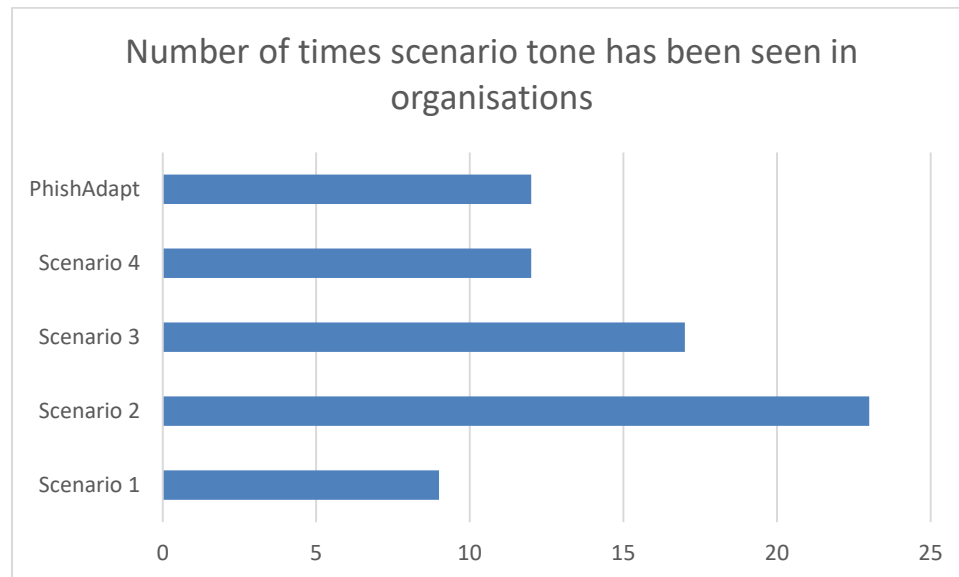


Figure 4.14: Number of times scenario tone has been seen in organisations

4.3.2 Qualitative data

Theme prevalence varied considerably across the five scenarios. *Emotional and psychological responses* was most common in Scenario 1 (54.7%) and declined significantly in Scenario 3 (12.2%), and the PhishAdapt evaluation (15.9%). In contrast, *Evaluation of Training Design and Delivery* was mentioned increasingly across scenarios, rising from 41.1% in Scenario 1 to 70.9% in Scenario 4. *Outcomes and Organisational Impacts of Phishing Simulation Training* also featured prominently, with consistently high frequencies, ranging from 40.2% in Scenario 2 to 70.5% in PhishAdapt evaluations. A full comparison is visualised in Figure 4.15.

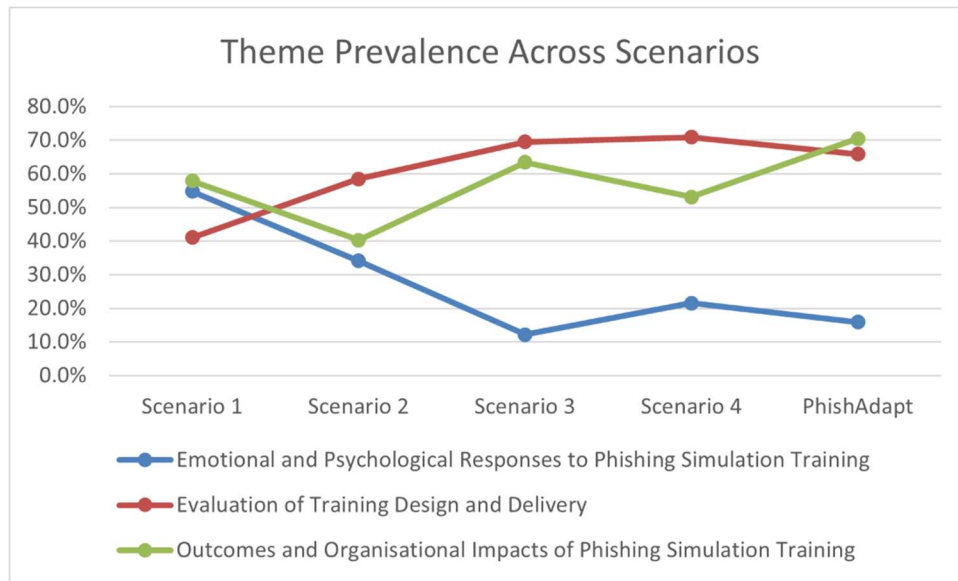


Figure 4.15: Theme prevalence across scenarios from preliminary and follow-up surveys

There is notable frequency of instances of the focussed code *Perspective and framing of responses*, which was observed in approximately one fifth of all scenario-based responses (20.6%). Further implications of this will be discussed in Chapter 5.

4.4 Interview Results

The interview was conducted with two participants responsible for the administration of phishing simulation training within a large organisation. The discussion's purpose was focussed primarily on PhishAdapt's personalisation feature and its delivery tone. Thematic analysis of the responses identified a dominant emphasis on *Evaluation of Training Design and Delivery*, accounting for 60% of coded excerpts, as seen in Figure 4.16.

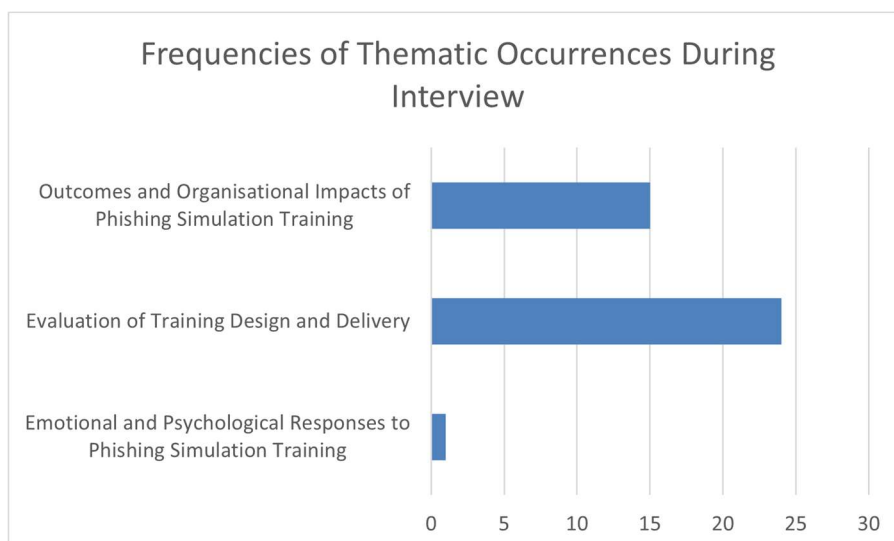


Figure 4.16: Chart of frequencies of thematic occurrences during interview

Platform functionality, in particular its reporting capabilities, was commented on often, emphasising the focus on needing to identify attitudes and behavioural trends within the user-base:

- P2: “Being able to have a really detailed report is really useful because then we can see exactly what they've done.”

Outcomes and Organisational Impacts of Phishing Training also contributed heavily, constituting 37.5% of codes applied to the responses. Both participants stressed the importance of the need to provide training in a non-punitive and supportive manner:

- P2: “[...] it's a supportive tone. That is what we strive for. We are using the simulations as a training tool. It is not a way to target our colleagues.”

User personalisation, while a guiding objective of the platform, appeared in less than 5% of coded responses. The value of group targeting and scenario tailoring was acknowledged, but participants generally steered the conversation towards operational features, such as whaling/spear phishing capabilities, pre-built templates, and user reporting behaviour. When asked specifically about the personalisation capabilities of PhishAdapt, P2 stated:

“[...] there's all the different formats for the training, which is great. I think that's really good because people learn in different ways. Give them that option. It doesn't, it doesn't then create a barrier for them”

In summary, the interview commentary was weighted heavily toward reporting mechanisms and operational design, with some discussion around tone of communications. These findings are further interpreted in Chapter 5 (Discussion). The interview transcript is presented in Appendix K.

5 Discussion

The previous chapter presented the results from empirical research conducted to support Research Objectives 2, 3, and 4 (reiterated below). This Discussion chapter builds upon the findings by continuing to explore those objectives, while also beginning to address objective 5.

2. **Explore the user-driven factors (e.g., participant satisfaction, age) that are believed to influence the effectiveness of delivered training programmes.**
3. **Critically assess current methods used to deliver training campaigns.**
4. **Evaluate the implementation of personalisation (through participant preferences) into a training campaign, including the impact this could have on desired outcomes.**
5. **Formulate recommendations to the cybersecurity community on how training approaches could be adjusted to achieve better outcomes.**

The chapter is structured thematically. It begins by examining user perceptions of training effectiveness and engagement, before exploring emotional responses to phishing simulation training. The roles of personalisation and demographic factors are then considered in turn. This is followed by a synthesis of trends observed across both qualitative and quantitative datasets, incorporating relevant findings from the Literature Review. The chapter concludes with a critical reflection on the study's limitations, ensuring transparency and methodological rigour.

5.1 Perceived Effectiveness and Engagement with Training

Participant feedback from the preliminary survey revealed a range of views on the perceived usefulness of phishing simulation training, with 85% of responses relating to the effectiveness of these programmes referring to them as “useful”. Despite this strong trend, almost a third of participants indicated they felt they had not learned anything from phishing training simulations that they had previously taken part in. This finding, which identifies a discrepancy of approximately 15% of participants who described the training as useful despite reporting no actual learning, highlights a potential disconnect between the expected value of phishing simulations and actual learning outcomes experienced by participants. Such a discrepancy has also been identified in existing research (Chen *et al.*, 2024; Ho *et al.*, 2024; Lain, Kostinen

and Čapkun, 2022) with some studies attributing this disconnect to time constraints and low perceived usefulness (Ho *et al.*, 2024; Lain, Kostinen and Čapkun, 2022). Whilst reasoning for the perceived lack of knowledge gain was not specifically requested in the preliminary survey, some participants suggested this was due to a lack of breadth of content and context:

P145: “They are much too basic and do not cover enough details about different types of phish. Also, they fail to make people understand the consequences of their actions, and how to effectively mitigate risks.”

The wider literature supports this call to make training content more realistic and specific (He and Zhang, 2019) but more generally credits the shortfall in user engagement to apathetic user attitudes (Berens *et al.*, 2024; Kumaraguru *et al.*, 2007a; Rizzoni *et al.*, 2022). The consequence of this lack of meaningful engagement contributes to limited long-term behavioural change and poor knowledge retention (Reinheimer *et al.*, 2020; Sarker *et al.*, 2024), which continues to undermine the efficacy of training interventions, aiding the ongoing success of phishing attacks in various forms.

Respondents repeatedly cited deadlines and mandatory completion as key motivators for participating in training programmes.

P342: “Sounds a bit like you could ignore it though if they don't give a deadline or check that you have completed the training.”

This sentiment was echoed during the interview, with interviewees stating that their training programme was mandatory as a way of enforcing user compliance. While such measures may ensure participation, they do not necessarily equate to effective training outcomes, evidenced not only in the lack of long-term behavioural changes already discussed, but also in a “knowing-doing gap” (where users knowledge does not translate into secure behaviour) (Buckley *et al.*, 2023; Canham, Posey and Constantino, 2022). In contrast to this, participants frequently mentioned that an attribute important to them for ensuring their engagement was that of a supportive tone. This finding is supported by existing research, which indicates that clear, non-punitive communications are likely to contribute to higher levels of engagement in users (Sarker *et al.*, 2024; Schiller *et al.*, 2024). Collectively, this evidence signifies a

need to shift towards a more holistic training model – one that prioritises emotional engagement and user experience alongside behavioural outcomes.

5.2 Emotional Responses and Organisational Support Perceptions

The most commonly used metric by organisations for evaluating simulation campaign success – the CTR – as reported by both the literature (Hillman, Harel and Toch, 2023) and interviewees, underscores a misalignment between organisational evaluation methods and user experience. This focus on binary outcomes overlooks the emotional nuance attached to user responses to simulations. When presented with a message designed to enforce training compliance (Scenario 1) – one emphasising mandatory training, short deadlines, and the possibility of disciplinary action – participants indicated that they were no more or less likely to engage with the training when compared to other message types. Furthermore, perceptions of knowledge gain from this message revealed notably lower scores for expected learning outcomes relative to other messages.

The unintended consequences of using blame or punishment when appealing for user engagement are documented in existing research (Kim, B., Lee and Kim, 2019; Renaud, Musarurwa and Zimmermann, 2021), with Proofpoint's State of the Phish report (2024) identifying punishment as being the least motivating factor for promoting desired security practices (Proofpoint, 2024). This was supported by qualitative data, with 96% of all coded responses to Scenario 1 centring on negative emotional reactions. Stress/worry was the most commonly mentioned response, with shame/embarrassment also noted frequently. Respondents also highlighted their perception that this negative emotional response would make them less likely to engage in the training programme:

P312: "Worried and unlikely to want to engage beyond what I absolutely had to"

Conversely, when presented with the most supportive message of the four in the preliminary survey, participants commonly reported that it fostered a sense of organisational support and empathy. Some indicated that it would motivate them to improve their knowledge or that it instilled a sense of collective collaboration:

P702: “I would feel supported by my employer to learn and improve my skills”

P312: “[...] the language used is positive and about working together and inviting feedback.”

This desire for a collaborative tone is echoed in the literature, where constructive approaches have been found to increase trust in an organisation and empower users to feel that they can be an important part of the solution to an issue (Renaud, Musarurwa and Zimmermann, 2021; Volkamer, Sasse and Boehm, 2020). These findings suggest that training programmes should adopt a more supportive and collaborative tone in their communications, foster a shared sense of responsibility, and recognise that emotional response and organisational trust can directly influence training outcomes (Silic and Lowry, 2020). With these adaptations, organisations are more likely to see behavioural change, as users better understand the real-world consequences of their actions (Lain *et al.*, 2024), ultimately reducing overall security risk.

5.3 Role of Personalisation in Training

Alongside the influence of a supportive tone, participant feedback - particularly when evaluating Scenario 4 and its realisation into the PhishAdapt platform – highlighted the role of personalisation in shaping engagement. When given the choice of all four scenarios in the preliminary survey, Scenario 4 was ranked highest as most likely to engage with, most likely to learn from, and as the overall preferred message. The preference for this scenario was notable, with it receiving at least twice as many votes as its nearest competitor. The dominant reason given for the preference, contributing over three quarters (76.9%) of coded responses, was the ability to choose the training delivery mode, with many stating they believed it would increase their capability to learn.

P733: “It is accessible. It can cater to what method of learning works best for me.”

P239: “There are choices of how to do more training, which means you can more effectively learn based on what works for you.”

Whilst the topic of personalisation was not heavily discussed during the interview, this significant finding is echoed in one comment from an interviewee, made after the demonstration of the PhishAdapt platform:

P2: “Of course, there's all the different formats for the training, which is great. I think that's really good because people learn in different ways. Give them that option. It doesn't, it doesn't then create a barrier for them because they're like, ‘I don't want to watch this video, I don't learn well from listening and watching videos’.”

Furthermore, Proofpoint's State of the Phish report indicates that the personalisation of training is the second highest scoring factor for user engagement (Proofpoint, 2024). In this report, training personalisation was marked as “motivating” by 88% of participants, second only to “making security easier for me” (94%) (Proofpoint, 2024). Despite this key insight, the role of user-driven personalisation in phishing simulation training is underexplored in existing research. Studies have been conducted that demonstrate the positive effects of customising training delivery based on user role (Oner, Cetin and Savas, 2025), user susceptibility (Sumner *et al.*, 2021), and existing user knowledge (Sarker *et al.*, 2024). Moreover, suggestions have been made that future research be conducted into the effects of providing targeted training campaigns to specific groups of users (Cranford *et al.*, 2022; Jampen *et al.*, 2020).

While there is clear interest in administratively driven personalisation, and evidence suggesting that any degree of customisation can improve user engagement and training outcomes (Abraham and Chengalur-Smith, 2019; Huang and Shiu, 2012; Klačnja-Milićević *et al.*, 2011), the benefits of user-stated personalisation have only recently begun to receive attention in the research literature. The recent study conducted by Schöni *et al.* (2025) showed participants had a clear preference for choosing their own training materials, as well as improved training outcomes for those that were given that option (Schöni *et al.*, 2025). With these findings (and others discussed in this section) in mind, there is a compelling case for further research to be conducted into the effects of user-driven personalisation on training outcomes.

5.4 Impact of Demographic Variables

Whilst the effects of user preferences may be underexplored in the literature, the impact of demographic variables is an aspect more frequently examined (Kim, B., Lee and Kim, 2019; Kumaraguru *et al.*, 2007b; Marshall, Sturman and Auton, 2023). When analysing participant responses, this empirical research found no evidence that any particular age group or gender showed a greater or lesser tendency towards scenario or training mode preference. This lack of correlation extended to sentiment scores for all presented scenarios. However, these findings should be treated with a degree of caution – whilst the overall sample size ($n=67$) is sufficient for broader analysis, disaggregation by age and gender yielded smaller subgroup sizes. Consequently, the expected frequency assumption was not met, producing potentially unreliable Chi Square test results (Field, 2018).

Although the statistical test results should be interpreted with care, the overall pattern aligns with previous research, which suggests that demographic factors exert limited influence on user engagement and training outcomes (Buckley *et al.*, 2023; Jampen *et al.*, 2020; Kim, B., Lee and Kim, 2019). Similarly, Schöni *et al.* (2025) found that age and gender had such minimal influence on training outcomes that demographics were deemed not worth considering when designing a training programme (Schöni *et al.*, 2025). Taken together, the findings outlined in this section may help to mitigate the ethical and practical challenges of tailoring training options based on protected characteristics, reinforcing the case for prioritising alternative factors, such as user-stated personalisation.

5.5 Code Trends and Theoretical Reflections

Beyond individual characteristics such as age and gender, a broader analysis of participant responses revealed several interpretative trends across the scenarios presented in the preliminary survey, offering insight into how phishing training is perceived by end users. As noted in the Results chapter, the occurrence of responses centring on emotional responses, whilst prominent in the evaluation of Scenario 1, fell sharply when participants evaluated progressive scenarios in the preliminary survey. Conversely, responses regarding an evaluation of training design and delivery steadily increased across the progression of scenarios. This shift in focus, combined with the preferred scenario results, suggests that perceived

relevance, content quality, and clarity of communication play central roles in driving user engagement and knowledge gain. This implication is further strengthened by the lack of significant difference between user perceptions of Scenario 4 and PhishAdapt, which were similar in tone and available features, differing only in delivery format – the former presented as a static email message; the latter, an interactive and flexible platform that allowed users to actively modify their training preferences.

Thematically, outcomes of training simulation remained consistently high as an important consideration across all scenarios. One participant offered a response that encapsulates these themes:

P702: “[...] it indicates a gap in knowledge, illustrates the risk, and offers option[s] to resolve and improve in a kinder way.”

Whilst these trends are mirrored in the wider research, which indicates that clear and timely feedback with concise and engaging training content significantly improves behavioural outcomes (Hull, Schuetz and Lowry, 2023; Kumaraguru *et al.*, 2007b; Rizzoni *et al.*, 2022; Singh *et al.*, 2023), they sit alongside a broader interpretative trend – a tendency for users to focus narrowly on personal experience, often overlooking the wider implications of phishing threats. Outside of the scope of the introductory questions about perceived usefulness of phishing simulation training, there was a notable absence of responses around the wider impact on user behaviour and reporting. Furthermore, any responses that did pertain to this theme largely focussed on negative effects, such as an increased reluctance to report genuine incidents.

P934: “[...] it may make people afraid to speak to teams if they need to report a real incident.”

This individualistic framing is echoed in the literature, which suggests that while phishing simulations may raise short-term awareness, they often fail to cultivate a deeper understanding of the broader organisational risks associated with successful phishing attacks (Ho *et al.*, 2024; Rizzoni *et al.*, 2022). Rather than fostering a comprehensive view of security, training simulations are often perceived as a test of individual compliance, reinforcing the idea that the primary consequence of “failure” is personal rather than organisational (Jampen *et al.*, 2020; Lain *et al.*, 2024; Rizzoni

et al., 2022). Future training initiatives should look to integrate personal learning objectives with organisational risk awareness, helping users understand not simply how to avoid phishing, but why doing so matters within a wider security context.

5.6 Limitations

Whilst this study presents meaningful insights, several limitations (identified in Chapter 3) must be considered in interpreting the findings and their broader implications. Potential avenues for addressing limitations noted here are considered in the Future Work section of the Conclusion chapter.

Research strategy – lack of interactivity: whilst participant feedback to the PhishAdapt artefact was generally open and engaged, the use of static images may have limited the emotional depth of responses (Marshall, Sturman and Auton, 2023). Additionally, this study could not capture the impact of personalisation on long-term behavioural changes. As such, the findings should be interpreted with caution when considering potential benefits of a fully developed version of the platform.

Data collection – sample size: whilst survey participation was sufficient for exploratory analysis, the relatively small sample (particularly when disaggregated by demographic group) limits the generalisability of certain findings (Jampen *et al.*, 2020). This is especially relevant to the Chi Square tests on gender/age correlations, as well as for interview data, which were based on only two participants.

Participant response – self-report and framing: the majority of the findings obtained from this empirical research relied upon self-reported data, which can introduce bias (Buckley *et al.*, 2023). Whilst there was little indication of socially desirable responding in this study, a notable portion of responses (20.7%) were framed from a third-party perspective. This limits the extent to which these responses can be considered valid reflections of individual participant views and behaviours.

5.7 Summary

This chapter has discussed the empirical findings in relation to the study's research objectives, drawing connections between participant perceptions, emotional responses, and the potential of personalised training delivery. Key patterns emerged around the importance of supportive messaging, the limited influence of

demographic factors, and the potential benefits of user-driven personalisation. While the findings offer promising directions for future training approaches, limitations in research strategy and project scope must be considered when interpreting results. The final chapter will now reflect on the conclusions drawn from this research, highlight the study's potential contributions to research and practice, and offer recommendations - both as endorsements to relevant stakeholders and for further work that could be undertaken to extend this study.

6 Conclusion

The overall aim of this research was to advance understanding of the effect of emotional response to phishing simulations, and to investigate how user-driven personalisation might influence training outcomes. The research was guided by the following objectives:

1. *Clarify* what outcomes are expected of phishing training campaigns and how they are measured.
2. *Explore* the user-driven factors (e.g., participant satisfaction, age) that are believed to influence the effectiveness of delivered training programmes.
3. *Critically assess* current methods used to deliver training campaigns.
4. *Evaluate* the implementation of personalisation (through participant preferences) into a training campaign, including the impact this could have on desired outcomes.
5. *Formulate recommendations* to the cybersecurity community on how training approaches could be adjusted to achieve better outcomes.

This chapter will firstly summarise any findings and offer conclusions to each research objective, before presenting a set of recommendations derived from these conclusions. Finally, the dissertation will outline opportunities for future work that could be undertaken to continue and build upon the empirical research conducted in this study.

6.1 Research Objectives: summary of findings and conclusions

The following section will address each of the five research objectives in turn, summarising the relevant findings and offering a concluding reflection for each.

6.1.1 Research Objective 1: Success Metrics and Training Outcomes

Organisational success in phishing training is typically measured using CTRs (NIST, 2023), though academic literature increasingly advocates for a more holistic, behaviour-focussed approach. There are suggestions that a more comprehensive measurement could incorporate both: (1) user-driven actions, such as reporting rates (Hillman, Harel and Toch, 2023) and file downloads (Ho *et al.*, 2024); and (2) user behavioural responses, such as false negative reporting (Al Maqousi, 2023; Sarker *et al.*, 2024) and knowledge gain (Al Maqousi, 2023; Lain *et al.*, 2024).

These findings suggest that if organisations are to minimise phishing risk effectively, they should adopt multifaceted metrics that include apathetic actions and user-centric behaviours, rather than relying solely on click rates in isolation.

6.1.2 Research Objective 2: Effects of User-specific Factors

User perceptions of phishing training campaigns can be said to be mixed – whilst the empirical research conducted indicated that phishing training is generally seen as useful, participants were less positive about training content relevance and training-induced emotional responses. These findings are reflected in the wider literature, which points to apathy (Hull, Schuetz and Lowry, 2023; Marshall, Sturman and Auton, 2023) and negative attitudes (Ho *et al.*, 2024; Lain *et al.*, 2024; Schöps *et al.*, 2024) as key barriers to meaningful engagement.

Demographic factors appear to have little, if any, effect on training outcomes. The results from the preliminary survey on this subject should be treated with caution due to the disaggregated sample sizes in use. However, they are supported by the wider literature with several studies suggesting that neither age (Buckley *et al.*, 2023; Jampen *et al.*, 2020; Tschakert and Ngamsuriyaroj, 2019) nor gender (Buckley *et al.*, 2023; Jampen *et al.*, 2020; Lain, Kostiainen and Čapkun, 2022) show a consistent correlation to training outcomes.

These results demonstrate that demographic profiling offers limited value; instead, research and design should focus on cultivating positive emotional engagement as a foundation for long-term behavioural change.

6.1.3 Research Objective 3: Training Delivery Techniques

While embedded training remains the most common delivery technique (NIST, 2023), its effectiveness is debated. Studies report mixed outcomes (Ho *et al.*, 2024; Jampen *et al.*, 2020; Lain *et al.*, 2024), with some suggesting it may lead to counterproductive behaviours (Lain, Kostiainen and Čapkun, 2022; Marshall, Sturman and Auton, 2023). These findings underscore a broader concern: while convenience and scalability make embedded training appealing to organisations, its often rigid or punitive delivery style can hinder long-term behavioural change in end-users.

Participant responses to the preliminary survey revealed a marked preference for training communications that were supportive, which aligns with a growing body of literature advocating for training models that promote engagement through clear and collaborative messaging (Chen *et al.*, 2024; Jampen *et al.*, 2020). Collectively, these insights challenge the one-size-fits-all paradigm prevalent across industry, and instead suggest the need for more flexible, user-centred delivery mechanisms that foster organisational trust and promote sustained, accessible learning for all.

6.1.4 Research Objective 4: Personalisation and its Influence

There is a strong body of research that demonstrates benefit to customising training based on user role (Canham, Posey and Constantino, 2022; Sarker *et al.*, 2024) or susceptibility (Oner, Cetin and Savas, 2025; Sumner *et al.*, 2021), with interview participants confirming that this is a feature considered desirable within organisational settings.

Whilst survey participants indicated very strong preferences towards the ability to choose their own training delivery mode, definitive influence of this feature cannot be determined within the scope of this study. The topic of user-stated personalisation remains relatively understudied in the wider literature, though interest in the subject is growing (Schöni *et al.*, 2025), with initial findings suggesting it is both well-received by participants and associated with improved training outcomes. More research is required to fully understand the implications of these findings, in particular a case study into whether long-term behavioural and attitudinal changes can be observed when user preferences are respected would be welcomed.

6.1.5 Research Objective 5: Recommendations to Improve Outcomes

The purpose of this research objective was to formulate recommendations based on the findings of this study. Given that this undertaking relies heavily and directly on the insights from the previous objectives detailed above, and frames the research through a forward-looking lens, these recommendations are presented in section 6.2 below.

6.2 Recommendations

This section addresses the fifth Research Objective, which was to formulate recommendations informed by the findings of this project.

6.2.1 Recommendation 1: Training Metrics

By default, organisations rely on CTRs as the primary measure of training success (NIST, 2023). However, the narrow focus of this metric neglects other more nuanced indicators, such as behavioural change and user engagement. It is recommended that training platforms incorporate additional metrics, such as reporting rates, user satisfaction, and the frequency of FP reporting, into the existing CTR model. This broader, more holistic, approach would inform a more accurate report of training effectiveness, aligning with recent literature and the findings of this study, which suggest that user-centric and behavioural indicators offer better insight than CTRs alone.

6.2.2 Recommendation 2: Communication Tone

Existing phishing training messaging often embraces an inflexible, or even punitive, tone, particularly when associated with embedded delivery methods. This approach can lead to feelings of negativity and disengagement amongst users, which may undermine desired training outcomes. Participant feedback in this study indicated a strong preference for clear, supportive, and collaborative communications.

Organisations should review and revise the tone of training messages to ensure they foster a learning-oriented environment (without focus on fault or blame), meeting the needs of end-users. Communications with a more positive tone can increase the likelihood of knowledge gain, resulting in long-term behavioural change. This approach is supported by both participant data from this empirical research and that of the wider literature (Jampen *et al.*, 2020; Sarker *et al.*, 2024) advocating for psychologically safe spaces in cybersecurity education.

6.2.3 Recommendation 3: Delivery Methods

The dominance of a “one-size-fits-all” approach to phishing training raises concerns about its long-term effectiveness. While convenient and scalable, it has been linked to apathetic or negative behaviours, with the literature reporting mixed results. Participants in this study expressed a strong desire for less disruptive and more

flexible training delivery, indicating that this may improve training outcomes. Training designers should move away from a rigid deployment and implement models that can incorporate user preferences in an automated manner. Offering varied delivery options empowers users and may increase engagement levels, both of which are linked to better behavioural outcomes (Hull, Schuetz and Lowry, 2023; Marshall, Sturman and Auton, 2023). This aligns with participant feedback from this study and reinforces trends in the literature calling for more user-centric approaches in cyber awareness training.

6.3 Future Work

The following avenues are proposed for future research and development, building upon the findings and limitations of this study:

- Specific to the PhishAdapt platform, or any other training platform in development, implement features described as desirable by interviewees:
 - A mechanism for users to report suspected phishing emails (e.g., an integrated add-in for graphical email clients).
 - Measures for decreasing user risk score (e.g., when training is completed).
 - Reporting capabilities with more granular options (e.g., track apathetic user behaviours).
- Conduct further research into how apathetic user behaviour (e.g., if an e-mail is forwarded, rather than reported) can be tracked and addressed, and how this could improve training outcomes.
- Undertake a case study to ascertain the real-world effects of user-stated personalisation, preferably with a larger sample size than used in this research. This would build upon the work already undertaken by Schöni *et al.* (2025), raise the possibility of further studying the effect of other demographic factors, and increase the generalisability of results.

With these future directions in mind, the research presented here lays a foundation for further research into delivering more adaptive, evidence-based, and impactful approaches to phishing awareness training, designed to meet the evolving needs of organisational resilience and end-user engagement.

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Appendix A – Surveys

Preliminary Survey

Background

This section will ask you some questions about your knowledge of phishing and anti-phishing training. These questions are not meant to test or trick you - please answer as honestly as you can.

Have you heard of "phishing" before now?

- ☐ Yes
- ☐ No
- ☐ Not sure/don't know

What do you understand "phishing" to be?

What do you think the risks of "phishing" are?

The UK's National Cyber Security Centre (NCSC) defines "phishing" as follows:

Phishing is when attackers send scam emails (or text messages) that contain links to malicious websites. The websites may contain malware (such as ransomware) which can sabotage systems and organisations. Or they might be designed to trick users into revealing sensitive information (such as passwords), or transferring money.

Based on this definition, would you say your understanding of "phishing" has now changed?

- ☐ Yes
- ☐ No
- ☐ Not sure/don't know

In what way?

Many organisations use "phishing simulation" campaigns as a training method for their staff. These exercises, controlled by an administrator, involve sending emails to staff members that contain a number of features that are intended to make it obvious to recipients that the email is not genuine (e.g. spelling mistakes). An employee's engagement with the email is recorded and, if their actions are deemed to have "failed" the test, training will be offered.

Were you aware of phishing simulations being used as training aids before now?

- ☐ Yes
- ☐ No
- ☐ Not sure/don't know

Have you ever (including in your current employment, if applicable) taken part in phishing simulation exercises at work?

- ☐ Yes
- ☐ No
- ☐ Not sure/don't know
- ☐ Not applicable

What size was the company? Please use your most recent (or current) employment.

- ☐ Micro business (less than 10 employees)
- ☐ Small business (10-49 employees)
- ☐ Medium business (50-249 employees)
- ☐ Large business (250-999 employees)
- ☐ Very large business/enterprise (more than 1000 employees)
- ☐ Not sure/don't know

What industry was/is the company in? Please use your most recent (or current) employment.

- ☐ Education and Public Services
- ☐ Healthcare and Social Care
- ☐ Technology and IT
- ☐ Finance, Legal, and Professional Services
- ☐ Retail, Hospitality, and Tourism
- ☐ Construction, Manufacturing, and Engineering
- ☐ Transport, Logistics, and Utilities
- ☐ Arts, Media, and Non-Profit
- ☐ Not sure/don't know

Do you believe you learned anything from the simulations?

- ☐ Yes
- ☐ No
- ☐ Not sure/don't know

What is your general opinion about phishing simulations as training exercises?

Scenario #1

For the following questions, you will be asked to imagine that you have "failed" a phishing simulation exercise at work. You will be presented with a possible response from your cyber security department in response to your actions with the phishing simulation and asked to consider how the wording in the response would make you feel.

How would you feel if, following a phishing simulation exercise, you received the following email from the cyber security department?

You failed the recent phish training when you clicked on the link in the unsolicited email sent to you. Your actions could have had dire consequences had the phish been genuine. You must complete some training as soon as possible. Failure to complete the training may result in system access being revoked. You may be subject to disciplinary action if your performance in future phishing training actions does not improve.

How likely would you be to follow up as the response requests?

- | | | | | | |
|-----------------------|-----------------------|-----------------------------|-----------------------|-----------------------|-----------------------|
| Very likely | Somewhat likely | Neither likely nor unlikely | Somewhat unlikely | Very unlikely | Not sure/don't know |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

How likely would you be to engage with future phishing simulation exercises (e.g. you would simply ignore or delete a message you suspected of being phishing)?

- | | | | | | |
|-----------------------|-----------------------|-----------------------------|-----------------------|-----------------------|-----------------------|
| Very likely | Somewhat likely | Neither likely nor unlikely | Somewhat unlikely | Very unlikely | Not sure/don't know |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

How positive/negative would this response make you feel about cyber security in your organisation?

- | | | | | | |
|-----------------------|-----------------------|-------------------------------|-----------------------|-----------------------|-----------------------|
| Very positive | Somewhat positive | Neither positive nor negative | Somewhat negative | Very negative | Not sure/don't know |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

How positive/negative would this response make you feel about your organisation as a whole?

- | | | | | | |
|-----------------------|-----------------------|-------------------------------|-----------------------|-----------------------|-----------------------|
| Very positive | Somewhat positive | Neither positive nor negative | Somewhat negative | Very negative | Not sure/don't know |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Please read the following statements and select how strongly you agree or disagree with each one.

	Strongly agree	Slightly agree	Neither agree nor disagree	Slightly disagree	Strongly disagree	Not sure/don't know
This response would make me feel angry	☐	☐	☐	☐	☐	☐
This response would make me feel bullied	☐	☐	☐	☐	☐	☐
This response would make me feel empowered	☐	☐	☐	☐	☐	☐
This response would make me feel motivated	☐	☐	☐	☐	☐	☐
This response would make me feel invested in cyber security	☐	☐	☐	☐	☐	☐
I would learn lessons from this response	☐	☐	☐	☐	☐	☐
This response would help me improve my skill set	☐	☐	☐	☐	☐	☐
This response would help build my trust in the organisation	☐	☐	☐	☐	☐	☐

Do you have any other thoughts about this response?

Have you ever worked at a company that used a response with this tone?

- ☐ Yes
- ☐ No
- ☐ Not sure/don't know
- ☐ Not applicable

Please read the following statements and select how strongly you agree or disagree with each one.

	Strongly agree	Slightly agree	Neither agree nor disagree	Slightly disagree	Strongly disagree	Not sure/don't know
This response would make me feel angry	☐	☐	☐	☐	☐	☐
This response would make me feel bullied	☐	☐	☐	☐	☐	☐
This response would make me feel empowered	☐	☐	☐	☐	☐	☐
This response would make me feel motivated	☐	☐	☐	☐	☐	☐
This response would make me feel invested in cyber security	☐	☐	☐	☐	☐	☐
I would learn lessons from this response	☐	☐	☐	☐	☐	☐
This response would help me improve my skill set	☐	☐	☐	☐	☐	☐
This response would help build my trust in the organisation	☐	☐	☐	☐	☐	☐

Do you have any other thoughts about this response?

Have you ever worked at a company that used a response with this tone?

- ☐ Yes
- ☐ No
- ☐ Not sure/don't know
- ☐ Not applicable

Scenario #2

For the following questions, you will be asked to imagine that you have "failed" a phishing simulation exercise at work. You will be presented with a possible response from your cyber security department in response to your actions with the phishing simulation and asked to consider how the wording in the response would make you feel.

How would you feel if, following a phishing simulation exercise, you received the following email from the cyber security department?

Unfortunately, you have failed the phish training exercise. You will need to do some further training on this topic. Your training must be completed within 3 working days of receipt of this email.

How likely would you be to follow up as the response requests?

- | | | | | | |
|-----------------------|-----------------------|-----------------------------|-----------------------|-----------------------|-----------------------|
| Very likely | Somewhat likely | Neither likely nor unlikely | Somewhat unlikely | Very unlikely | Not sure/don't know |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

How likely would you be to engage with future phishing simulation exercises (e.g. you would simply ignore or delete a message you suspected of being phishing)?

- | | | | | | |
|-----------------------|-----------------------|-----------------------------|-----------------------|-----------------------|-----------------------|
| Very likely | Somewhat likely | Neither likely nor unlikely | Somewhat unlikely | Very unlikely | Not sure/don't know |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

How positive/negative would this response make you feel about cyber security in your organisation?

- | | | | | | |
|-----------------------|-----------------------|-------------------------------|-----------------------|-----------------------|-----------------------|
| Very positive | Somewhat positive | Neither positive nor negative | Somewhat negative | Very negative | Not sure/don't know |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

How positive/negative would this response make you feel about your organisation as a whole?

- | | | | | | |
|-----------------------|-----------------------|-------------------------------|-----------------------|-----------------------|-----------------------|
| Very positive | Somewhat positive | Neither positive nor negative | Somewhat negative | Very negative | Not sure/don't know |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Please read the following statements and select how strongly you agree or disagree with each one.

	Strongly agree	Slightly agree	Neither agree nor disagree	Slightly disagree	Strongly disagree	Not sure/don't know
This response would make me feel angry	☐	☐	☐	☐	☐	☐
This response would make me feel bullied	☐	☐	☐	☐	☐	☐
This response would make me feel empowered	☐	☐	☐	☐	☐	☐
This response would make me feel motivated	☐	☐	☐	☐	☐	☐
This response would make me feel invested in cyber security	☐	☐	☐	☐	☐	☐
I would learn lessons from this response	☐	☐	☐	☐	☐	☐
This response would help me improve my skill set	☐	☐	☐	☐	☐	☐
This response would help build my trust in the organisation	☐	☐	☐	☐	☐	☐

Do you have any other thoughts about this response?

Have you ever worked at a company that used a response with this tone?

- ☐ Yes
- ☐ No
- ☐ Not sure/don't know
- ☐ Not applicable

Scenario #3

For the following questions, you will be asked to imagine that you have "failed" a phishing simulation exercise at work. You will be presented with a possible response from your cyber security department in response to your actions with the phishing simulation and asked to consider how the wording in the response would make you feel.

How would you feel if, following a phishing simulation exercise, you received the following email from the cyber security department?

It looks like you might benefit from gaining some knowledge about how to spot and deal with phishing emails. We recommend the training resources listed below and would really appreciate it if you would complete the corporate training on the topic at your earliest convenience.

How likely would you be to follow up as the response requests?

Very likely	Somewhat likely	Neither likely nor unlikely	Somewhat unlikely	Very unlikely	Not sure/don't know
☐	☐	☐	☐	☐	☐

How likely would you be to engage with future phishing simulation exercises (e.g. you would simply ignore or delete a message you suspected of being phishing)?

Very likely	Somewhat likely	Neither likely nor unlikely	Somewhat unlikely	Very unlikely	Not sure/don't know
☐	☐	☐	☐	☐	☐

How positive/negative would this response make you feel about cyber security in your organisation?

Very positive	Somewhat positive	Neither positive nor negative	Somewhat negative	Very negative	Not sure/don't know
☐	☐	☐	☐	☐	☐

How positive/negative would this response make you feel about your organisation as a whole?

Very positive	Somewhat positive	Neither positive nor negative	Somewhat negative	Very negative	Not sure/don't know
☐	☐	☐	☐	☐	☐

Please read the following statements and select how strongly you agree or disagree with each one.

	Strongly agree	Slightly agree	Neither agree nor disagree	Slightly disagree	Strongly disagree	Not sure/don't know
This response would make me feel angry	☐	☐	☐	☐	☐	☐
This response would make me feel bullied	☐	☐	☐	☐	☐	☐
This response would make me feel empowered	☐	☐	☐	☐	☐	☐
This response would make me feel motivated	☐	☐	☐	☐	☐	☐
This response would make me feel invested in cyber security	☐	☐	☐	☐	☐	☐
I would learn lessons from this response	☐	☐	☐	☐	☐	☐
This response would help me improve my skill set	☐	☐	☐	☐	☐	☐
This response would help build my trust in the organisation	☐	☐	☐	☐	☐	☐

Do you have any other thoughts about this response?

Have you ever worked at a company that used a response with this tone?

- ☐ Yes
- ☐ No
- ☐ Not sure/don't know
- ☐ Not applicable

Scenario #4

For the following questions, you will be asked to imagine that you have "failed" a phishing simulation exercise at work. You will be presented with a possible response from your cyber security department in response to your actions with the phishing simulation and asked to consider how the wording in the response would make you feel.

How would you feel if, following a phishing simulation exercise, you received the following email from the cyber security department?

It seems like our existing training has not been successful in communicating the risks that phishing emails can pose, not just to us as a business but also to individuals. We are committed to improving the skills our employees have in this area to make our cyber space safer. We would like to offer some training that could help. Please could you let us know which of the following you would find most useful:

- E-training.
- Video.
- Classroom training.
- 1-on-1 coaching.
- Quiz.
- Nothing.

How likely would you be to follow up as the response requests?

- | | | | | | |
|-----------------------|-----------------------|-----------------------------|-----------------------|-----------------------|-----------------------|
| Very likely | Somewhat likely | Neither likely nor unlikely | Somewhat unlikely | Very unlikely | Not sure/don't know |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

How likely would you be to engage with future phishing simulation exercises (e.g. you would simply ignore or delete a message you suspected of being phishing)?

- | | | | | | |
|-----------------------|-----------------------|-----------------------------|-----------------------|-----------------------|-----------------------|
| Very likely | Somewhat likely | Neither likely nor unlikely | Somewhat unlikely | Very unlikely | Not sure/don't know |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

How positive/negative would this response make you feel about cyber security in your organisation?

- | | | | | | |
|-----------------------|-----------------------|-------------------------------|-----------------------|-----------------------|-----------------------|
| Very positive | Somewhat positive | Neither positive nor negative | Somewhat negative | Very negative | Not sure/don't know |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

How positive/negative would this response make you feel about your organisation as a whole?

- | | | | | | |
|-----------------------|-----------------------|-------------------------------|-----------------------|-----------------------|-----------------------|
| Very positive | Somewhat positive | Neither positive nor negative | Somewhat negative | Very negative | Not sure/don't know |
| ☐ | ☐ | ☐ | ☐ | ☐ | ☐ |

Please read the following statements and select how strongly you agree or disagree with each one.

	Strongly agree	Slightly agree	Neither agree nor disagree	Slightly disagree	Strongly disagree	Not sure/don't know
This response would make me feel angry	☐	☐	☐	☐	☐	☐
This response would make me feel bullied	☐	☐	☐	☐	☐	☐
This response would make me feel empowered	☐	☐	☐	☐	☐	☐
This response would make me feel motivated	☐	☐	☐	☐	☐	☐
This response would make me feel invested in cyber security	☐	☐	☐	☐	☐	☐
I would learn lessons from this response	☐	☐	☐	☐	☐	☐
This response would help me improve my skill set	☐	☐	☐	☐	☐	☐
This response would help build my trust in the organisation	☐	☐	☐	☐	☐	☐

Do you have any other thoughts about this response?

Have you ever worked at a company that used a response with this tone?

- ☐ Yes
- ☐ No
- ☐ Not sure/don't know
- ☐ Not applicable

Wrap up questions

Thank you so much for your participation up to this point. There are just a few more questions about the responses you have seen in general.

Of the four scenarios you saw, which do you prefer? As a reminder, the opening sentence of each scenario can be seen (numbered) below.

1. You failed the recent phish training when you clicked on the link in the unsolicited email sent to you.
2. Unfortunately, you have failed the phish training exercise.
3. It looks like you might benefit from gaining some knowledge about how to spot and deal with phishing emails.
4. It seems like our existing training has not been successful in communicating the risks that phishing emails can pose, not just to us as a business but also to individuals.

- ☐ Response #1
- ☐ Response #2
- ☐ Response #3
- ☐ Response #4
- ☐ Not sure/don't know

What is it about that particular response you prefer?

Of the four responses, which do you think would be most likely to make you want to engage in your organisational cyber security?

- ☐ Response #1
- ☐ Response #2
- ☐ Response #3
- ☐ Response #4
- ☐ Not sure/don't know

What is it about that particular response that makes you feel that way?

Of the four responses, which do you think would be most effective at improving your knowledge/skill set?

- ☐ Response #1
- ☐ Response #2
- ☐ Response #3
- ☐ Response #4
- ☐ Not sure/don't know

What is it about that particular response that makes you feel that way?

If given the choice, what types of training would you like to be given the option(s) of?

- ☐ Videos
- ☐ Quizzes
- ☐ In person training
- ☐ Discussion group/workshop
- ☐ Games
- ☐ Reading material
- ☐ e-Learning modules
- ☐ Other

Demographic information

Please provide your email address. This will not be shared with anyone and will not form part of your survey responses. It will only be used should the researcher wish to send you a follow up questionnaire.

Where are you located? Please provide your town/city, state (if applicable), and country.

What is your age?

- ☐ 18-24
- ☐ 25-34
- ☐ 35-44
- ☐ 45-54
- ☐ 55-64
- ☐ 65 and over

What gender do you identify as?

- ☐ Male
- ☐ Female
- ☐ Non binary / Third gender
- ☐ Prefer to self-describe
- ☐ Prefer not to say

Self-described gender:

What is your employment status?

- ☐ Full-time employed
- ☐ Part-time employed
- ☐ Self-employed
- ☐ Unemployed
- ☐ Student
- ☐ Retired
- ☐ Homemaker/carer
- ☐ Unable to work due to disability or illness
- ☐ Other

What is the highest level of education you have completed?

- ☐ No formal education
- ☐ Primary education
- ☐ Secondary education
- ☐ Vocational/technical qualification
- ☐ College/further education
- ☐ Bachelor's degree or equivalent
- ☐ Postgraduate degree (Master's, PhD)

What is your first language?

Follow-up Survey

Project title: EMS10468 (EMS Approval Code) Follow-up

Researcher name: Jane Kocan-Payne

* Required

Project title: EMS10468 (EMS Approval Code)

1

What is the research about?

We invite you to participate in a research project about how people feel about phishing simulations when used for training purposes, and how those perceptions may impact upon the success of a training campaign. Your responses will be used as input for a proposed methodology for future training campaigns that would allow for customisable training options with the aim of improving the success of future campaigns.

2

Do I have to take part?

This form has been written to help you decide if you would like to take part. It is up to you and you alone whether you wish to take part. If you do decide to take part you will be free to withdraw at any time without providing a reason and without penalty.

3

What will I be required to do?

The questions contained within this survey will ask you about your opinions of phishing training methods/approaches and your knowledge of phishing in general. You will be asked to provide your name and email address, which may be used to contact you with the opportunity to partake in a follow-up questionnaire – these details will NOT be used as part of the research. You will also be asked to provide some demographic information. The questionnaire should take approximately 20 minutes to complete. Questions may be skipped if you prefer not to answer. You have the right to withdraw from all or part of the research at any time.

4

How will you handle my data?

Your data will be stored in a fully identifiable form and will only be accessible to Jane Kocan-Payne (contact email: 2300711@uad.ac.uk). This means that your data will be fully identifiable as yours. Your data will be stored in OneDrive (uad.ac.uk domain), with data fully anonymised at the earliest opportunity (i.e., when data that could identify you is no longer necessary for the purposes of the research. Your responses are treated in the strictest confidence - it will be impossible to identify individuals within a dataset when any of the research is disseminated (e.g., in publications/presentations/datasets). Abertay University acts as Data Controller (DataProtectionOfficer@abertay.ac.uk).

5

Retention of research data

Researchers are obliged to retain research data for up to 10 years' post-publication, however your anonymised research data may be retained indefinitely (e.g., so that researchers engage in open research, and other researchers can access their data to confirm the conclusions of published work). Consistent with our data retention policy, researchers retain consent forms for as long as we continue to hold information about a data subject and for 10 years for published research (including Research Degree thesis).

Consent statement: *

Abertay University attaches high priority to the ethical conduct of research. Please consider the following before indicating your consent on this form. Indicating your consent confirms that you are willing to participate in the research, however, indicating consent does not commit you to anything you do not wish to do and you are free to withdraw your participation at any time. You are indicating consent under the following assumptions (please check all boxes):

- ☐ I understand the contents of the participant information sheet and consent form.
- ☐ I have been given the opportunity to ask questions about the research and have had them answered satisfactorily.
- ☐ I understand that my participation is entirely voluntary and that I can withdraw from the research (parts of the project or the entire project) at any time without penalty and without having to provide an explanation.
- ☐ I understand who has access to my data and how it will be handled at all stages of the research project.

7

I consent to take part in this study conducted by Jane Kocan-Payne who intends to use my data for further research examining the perceptions of phishing simulations as a training aid. *

- ☐ I confirm that I am willing to take part in this research.

8

NAME: *

9

ANONYMISED PARTICIPANT ID (found in your invitation email): *

The value must be a number

10

TODAY'S DATE: *

11

You may take a print-out/screenshot of this consent form as no hard copy will be provided to you.

You can find our procedure for complaints (regarding research projects) and our privacy notice and legal basis for processing research data at: <https://www.abertay.ac.uk/legal/privacy-notice-for-research-participants/>

Scenario

For the following questions, you will be asked to imagine that you have "failed" a phishing simulation exercise at work. You will be presented with a possible response from your cyber security department in response to your actions with the phishing simulation and asked to consider how the wording in the response would make you feel.

12

Imagine that you are presented with the following web page upon clicking on a link that turns out to be part of a phishing training campaign.

We're sorry to tell you this but...

That email was a phish.

It seems like our existing training has not been successful in communicating the risks that phishing emails can pose, not just to us as a business but also to individuals. We are committed to improving the skills our employees have in this area to make our cyber space safer.

We would like to offer some training that could help. Please could you let us know which of the following you would find most useful:

Videos

If you like the idea of a lecture but can't commit to a specific session, our video library is a solid alternative.

Choose this option and we'll send you a link with a list of video resources created by the experts that can help you get the most out of your phishing training.

Tick here to add Videos to your training preferences: ☐

Articles

Maybe you want to go old-fashioned with your learning and just read all about it?

Our training library is packed with articles of all lengths that can help you get to grips with the practical actions you can take to improve your cybersecurity skills.

Tick here to add Articles to your training preferences: ☐

You can choose as many of the training options as you want, or just stick with one if you prefer. You can even access all of the training material from our [Training Content page](#), at any time.

Once you're happy with your choices, hit the "Submit" button above, and we'll take care of the rest! And if you change your mind, you can change your preferences at any time from your PhishAdapt profile page.

After completing the form above, you receive the following email (with the content tailored to the choices you set in the form above).

In the event that you do not complete the form above, an email would still be sent to you but the content would not be personalised to your preferences - you would be offered the default training that your company uses.

Dear

We recently ran a phishing training campaign. Our records suggest that you may have clicked on the link we included in the fake phish. It seems like our existing training has not been successful in communicating the risks that phishing emails can pose, not just to us as a business but also to individuals.

We are committed to improving the skills our employees have in this area to make our cyber space safer.

Please see below for your personalised training recommendations.

Looking for all of our training material? [Click here.](#)

Videos

Klik Solutions - How to Identify and Avoid Phishing Scams

In this video, we break down how phishing works, the red flags to watch for, and the best ways to protect yourself. Learn how to spot fake emails, avoid falling for scams, and what to do if you've clicked a phishing link.

Length: approx. 5 minutes

[How to Identify and Avoid Phishing Scams](#)

Defense - Phishing: 3 Tips

A short video with three actionable tips for spotting phishing.

Length: approx. 1 minute

[Phishing: 3 Tips](#)

Articles

[Phishing: Spot and report scam emails, texts, websites and calls \(BPSG\)](#)

[How to Spot a Phishing Email \(CrowdStrike\)](#)

We appreciate that sometimes these campaigns can go wrong, and sometimes we might not do a great job of communicating to you why these things are important.

If you have any feedback or have run in to any problems with the campaign or the material we've sent you, please don't hesitate to contact us at

Thank you, and happy training!

Murphy and the PhishGuard Team

14

How would you feel if, following a phishing simulation exercise, you received the response detailed above from the cyber security department?

15

How likely would you be to follow up as the response requests?

Very likely	Somewhat likely	Neither likely nor unlikely	Somewhat unlikely	Very unlikely	Not sure/don't know
☐	☐	☐	☐	☐	☐

16

How likely would you be to engage with future phishing simulation exercises (e.g. you would not simply ignore or delete a message you suspected of being phishing)?

Very likely	Somewhat likely	Neither likely nor unlikely	Somewhat unlikely	Very unlikely	Not sure/don't know
☐	☐	☐	☐	☐	☐

17

How positive/negative would this response make you feel about cyber security in your organisation?

Very positive	Somewhat positive	Neither positive nor negative	Somewhat negative	Very negative	Not sure/don't know
☐	☐	☐	☐	☐	☐

18

How positive/negative would this response make you feel about your organisation as a whole?

Very positive	Somewhat positive	Neither positive nor negative	Somewhat negative	Very negative	Not sure/don't know
☐	☐	☐	☐	☐	☐

19

Please read the following statements and select how strongly you agree or disagree with each one.

	Strongly agree	Slightly agree	Neither agree nor disagree	Slightly disagree	Strongly disagree	Not sure/don't know
This response would make me feel angry	☐	☐	☐	☐	☐	☐
This response would make me feel bullied	☐	☐	☐	☐	☐	☐
This response would make me feel empowered	☐	☐	☐	☐	☐	☐
This response would make me feel motivated	☐	☐	☐	☐	☐	☐
This response would make me feel invested in cyber security	☐	☐	☐	☐	☐	☐
I would learn lessons from this response	☐	☐	☐	☐	☐	☐
This response would help me improve my skill set	☐	☐	☐	☐	☐	☐
This response would help build my trust in the organisation	☐	☐	☐	☐	☐	☐

Do you have any other thoughts about this response?

21

Have you ever worked at a company that used a response with this tone?

- ☐ Yes
- ☐ No
- ☐ Not sure/don't know
- ☐ Not applicable

Appendix B – Ethics Paperwork

Preliminary Survey Participant Information Sheet

Project title: EMS10468 (EMS Approval Code)

Researcher name: Jane Kocan-Payne

* Required

Project title: EMS10468 (EMS Approval Code)

What is the research about?

We invite you to participate in a research project about how people feel about phishing simulations when used for training purposes, and how those perceptions may impact upon the success of a training campaign. Your responses will be used as input for a proposed methodology for future training campaigns that would allow for customisable training options with the aim of improving the success of future campaigns.

Do I have to take part?

This form has been written to help you decide if you would like to take part. It is up to you and you alone whether you wish to take part. If you do decide to take part you will be free to withdraw at any time without providing a reason and without penalty.

What will I be required to do?

The questions contained within this survey will ask you about your opinions of phishing training methods/approaches and your knowledge of phishing in general. You will be asked to provide your name and email address, which may be used to contact you with the opportunity to partake in a follow-up questionnaire – these details will NOT be used as part of the research. You will also be asked to provide some demographic information. The questionnaire should take approximately 20 minutes to complete. Questions may be skipped if you prefer not to answer. You have the right to withdraw from all or part of the research at any time.

How will you handle my data?

Your data will be stored in a fully identifiable form and will only be accessible to Jane Kocan-Payne (contact email: 2300711@uad.ac.uk). This means that your data will be fully identifiable as yours. Your data will be stored in OneDrive (uad.ac.uk domain), with data fully anonymised at the earliest opportunity (i.e., when data that could identify you is no longer necessary for the purposes of the research. Your responses are treated in the strictest confidence - it will be impossible to identify individuals within a dataset when any of the research is disseminated (e.g., in publications/presentations/datasets). Abertay University acts as Data Controller (DataProtectionOfficer@abertay.ac.uk).

Retention of research data

Researchers are obliged to retain research data for up to 10 years' post-publication, however your anonymised research data may be retained indefinitely (e.g., so that researchers engage in open research, and other researchers can access their data to confirm the conclusions of published work). Consistent with our data retention policy, researchers retain consent forms for as long as we continue to hold information about a data subject and for 10 years for published research (including Research Degree thesis).

Consent statement: *

Abertay University attaches high priority to the ethical conduct of research. Please consider the following before indicating your consent on this form. Indicating your consent confirms that you are willing to participate in the research, however, indicating consent does not commit you to anything you do not wish to do and you are free to withdraw your participation at any time. You are indicating consent under the following assumptions (**please check all boxes**):

- ☐ I understand the contents of the participant information sheet and consent form.
- ☐ I have been given the opportunity to ask questions about the research and have had them answered satisfactorily.
- ☐ I understand that my participation is entirely voluntary and that I can withdraw from the research (parts of the project or the entire project) at any time without penalty and without having to provide an explanation.
- ☐ I understand who has access to my data and how it will be handled at all stages of the research project.

I consent to take part in this study conducted by Jane Kocan-Payne who intends to use my data for further research examining the perceptions of phishing simulations as a training aid. *

- ☐ I confirm that I am willing to take part in this research.

NAME: *

DATE: *

You may take a print-out/screenshot of this consent form as no hard copy will be provided to you.

You can find our procedure for complaints (regarding research projects) and our privacy notice and legal basis for processing research data at: <https://www.abertay.ac.uk/legal/privacy-notice-for-research-participants/>

Preliminary Survey Participant Debrief Sheet

Nature of research

This research is part of a Master's thesis project exploring how people perceive phishing awareness training methods, and whether those perceptions influence how effective the training is.

Phishing emails are a common cybersecurity threat, and many organisations use training to help staff recognise and respond to them.

You were invited to take part because your experiences and opinions can provide valuable insight into how such training is received and whether it leads to meaningful outcomes, such as increased awareness or changes in behaviour. Your responses will contribute to a better understanding of what types of training might be more effective and relevant for different types of employees.

Data

Your data will be stored, shared and processed as outlined in the Participant Information Sheet and Consent form (page 1 of this questionnaire) for this project (EMS10468). If you no longer wish to participate in the research, you are free to withdraw at any time. **You will be able to withdraw your data before Sunday 20th July, 23:59.** If your information (data) is anonymous at the point of collection or subsequently anonymised, we will not be able to withdraw it after that point because we will no longer know which information (data) is yours.

Sources of support

If any of the questions in this survey have caused you distress or brought up difficult emotions, please know that support is available. Your well-being is important, and we encourage you to reach out to someone you trust or contact one of the support services detailed below.

For students studying at Abertay University, details for the Student Services/Student Counselling teams can be found at <https://www.abertay.ac.uk/life/student-support-and-services/counselling/>.

For non-students: Samaritans (UK) – Free 24/7 emotional support: Call 116 123 or visit www.samaritans.org / Mind (Mental Health Support) – Information and support: Call 0300 123 3393 or visit www.mind.org.uk / Your GP or local healthcare provider – For personal mental or physical health concerns

If you're outside the UK, similar mental health and crisis support services are available in most countries. Please seek local resources or speak with someone you trust.

Contact

If you have any further questions you may contact the researcher or their supervisor on the details below.

Researcher: 2300711@uad.ac.uk (Jane Kocan-Payne)

Supervisors: r.dewar@abertay.ac.uk (Dr. Robert Dewar) / s.prior@abertay.ac.uk (Dr. Suzanne Prior)

Ethics Approval Certificate



Name: JANE KOCAN-PAYNE

Project Title: Assessing the Perceptions of Phishing Awareness Campaigns: A Methodology for Customised Delivery to Improve Training Outcomes

Reference: EMS10468

Status: Full Approval

Approval Date: 24.06.25

The Standard Conditions below apply to all approved student Research Ethics applications:

- i. If any substantive changes to the proposed project are made, a new ethical approval application must be submitted to the Committee.
- ii. The Proposer must remain in regular contact with the project supervisor.
- iii. The Supervisor must see a copy of all materials and procedures prior to commencing data collection.
- iv. Any changes to the agreed procedures must be negotiated with the project supervisor.

Interview Information Sheet

Project title: [Project Title]

Researcher name: [Researcher Name]

What is the research about?

We invite you to participate in a research project about how people feel about phishing simulations when used for training purposes, and how those perceptions may impact upon the success of a training campaign. Your responses will be used as input for a proposed methodology for future training campaigns that would allow for customisable training options with the aim of improving the success of future campaigns.

Do I have to take part?

This form has been written to help you decide if you would like to take part. It is up to you and you alone whether you wish to take part. If you do decide to take part you will be free to withdraw at any time without providing a reason and without penalty.

What will I be required to do?

The questions contained within this interview will ask you about your opinions of a phishing training platform that has been developed. The interview should take approximately 30 minutes to complete. Questions may be skipped if you prefer not to answer. You have the right to withdraw from all or part of the research at any time.

How will you handle my data?

Your data will be stored in a fully identifiable form and will only be accessible to [Researcher Name] (contact email: [Researcher Email Address]). This means that your data will be fully identifiable as yours. Your data will be stored in OneDrive (uad.ac.uk domain), with data fully anonymised at the earliest opportunity (i.e., when data that could identify you is no longer necessary for the purposes of the research. Your responses are treated in the strictest confidence - it will be impossible to identify individuals within a dataset when any of the research is disseminated (e.g., in publications/presentations/datasets). [Research Institution] acts as Data Controller ([Data Controller Email Address]).

Retention of research data

Researchers are obliged to retain research data for up to 10 years' post-publication, however your anonymised research data may be retained indefinitely (e.g., so that researchers engage in open research, and other researchers can access their data to confirm the conclusions of published work). Consistent with our data retention policy, researchers retain consent forms for as long as we continue to hold information about a data subject and for 10 years for published research (including Research Degree thesis).

Consent statement:

[Research Institution] attaches high priority to the ethical conduct of research. Please consider the following before indicating your consent on this form. Indicating your consent confirms that you are willing to participate in the research, however, indicating consent does not commit you to anything you do not wish to do and you are free to withdraw your participation at any time. You are indicating consent under the following assumptions:

- I understand the contents of the participant information sheet and consent form.

- I have been given the opportunity to ask questions about the research and have had them answered satisfactorily.
- I understand that my participation is entirely voluntary and that I can withdraw from the research (parts of the project or the entire project) at any time without penalty and without having to provide an explanation.
- I understand who has access to my data and how it will be handled at all stages of the research project.

I consent to take part in this study conducted by [Researcher Name] who intends to use my data for further research examining the perceptions of phishing simulations as a training aid.

I confirm that I am willing to take part in this research:

NAME:

DATE:

You can find our procedure for complaints (regarding research projects) and our privacy notice and legal basis for processing research data at:

<https://www.abertay.ac.uk/legal/privacy-notice-for-research-participants/>

Interview Debrief Form

Project title: [Project Title]

Researcher name: [Researcher Name]

Thank you for taking part in this research project; your contribution is valuable.

Nature of research

This research is part of a Master's thesis project exploring how people perceive phishing awareness training methods, and whether those perceptions influence how effective the training is. Phishing emails are a common cybersecurity threat, and many organisations use training to help staff recognise and respond to them.

You were invited to take part because your experiences and opinions can provide valuable insight into how such training is received and whether it leads to meaningful outcomes, such as increased awareness or changes in behaviour. Your responses will contribute to a better understanding of what types of training might be more effective and relevant for different types of employees.

Data

Your data will be stored, shared and processed as outlined in the Participant Information Sheet and Consent form for this project ([Ethical Consent Reference]). If you no longer wish to participate in the research, you are free to withdraw at any time. You will be able to withdraw your data before [Deadline]. If your information (data) is anonymous at the point of collection or subsequently anonymised, we will not be able to withdraw it after that point because we will no longer know which information (data) is yours.

Sources of support

If any of the questions in this survey have caused you distress or brought up difficult emotions, please know that support is available. Your well-being is important, and we encourage you to reach out to someone you trust or contact one of the support services detailed below.

For students studying at [Research Institution], details for the Student Services/Student Counselling teams can be found at [URL].

For non-students:

Samaritans (UK) – Free 24/7 emotional support: Call 116 123 or visit www.samaritans.org

Mind (Mental Health Support) – Information and support: Call 0300 123 3393 or visit www.mind.org.uk

Your GP or local healthcare provider – For personal mental or physical health concerns

If you're outside the UK, similar mental health and crisis support services are available in most countries. Please seek local resources or speak with someone you trust.

Further reading

If you'd like to know more about phishing and the risks it poses to businesses or to you as an individual, please see the following links:

NCSC guidance on phishing: <https://www.ncsc.gov.uk/guidance/phishing>

Phishing quiz provided by Google: <https://phishingquiz.withgoogle.com/>

Stay Safe Online (guidance for staying safe in cyber safe provided by the National Cybersecurity Alliance): <https://www.staysafeonline.org/>

Have I Been Pwned (provides information for whether your details may have been leaked as part a data breach, which can be caused by a successful phish):

<https://haveibeenpwned.com/>

Contact

If you have any further questions, you may contact the researcher or my supervisor on the details below.

Researcher:

[Researcher Contact Details]

Supervisor (primary):

[Primary Supervisor Contact Details]

Supervisor (secondary):

[Secondary Supervisor Contact Details]

Thank you once again for taking part in the research!

Appendix C – Virtual Machine Setup and Configuration

VM setup

Core platform components

Platform name	Version number	Purpose
Windows 11	24H2 build 26100.4946	Hosting the Oracle VirtualBox virtualisation software.
Oracle VirtualBox	7.1.10 r169112 (Qt6.5.3)	Hosting the Ubuntu server used for the PhishAdapt platform.

Installation procedure

Oracle VirtualBox

Oracle VirtualBox is free software application. PhishAdapt's dependencies apply not to the virtualisation software hosting its server but to the server itself. As such its installation is out of scope for this paper. The installer media can be obtained from the Oracle website:

<https://www.virtualbox.org/wiki/Downloads>

The installation was a simple process and all the defaults offered by the installer were selected. The installer was run as an administrator on the Windows machine.

Ubuntu Server

Due to an error when trying to install Ubuntu Server on to a bare metal virtual machine, a pre-made out-of-the-box virtual machine was downloaded from osboxes.org:

<https://www.osboxes.org/ubuntu-server/>

Importing the hard drive into a newly created virtual machine is outside of the scope of this report, however it should be noted that the amount of RAM allocated to the Ubuntu Server instance was increased to 8GB from the default value to 2GB. All other default values were accepted.

Graphical desktop environment

Ubuntu Server is, by default, a command line-based operating system. In order to be able to develop and interact with the PhishAdapt platform, a graphical user interface

was required. This was installed, from the command line, with the following commands:

```
sudo apt update && sudo apt upgrade  
sudo apt install slim  
sudo apt install kde-plasma-desktop
```

A reboot was required following the installations of SLiM and KDE Plasma. Following the installation of Plasma, the login screen displayed a graphical interface:



Python environment

To ensure that the PhishAdapt platform was able to manage its dependencies free of the constraints of those from the system or other application, a virtual environment was required. This would provide a containerised Python instance for developing the application. The installation of the components necessary to meet this requirement was achieved using the process outlined below.

- Install Python language and the pip tool (for managing Python software packages) with the following command:
- Install Python packages and development tools necessary to ensure a robust programming environment:

```
sudo apt install -y build-essential libssl-dev libffi-dev python3-dev
```

- Install Python venv module, enabling the creation of a virtual environment:

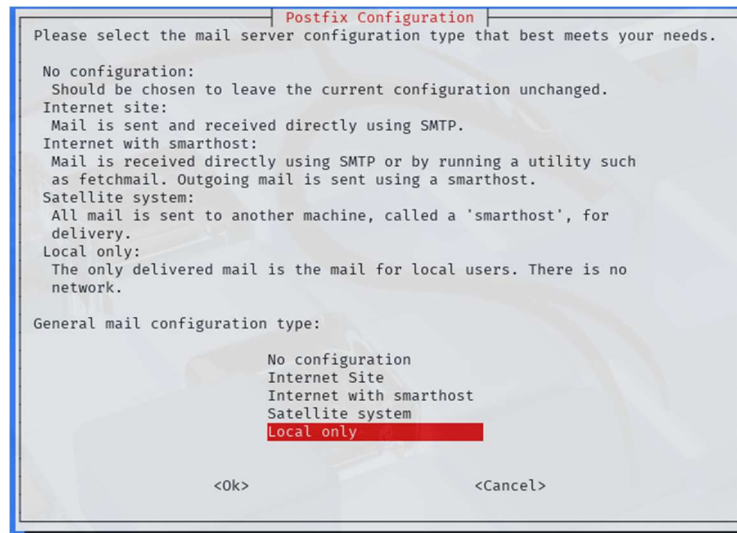
```
sudo apt install -y python3-venv
```

SMTP Server

Postfix is a mail server and mail transfer agent. Its website advocates that it is fast and easy to administer. The documentation for Ubuntu Server includes instructions for its installation. It was installed with the following command at the command line:

```
sudo apt install postfix
```

Installation of Postfix is guided by a wizard. It was installed in “Local only” mode when prompted, as shown below (owing to PhishAdapt being a proof-of-concept platform at this time):



When prompted, a system mail name was entered, as can be seen in the image below. The name chosen at this point is arbitrary (“PhishAdapt.msc” was chosen in this instance, but this is arbitrary).



Once the installation was completed, the mailbox directory was configured (as per the Ubuntu Server documentation) using the following command:

```
sudo postconf -e 'home_mailbox = Maildir/'
```

There was no need to implement the authentication, TLS, or SASL configuration detailed in the document, owing to the local setup implemented.

Upon installation completion, an entry was added to the `/etc/hosts` file to ensure that the chosen system mail name resolved to the local host.

IMAP server

Dovecot is an IMAP server, enabling users to access their mailboxes held on an SMTP server through a graphical e-mail client. Its website advocates that it is fast and secure. The documentation for Ubuntu Server includes instructions for its installation. It was installed with the following command at the command line:

```
sudo apt install dovecot-imapd
```

Once the installation was completed, the mailbox directory was configured (as per the Ubuntu Server documentation) by adjusting the `/etc/dovecot/conf.d/10-mail.conf` file as shown below:

Original configuration

```
# <doc/wiki/MailLocation.txt>
#
mail_location = mbox:~/mail:INBOX=/var/mail/%u
```

Customised configuration

```
# <doc/wiki/MailLocation.txt>
#
mail_location = maildir:~/Maildir
```

After the configuration file had been updated, the Dovecot service was restarted with the following command:

```
sudo systemctl restart dovecot
```

There was no need to implement the SSL or firewall configuration detailed in the document, owing to the local setup implemented.

User accounts

User accounts were created to be able to collect a variety of responses to the post-training campaign questionnaire. User account details were generated using an online random name generator²⁰.

An administrator account/mailbox was also required, to send and receive e-mails for the phishing training campaign and its user questionnaire responses.

Each account was created using the following command:

```
adduser <username>
```

A basic username structure of initial+surname was implemented (e.g., Matthew Bishop became mbishop), though this is arbitrary. A simple password was chosen for each user account which, although arbitrary, was required as part of the sign-in process when logging in to the e-mail client.

E-mail client

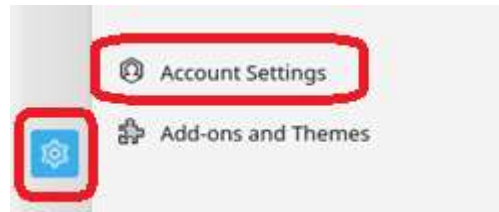
A graphical e-mail client was required to allow users to interact with e-mails sent out by the PhishAdapt platform. Thunderbird is a free open-source e-mail client. It is easy to install and configure with user accounts. It was installed from the command line with the following command:

```
sudo apt install thunderbird
```

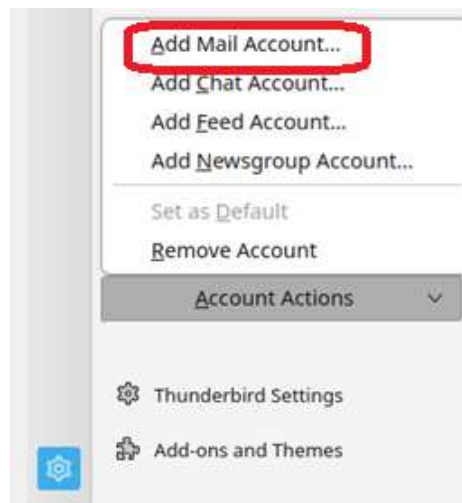
²⁰ <https://www.fakenamegenerator.com/>

It was necessary to log each user account into Thunderbird individually. This was achieved with the procedure detailed below.

- Once Thunderbird is open, click settings (bottom left corner of application window) > Account Settings:



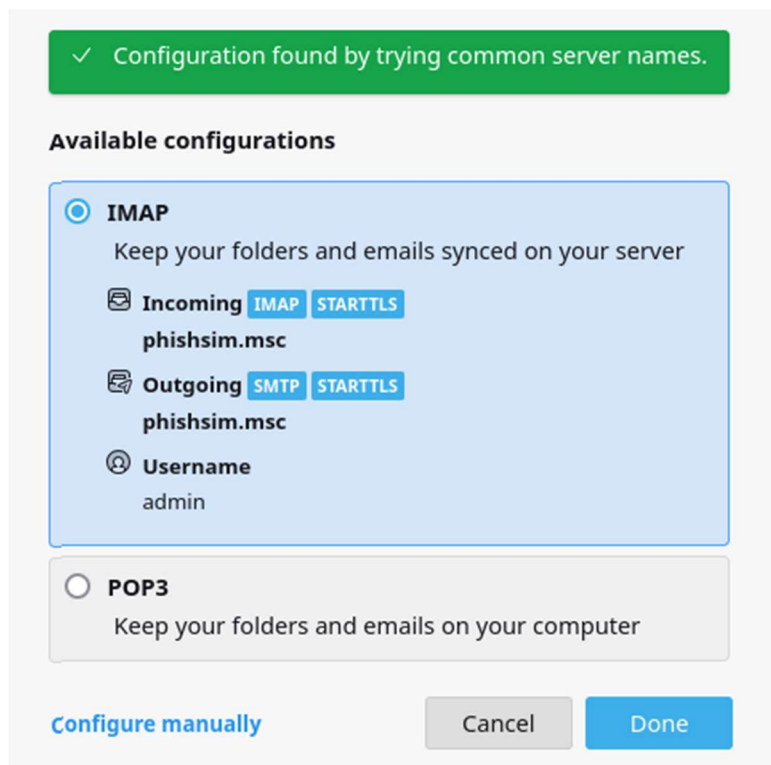
- Click Account Actions and select Add Mail Account... from the drop-down menu:



Enter the user details as set up with the `adduser` command and click Continue:

A screenshot of the Thunderbird account setup form. The 'Your full name' field contains 'PhishAdapt Admin'. The 'Email address' field contains 'phishadaptadmin@phishadapt.msc'. The 'Password' field is masked with dots. The 'Remember password' checkbox is checked. The 'Continue' button is highlighted with a red rectangle. The 'Configure manually' link is also visible. At the bottom, a note states: 'Your credentials will only be stored locally on your computer.'

Thunderbird completes the configuration automatically, at which point the mailbox for that user can be viewed within the e-mail client:



Internet browser

A graphical internet browser was required, which would allow an administrator to interact the PhishAdapt platform, and as support for users interacting with e-mails sent out from it. Firefox is a free open-source internet browser commonly found on Linux operating systems. It was installed from the command line with the following command:

```
sudo apt install firefox
```

Visual Studio Code

A development environment was required, which would allow for the design, development, and implementation of the PhishAdapt web application. Microsoft Visual Studio Code is a cross-platform code editor that enables running applications within a virtual environment. It is a free open-source application commonly used by web developers. The documentation for installation provided by Microsoft²¹ is comprehensive, yet easy to follow and as such, installation instructions will not be replicated here.

²¹ <https://code.visualstudio.com/docs/setup/setup-overview>

Software components of PhishAdapt platform (versioning)

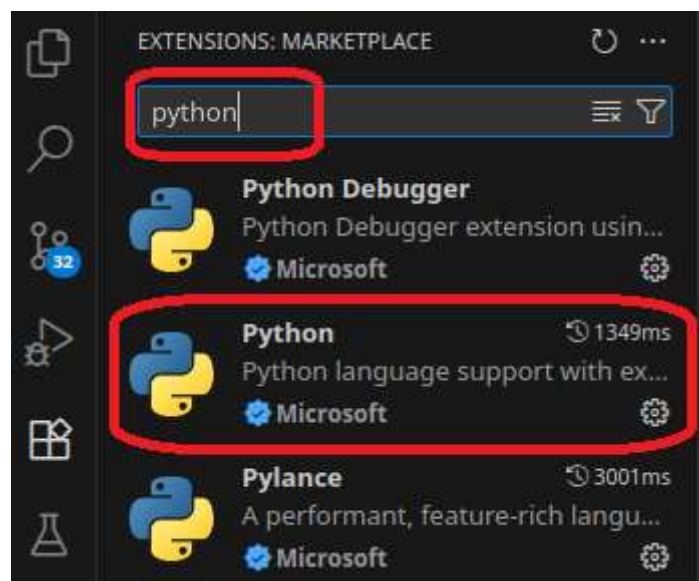
Software name	Version	Function
Ubuntu Server	6.11.0-29-generic	Core operating system of the PhishAdapt server
SLiM	1.3.6	Login manager for the PhishAdapt server
KDE Plasma	6.1.5	Graphical desktop environment for the PhishAdapt server
Python	3.12.7	Scripting language for web application functions
pip	24.2	Software package manager for Python
build-essential	12.10ubuntu1	Meta-package containing essentials software tools and libraries required for development work
libssl-dev	3.3.1-2ubuntu2.1	Development library
libffi-dev	3.4.6-1build1	Development library
python3-dev	3.12.6-0ubuntu1	Python development library
venv	3.12.6-0ubuntu1	Python virtual environment module
Postfix	3.9.0-3	SMTP mail server
Dovecot	2.3.21.1	IMAP server - provides access to user mailboxes via an e-mail client
Thunderbird	128.11.1esr	Graphical e-mail client
Firefox	141.0.3	Graphical internet browser
Visual Studio Code	1.102.1	Development and running of PhishAdapt web application

After installation was complete, the Python extension was installed with the following process:

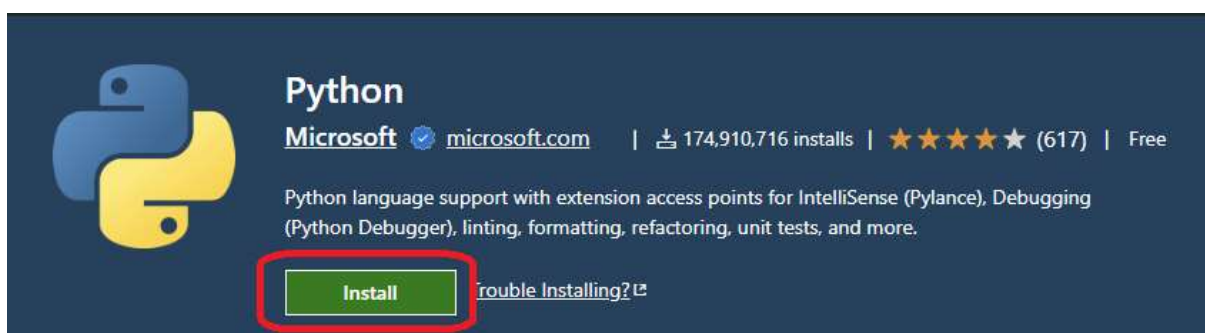
- Click on the Extension button within Visual Studio Code:



- Search for the Python extension:



- Click on the official Microsoft Python extension and click “Install” on the subsequent page:



Appendix D – PhishAdapt Architecture

PhishAdapt Directory Tree

PhishAdapt

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Table of PhishAdapt Functions

Element	Files involved	Function/class name
Initialise web application	app/__init__.py phishadapt.py	create_app()
Create database entries/web application objects (core functionality)	app/models.py	N/A
Base template for web pages	templates/base.html	N/A
CSS style sheet for web pages	app/static/styles.css	N/A
Send email function	app/email.py	app.email.send_email()
Web application config settings	config.py	N/A
Web application environmental variables	.flaskenv	N/A
Blueprint initialisation	app/admin/__init__.py app/auth/__init__.py app/errors/__init__.py app/main/__init__.py app/training/__init__.py	N/A
Child templates for training preference customised content pages	app/templates/_personalised_training.html app/templates/_training_preferences.html	N/A

Child templates for pages to run training campaigns	app/templates/admin/_baseline_send.html app/templates/admin/_baseline_warning.html app/templates/admin/_preview.html	N/A
Error handling	app/errors/handlers.py app/templates/errors/403.html app/templates/errors/404.html app/templates/errors/500.html	app.errors.handlers.not_found_error()
Landing (login) page + login function	app/main/routes.py app/auth/routes.py app/auth/forms.py app/templates/auth/login.html app/static/images//PhishAdapt logo (transparent).png app/models.py	app.main.index() app.auth.routes.login() app.auth.forms.LoginForm() app.models.load_user()
Logout function (link in navigation bar)	app/main/routes.py templates/base.html	app.auth.routes.logout()
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User self-service reset password function	app/auth/routes.py app/auth/forms.py app/templates/auth/login.html app/auth/email.py app/models.py templates/email/reset_password.html	app.auth.routes.reset_password_request() app.auth.ResetPasswordRequestForm() app.auth.email.send_password_reset_email() app.auth.routes.reset_password() app.auth.forms.ResetPasswordForm()
Admin console page	app/admin/routes.py app/admin/forms.py app/templates/admin/console.html	app.admin.routes.console() app.admin.forms.UserSearch()
Admin register new user function	app/admin/routes.py app/admin/forms.py app/templates/admin/register.html	app.admin.routes.register() app.admin.forms.RegistrationForm()
Admin reset user password function	app/admin/routes.py app/admin/forms.py app/templates/admin/reset_user_password.html	app.admin.routes.reset_user_password() app.admin.forms.ResetUserPassword()
Admin edit user function	app/admin/routes.py app/admin/forms.py app/templates/admin/target_edit.html	app.admin.routes.edit_user() app.admin.forms.EditUser()

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User edit training preferences function	app/main/routes.py app/main/forms.py app/templates/edit_training_preferences.html	app.main.routes.training_preferences() app.main.forms.EditTrainingPreferences()
User personalised training suggestions page	app/main/routes.py app/templates/personalised_training.html	app.main.routes.personalised_training()
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Admin targeted campaigns options page	app/admin/routes.py app/templates/admin/targeted_campaigns.html	app.admin.routes.targeted_campaigns()
Admin run departmental targeted voicemail campaign function	app/admin/routes.py app/admin/forms.py app/admin/email.py app/utils/jwt_tokens.py app/templates/admin/group_voicemail.html app/templates/email/baseline_voicemail.html	app.admin.routes.group_voicemail() app.admin.forms.DepartmentalGroups() app.admin.email.send_voicemail() app.utils.jwt_tokens.get_training_token()
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Page to display if user clicks on link in training campaign email	app/training/routes.py app/main/forms.py app/templates/training/clicked.html	app.training.routes.click() app.main.forms.EditTrainingPreferences()
Admin show list of executed campaigns function	app/admin/routes.py app/admin/forms.py app/templates/admin/executed_campaigns.html	app.admin.routes.executed_campaigns() app.admin.forms.FilterCampaigns()
Admin send training invitation to risky users function	app/admin/routes.py app/admin/forms.py app/admin/email.py app/templates/admin/training_invitation.html app/templates/email/training_invite.html	app.admin.routes.training_invitation() app.admin.forms.FilterCampaigns() app.admin.email.send_training_invitation()

Appendix E – PhishAdapt Code

.flaskenv

```
FLASK_APP=phishadapt.py
```

config.py

```
# Import libraries
import os

# Set file path for SQL db file
basedir = os.path.abspath(os.path.dirname(__file__))

class Config:
    # Secret key required forms (anti-CSRF)
    SECRET_KEY = os.environ.get('SECRET_KEY') or 'you-wil-never-guess'
    # Set db file path
    SQLALCHEMY_DATABASE_URI = os.environ.get('DATABASE_URL') or 'sqlite:/// ' +
os.path.join(basedir, 'app.db')

    # Mail server settings
    MAIL_SERVER = os.environ.get('MAIL_SERVER')
    MAIL_PORT = int(os.environ.get('MAIL_PORT') or 25)
    # MAIL_PORT = 8025 # << this for local debugging testing
    MAIL_USE_TLS = os.environ.get('MAIL_USE_TLS') is not None
    MAIL_USERNAME = os.environ.get('MAIL_USERNAME')
    MAIL_PASSWORD = os.environ.get('MAIL_PASSWORD')
    # ADMINS = [''] << add admin email addresses here
```

phishadapt.py

```
# Import app and db class instances
from app import create_app

# Create web application instance
app = create_app()
```

app/__init__.py

```
# Import libraries
from flask import Flask
import os # For use with logging
from config import Config
from flask_sqlalchemy import SQLAlchemy
from flask_migrate import Migrate
from flask_login import LoginManager
import logging
```



```

from logging.handlers import SMTPHandler, RotatingFileHandler
from flask_mail import Mail

# Instantiate DB class and migration config
db = SQLAlchemy()
migrate = Migrate()

# Instantiate login class and set class methods
login = LoginManager()
login.login_view = 'auth.login'
login.login_message = 'Please log in to access this page.'

# Instantiate mail class
mail = Mail()

# Main web application function
def create_app(config_class=Config):
    # Instantiate web application class instance
    app = Flask(__name__)
    # Set configuration options for class
    app.config.from_object(config_class)

    # Initialise classes
    db.init_app(app)
    migrate.init_app(app, db)
    login.init_app(app)
    mail.init_app(app)

    # Register blueprints with web application
    from app.main import bp as main_bp
    app.register_blueprint(main_bp)

    from app.auth import bp as auth_bp
    app.register_blueprint(auth_bp, url_prefix='/auth')

    from app.admin import bp as admin_bp
    app.register_blueprint(admin_bp, url_prefix='/admin')

    from app.errors import bp as errors_bp
    app.register_blueprint(errors_bp)

    from app.training import bp as training_bp
    app.register_blueprint(training_bp, url_prefix='/training')

    # Set error logging to email administrators when web app not in debug mode

```

```

if not app.debug and not app.testing:
    if app.config['MAIL_SERVER']:
        auth = None
        if app.config['MAIL_USERNAME'] or app.config['MAIL_PASSWORD']:
            auth = (app.config['MAIL_USERNAME'],
                    app.config['MAIL_PASSWORD'])
        secure = None
        if app.config['MAIL_USE_TLS']:
            secure = ()
        mail_handler = SMTPHandler(
            mailhost=(app.config['MAIL_SERVER'], app.config['MAIL_PORT']),
            fromaddr='no-reply@' + app.config['MAIL_SERVER'],
            toaddrs=app.config['ADMINS'], subject='PhishAdapt Failure',
            credentials=auth, secure=secure)
        mail_handler.setLevel(logging.ERROR)
        app.logger.addHandler(mail_handler)

    # Create logs folder if it doesn't already exist
    if not os.path.exists('logs'):
        os.mkdir('logs')
    file_handler = RotatingFileHandler('logs/phishadapt.log',
maxBytes=10240, backupCount=10)
    file_handler.setFormatter(logging.Formatter(
        '%(asctime)s %(levelname)s: %(message)s '
        '[in %(pathname)s:%(lineno)d]'))
    file_handler.setLevel(logging.INFO)
    app.logger.addHandler(file_handler)

    # Set logger default logging level to INFO
    app.logger.setLevel(logging.INFO)

    # Set default text for logger log entries
    app.logger.info('PhishAdapt startup')

return app

```

app/email.py

```

# Import required libraries
from flask_mail import Message

# Import mail classes from app function
from app import mail

def send_email(subject, sender, recipients, html_body): # add msg.body back to
arguments if text option required

```

```

# Instantiate Message object
msg = Message(subject, sender=sender, recipients=recipients)
## Add text email option if email server/client requires
# msg.body = text_body
msg.html = html_body
mail.send(msg)

```

app/models.py

```

# Import required libraries
import sqlalchemy as sa
import sqlalchemy.orm as so
from sqlalchemy.orm import relationship
from flask_login import UserMixin
from typing import Optional, List
from werkzeug.security import generate_password_hash, check_password_hash #
For use in User model
from hashlib import md5 # For use in User model
from time import time # For use in User model
import jwt # For use in User model
from flask import current_app # For use in User model
from datetime import datetime # For use in User model

# Import classes from other functions
from app import db, login

# Create User class
class User(UserMixin, db.Model):
    # Set data types for db entries
    id: so.Mapped[int] = so.mapped_column(primary_key=True)
    username: so.Mapped[str] = so.mapped_column(sa.String(64), index=True,
unique=True)
    firstname: so.Mapped[str] = so.mapped_column(sa.String(16), index=True)
    lastname: so.Mapped[str] = so.mapped_column(sa.String(16), index=True)
    email: so.Mapped[str] = so.mapped_column(sa.String(120), index=True,
unique=True)
    jobtitle: so.Mapped[str] = so.mapped_column(sa.String(32), index=True)
    team: so.Mapped[str] = so.mapped_column(sa.String(32), index=True)
    department: so.Mapped[str] = so.mapped_column(sa.String(32), index=True)
    password_hash: so.Mapped[Optional[str]] = so.mapped_column(sa.String(256))
    is_admin: so.Mapped[bool] = so.mapped_column(sa.Boolean)

    # Map relationship to profile table
    profile: so.Mapped["Profile"] = relationship(back_populates="user",
uselist=False)

    # Map relationship to campaign_results table

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```

        campaign_results: so.Mapped[List["CampaignResult"]] =
relationship(back_populates="user")

# Method to set user password
def set_password(self, password):
    # Generate password hash for insertion to db
    self.password_hash = generate_password_hash(password)

# Method to check user password
def check_password(self, password):
    # Check provided password hash against hash held in db
    return check_password_hash(self.password_hash, password)

# Set string representation of for object
def __repr__(self):
    return '<User {}>'.format(self.username)

# Method to display user avatar
def avatar(self, size):
    # Generate MD5 hash of user email address
    digest = md5(self.email.lower().encode('utf-8')).hexdigest()
    # Generate avatar using hashed email address. Third party service
    return
f'https://www.gravatar.com/avatar/{digest}?d=identicon&s={size}'

# Method to generate JWT token for user self-service reset password
function
def get_reset_password_token(self, expires_in=600):
    # Generate JWT token
    return jwt.encode({'reset_password': self.id, 'exp': time() +
expires_in}, current_app.config['SECRET_KEY'], algorithm='HS256')

# Class method to verify validity of a user reset JWT token
@staticmethod
def verify_reset_password_token(token):
    # Decode JWT, verify decoded contents match format for encoding the
original token
    try:
        id = jwt.decode(token, current_app.config['SECRET_KEY'],
algorithms=['HS256'])['reset_password']
        # Skip functionality if JWT does not match encoding format
    except:
        return
    # Instantiate a User object with user information from db if decoded
JWT matches encoding format

```

```

        return db.session.get(User, id)

# User login function
@login.user_loader
def load_user(id):
    # Instantiate a User object with user information from db
    return db.session.get(User, int(id))

# Create Organisation (for team/department info) class
class Organisation(db.Model):
    # Set data types for db entries
    id: so.Mapped[int] = so.mapped_column(primary_key=True)
    department: so.Mapped[str] = so.mapped_column(sa.String(64), index=True)
    team: so.Mapped[str] = so.mapped_column(sa.String(64), index=True)

# Create Profile class
class Profile(db.Model):
    # Set data types for db entries
    id: so.Mapped[int] = so.mapped_column(primary_key=True)
    user_id: so.Mapped[int] = so.mapped_column(sa.ForeignKey(User.id),
index=True, unique=True)
    user: so.Mapped["User"] = relationship(back_populates="profile")
    instructor: so.Mapped[bool] = so.mapped_column(sa.Boolean, nullable=True)
    group: so.Mapped[bool] = so.mapped_column(sa.Boolean, nullable=True)
    game: so.Mapped[bool] = so.mapped_column(sa.Boolean, nullable=True)
    elearn: so.Mapped[bool] = so.mapped_column(sa.Boolean, nullable=True)
    quiz: so.Mapped[bool] = so.mapped_column(sa.Boolean, nullable=True)
    demo: so.Mapped[bool] = so.mapped_column(sa.Boolean, nullable=True)
    video: so.Mapped[bool] = so.mapped_column(sa.Boolean, nullable=True)
    text: so.Mapped[bool] = so.mapped_column(sa.Boolean, nullable=True)
    visual: so.Mapped[bool] = so.mapped_column(sa.Boolean, nullable=True)
    coach: so.Mapped[bool] = so.mapped_column(sa.Boolean, nullable=True)
    audio: so.Mapped[bool] = so.mapped_column(sa.Boolean, nullable=True)
    risk: so.Mapped[int] = so.mapped_column(sa.Integer, nullable=True)

# Create Campaign class
class Campaign(db.Model):
    # Set data types for db entries
    id: so.Mapped[int] = so.mapped_column(primary_key=True)
    name: so.Mapped[str] = so.mapped_column(sa.String(128), index=True)
    sent_at: so.Mapped[datetime] = so.mapped_column(sa.DateTime,
default=datetime.now)

    # Map relationship to campaign result table

```

```

        results: so.Mapped[List["CampaignResult"]] =
relationship(back_populates="campaign", cascade="all, delete-orphan")

# Create CampaignResult class
class CampaignResult(db.Model):
    # Set data types for db entries
    id: so.Mapped[int] = so.mapped_column(primary_key=True)
    campaign_id: so.Mapped[int] =
so.mapped_column(sa.ForeignKey('campaign.id'), index=True, nullable=False)
    user_id: so.Mapped[int] = so.mapped_column(sa.ForeignKey("user.id",
name="fk_campaign_result_user_id"), nullable=False)
    username: so.Mapped[str] = so.mapped_column(sa.String(64), index=True)
    clicked: so.Mapped[bool] = so.mapped_column(sa.Boolean, default=False)

    # Map relationship to campaign table
    campaign: so.Mapped["Campaign"] = relationship(back_populates="results")
    # Map relationship to user table
    user: so.Mapped["User"] = relationship(back_populates="campaign_results")

```

app/admin/__init__.py

```

# Import Blueprint library
from flask import Blueprint

# Initialise admin Blueprint
bp = Blueprint('admin', __name__)

# Import routes (required here, prevents circular import issues)
from app.admin import routes

```

app/admin/email.py

```

# Import required libraries
from flask import render_template, current_app

# Import functions from other Blueprints
from app.email import send_email
from app.utils.jwt_tokens import get_training_token

# Import classes from other functions
from app.models import Profile

# Function for sending voicemail training campaign email to user with user-
specific link
def send_voicemail(user, campaign):
    # Create JWT token
    token = get_training_token(user.id, campaign.id)

```

```

    # Set variables for send_email function
    send_email(f'Voicemail for {user.username}',
               sender='voicemail@company.com',
               recipients=[user.email],
               html_body=render_template('email/baseline_voicemail.html',
user=user, token=token))

# Function for sending training invitation email to user with user-specific
link
def send_training_invitation(user):
    # Obtain user profile details
    profile = Profile.query.filter_by(user_id=user.id).first()
    # Set variables for send_email function
    send_email(f'Training recommendations for {user.username}',
               sender='voicemail@company.com',
               recipients=[user.email],
               html_body=render_template('email/training_invite.html',
user=user, profile=profile))

```

app/admin/forms.py

```

# Import required libraries
from flask_wtf import FlaskForm
from wtforms import StringField, PasswordField, BooleanField, SubmitField,
SelectField, IntegerField
from wtforms.validators import DataRequired, Email, EqualTo, NumberRange

# Form for user search on admin console page
class UserSearch(FlaskForm):
    username = StringField('Username:', validators=[DataRequired()])
    search = SubmitField('Search')

# Form for registering new users
class RegistrationForm(FlaskForm):
    username = StringField('Username', validators=[DataRequired()])
    firstname = StringField('First Name', validators=[DataRequired()])
    lastname = StringField('Last Name', validators=[DataRequired()])
    email = StringField('Email address', validators=[DataRequired(), Email()])
    jobtitle = StringField('Job Title', validators=[DataRequired()])
    team = SelectField('Team', choices = [], validators=[DataRequired()])
    department = SelectField('Department', choices = [],
validators=[DataRequired()])
    password = PasswordField('Password', validators=[DataRequired()])
    password2 = PasswordField('Repeat Password', validators=[DataRequired(),
EqualTo('password')])
    isadmin = BooleanField('Admin?')

```

```

        submit = SubmitField('Register')

# Form for running baseline campaign
class BaselineCampaign(FlaskForm):
    send = SubmitField('Send Campaign!')

# Form to generate departmental dropdown list for Campaign filter
class DepartmentalGroups(FlaskForm):
    department = SelectField(choices = [], validators=[DataRequired()])
    send = SubmitField('Send Campaign!')

# Form to generate risk score dropdown lists for Campaign filter
class RiskGroups(FlaskForm):
    operator = SelectField(choices = [], validators=[DataRequired()])
    score = IntegerField(validators=[DataRequired(), NumberRange(min=0,
max=100, message='Enter a value between 0 and 100')])
    send = SubmitField('Send Campaign!')

# Form to generate dropdown lists for user list filter
class FilterUsers(FlaskForm):
    department = SelectField('Department', choices = [])
    score_operator = SelectField('Risk score', choices = [])
    score = IntegerField()
    training_preference = SelectField('Training Preference')
    submit = SubmitField('Filter')

# Form for editing existing user information
class EditUser(FlaskForm):
    username = StringField('Username:', validators=[DataRequired()])
    firstname = StringField('First Name', validators=[DataRequired()])
    lastname = StringField('Last Name', validators=[DataRequired()])
    email = StringField('Email address', validators=[DataRequired(), Email()])
    jobtitle = StringField('Job Title', validators=[DataRequired()])
    current_team = StringField('Current Team')
    new_team = SelectField('New Team', choices = [],
validators=[DataRequired()])
    current_department = StringField('Current Department')
    new_department = SelectField('New Department', choices = [],
validators=[DataRequired()])
    password = PasswordField('Password', validators=[DataRequired()])
    password2 = PasswordField('Repeat Password', validators=[DataRequired(),
EqualTo('password')])
    isadmin = BooleanField('Admin?')
    update = SubmitField('Update user')

```



```

# Form for deleting existing user
class DeleteUser(FlaskForm):
    username = StringField('Username:', validators=[DataRequired()])
    delete = SubmitField('Delete User')

# Form to generate name dropdown list for Campaign Result filter
class FilterCampaigns(FlaskForm):
    campaign = SelectField('Campaign Name', choices=[])
    submit = SubmitField('Filter')
    send = SubmitField('Send')

# Form for admin reset password function
class ResetUserPassword(FlaskForm):
    username = StringField('Username:', validators=[DataRequired()])
    password = PasswordField('Password', validators=[DataRequired()])
    password2 = PasswordField('Repeat Password', validators=[DataRequired(),
    EqualTo('password')])
    update = SubmitField('Update password')

```

app/admin/routes.py

```

# Import Blueprint class instance
from app.admin import bp

# Import required libraries
from flask import render_template, flash, redirect, url_for, abort, request
from flask_login import current_user, login_required
import operator
from datetime import datetime
from sqlalchemy import func

# Import required forms
from app.admin.forms import UserSearch, RegistrationForm, ResetUserPassword,
EditUser, DeleteUser, BaselineCampaign, DepartmentalGroups, RiskGroups,
FilterUsers, FilterCampaigns

# Import classes from other functions
from app import db
from app.models import User, Organisation, Profile, Campaign, CampaignResult

# Import functions from other Blueprints
from app.admin.email import send_voicemail, send_training_invitation

# Blueprint route for admin console page
@bp.route('/console', methods=['GET', 'POST'])

```

```

@login_required
def console():
    # Check if logged-in user is registered as an admin and return 403
    (Forbidden) error message if not
    if not current_user.is_admin:
        abort(403)

    # Instantiate instance of user search form
    form = UserSearch()

    # Attempt to set a user object to edit if request to page submitted
    includes a completed form
    if form.validate_on_submit():
        target = form.username.data
        # Redirect to admin edit user
        return redirect(url_for('admin.edit_user', target=target))

    # Render admin console page if request contains no form data
    return render_template('admin/console.html', title='Admin console',
form=form)

# Blueprint route for admin register new user page
@bp.route('/register', methods=['GET', 'POST'])
@login_required
def register():
    # Check if logged-in user is registered as an admin and return 403
    (Forbidden) error message if not
    if not current_user.is_admin:
        abort(403)

    # Instantiate instance of user registration form
    form = RegistrationForm()

    # Generate options for dropdown Department list and set to form field
    departments = db.session.query(Organisation.department).distinct().all()
    form.department.choices = [(dept.department, dept.department) for dept in
departments]

    # Generate options for dropdown Team list and set to form field
    teams = db.session.query(Organisation.team).distinct().all()
    form.team.choices = [(team.team, team.team) for team in teams]

    # Instantiate new User object and attributes using form data if request to
page submitted includes a completed form
    if form.validate_on_submit():
        user = User(
            username=form.username.data,
            firstname=form.firstname.data,
            lastname=form.lastname.data,
            email=form.email.data,
            jobtitle=form.jobtitle.data,
            team=form.team.data,

```

```

        department=form.department.data,
        is_admin=form.isadmin.data)
    # Set user password as per form data
    user.set_password(form.password.data)
    # Enter user information into user table
    db.session.add(user)
    db.session.commit()
    # Create user profile for new user
    profile = Profile(user_id=user.id, risk=0)
    db.session.add(profile)
    db.session.commit()
    # Display info message on completion
    username = form.username.data
    flash(f'{username} is now a registered user.')
    # Redirect to main admin console page on completion
    return redirect(url_for('admin.console'))

# Render register new user page if request contains no form data
return render_template('admin/register.html', title='Register', form=form)

# Blueprint route for admin reset password functionality
@bp.route('/reset_user_password', methods=['GET', 'POST'])
@login_required
def reset_user_password():
    # Check if logged-in user is registered as an admin and return 403
    (Forbidden) error message if not
    if not current_user.is_admin:
        abort(403)

    # Instantiate reset user password form
    form = ResetUserPassword()
    if form.validate_on_submit():
        # Format user input from form
        username = form.username.data.strip()
        # Set provided user name as variable
        user = db.session.query(User).filter_by(username=username).first()
        # Display error message if user not found in user db
        if not user:
            flash(f'User {username} not found.', 'warning')
            # Redirect to admin console
            return redirect(url_for('admin.console'))
        # Set password for supplied user
        user.set_password(form.password.data)
        db.session.commit()
        # Display info message on completion
        flash(f"The password for '{username}' has been changed!")
        # Redirect to admin console page on completion

```

```

        return redirect(url_for('admin.console'))

    # Render admin reset user password page if request contains no form data
    return render_template('admin/reset_user_password.html', title='Reset User
password', form=form)

# Blueprint route for admin edit user functionality
@bp.route('/edit_user/<target>', methods=['GET', 'POST'])
@login_required
def edit_user(target):
    # Check if logged-in user is registered as an admin and return 403
    (Forbidden) error message if not
    if not current_user.is_admin:
        abort(403)

    # Set user for editing as variable
    user = db.session.query(User).filter_by(username=target).first()

    # Display error message if user name provided (from UserSearch form on
    Console page) is invalid
    if not user:
        flash(f'User {target} not found.', 'warning')
        # Redirect to admin console page
        return redirect(url_for('admin.console'))

    # Instantiate admin edit user form
    form=EditUser()

    # Generate options for dropdown Department list and set to form field
    departments = db.session.query(Organisation.department).distinct().all()
    form.new_department.choices = [(dept.department, dept.department) for dept
in departments]

    # Generate options for dropdown Team list and set to form field
    teams = db.session.query(Organisation.team).distinct().all()
    form.new_team.choices = [(team.team, team.team) for team in teams]

    # Set attributes for user as per form if request made via submit button
    if request.method=='POST':
        if form.update.data:
            user.firstname = form.firstname.data
            user.lastname = form.lastname.data
            user.email = form.email.data
            user.jobtitle = form.jobtitle.data
            user.team = form.new_team.data
            user.department = form.new_department.data
            user.is_admin = form.isadmin.data
            # Update db values with form data
            db.session.commit()
            # Display info message on completion
            flash(f"User '{user.username}' has been updated successfully.",
'success')

```

```

        # Redirect to admin console page on completion
        return redirect(url_for('admin.console'))
    # Autofill form with existing user data if request made via user search
function
    else:
        form.username.data = user.username
        form.firstname.data = user.firstname
        form.lastname.data = user.lastname
        form.email.data = user.email
        form.jobtitle.data = user.jobtitle
        form.current_team.data = user.team
        form.new_team.data = user.team
        form.current_department.data = user.department
        form.new_department.data = user.department
        form.isadmin.data = user.is_admin
        # Render admin edit user page if request is made via user search function
        return render_template('admin/target_edit.html', title=f'Edit User
{user.username}', form=form, user=user)

# Blueprint route for delete user functionality
@bp.route('/delete_user', methods=['GET', 'POST'])
@login_required
def delete_user():
    # Check if logged-in user is registered as an admin and return 403
(Forbidden) error message if not
    if not current_user.is_admin:
        abort(403)
    # Instantiate admin delete user form
    form = DeleteUser()
    if form.validate_on_submit():
        # Format user input from form
        username = form.username.data.strip()
        # Set provided user name as variable
        user = db.session.query(User).filter_by(username=username).first()
        # Display error message if user not found in user db
        if not user:
            flash(f'User {username} not found.', 'warning')
            # Redirect to admin console
            return redirect(url_for('admin.console'))
        # Set profile information if user name valid
        profile = db.session.query(Profile).filter_by(user_id=user.id).first()
        # Delete profile if it exists
        if profile:
            db.session.delete(profile)
        # Delete user and commit db changes

```

```

        db.session.delete(user)
        db.session.commit()
        # Display info message on completion
        flash(f"User '{username}' and associated profile deleted
successfully.", 'success')
        # Redirect to admin console page
        return redirect(url_for('admin.console'))
    # Render admin delete user page if request contains no form data
    return render_template('admin/delete_user.html', title='Delete User',
form=form)

# Blueprint route for displaying list of users and associated profile
information
@bp.route('/users', methods=['GET', 'POST'])
@login_required
def users():
    # Check if logged-in user is registered as an admin and return 403
(Forbidden) error message if not
    if not current_user.is_admin:
        abort(403)
    # Instantiate form for user filtering
    form = FilterUsers()
    # Generate options for dropdown Department list and set to form field
    departments = db.session.query(Organisation.department).distinct().all()
    form.department.choices = [('All', 'All')] + [(dept.department,
dept.department) for dept in departments]
    # Generate options for dropdown risk score operator list and set to form
field
    form.score_operator.choices = [('All', 'All'), ('>', '>'), ('>=', '>='),
('<', '<'), ('<=', '<='), ('==', '==')]
    # Generate options for user training preference training from available
options in profile table and set to form field
    preference_choices = [
        ('instructor', 'Instructor-led'),
        ('group', 'Group-based'),
        ('game', 'Games'),
        ('elearn', 'e-Learning'),
        ('quiz', 'Quizzes'),
        ('demo', 'Demonstration'),
        ('video', 'Video'),
        ('text', 'Articles'),
        ('visual', 'Visual-based'),
        ('coach', 'Coaching'),
        ('audio', 'Audio-based')
    ]

```

```

    form.training_preference.choices = [('All', 'All')] + preference_choices
    # Query db to get profile info for user
    query = db.session.query(User).join(Profile)
    # Apply filters set to user list if submit button used to make request to
route
    if request.method == 'POST':
        if form.department.data != 'All':
            query = query.filter(User.department == form.department.data)
        if form.score_operator.data != 'All' and form.score.data is not None:
            # Map risk score operator dropdown list choices to arithmetic
expressions
            op_map = {'>': operator.gt, '>=': operator.ge, '<': operator.lt,
'<=': operator.le, '==': operator.eq}
            query =
query.filter(op_map[form.score_operator.data](Profile.risk, form.score.data))
        if form.training_preference.data != 'All':
            pref_field = getattr(Profile, form.training_preference.data, None)
            if pref_field is not None:
                query = query.filter(pref_field == True)
        users = query.all()
    # Return list of all users if no filters set
    else:
        users = query.all()
    # Render targeted voicemail campaign page if request contains no filtering
data
    return render_template('admin/users.html', title='Users', users=users,
form=form)

# Blueprint route for Campaign page
@bp.route('/campaigns')
@login_required
def campaigns():
    # Check if logged-in user is registered as an admin and return 403
(Forbidden) error message if not
    if not current_user.is_admin:
        abort(403)
    # Render campaigns page
    return render_template('admin/campaigns.html', title='Training Campaigns')

# Blueprint route for baseline campaigns page
@bp.route('/campaigns/baselines')
@login_required
def baselines():
    # Check if logged-in user is registered as an admin and return 403
(Forbidden) error message if not

```

```

        if not current_user.is_admin:
            abort(403)
        # Render baseline campaigns page
        return render_template('admin/baselines.html', title='Baseline Training
Campaigns')

# Blueprint route for voicemail baseline campaign page
@bp.route('/campaigns/baselines/voicemail', methods=['GET', 'POST'])
@login_required
def baseline_voicemail():
    # Check if logged-in user is registered as an admin and return 403
    (Forbidden) error message if not
    if not current_user.is_admin:
        abort(403)

    # Instantiate instance of baseline campaign form
    form = BaselineCampaign()

    # Create new Campaign object and add to campaign table if request to page
    submitted includes a completed form
    if form.validate_on_submit():
        campaign = Campaign(name=f'Baseline Voicemail
{datetime.now().strftime("%d%m%Y")}')
        db.session.add(campaign)
        db.session.commit()

        # Get list of all users in db
        users = db.session.query(User).all()

        # Send voicemail baseline campaign email to all users in list
        for user in users:
            send_voicemail(user, campaign)

            # Add entry to campaign result table for user
            result = CampaignResult(campaign_id=campaign.id, user_id=user.id,
username=user.username)
            db.session.add(result)
            db.session.commit()

        # Display info message on completion
        flash('Baseline voicemail campaign has been sent to all users!')

        # Redirect to main admin console page on completion
        return redirect(url_for('admin.console'))

    # Render baseline voicemail campaign page if request contains no form data
    return render_template('admin/baseline_voicemail.html', title='Baseline
Voicemail Campaign', form=form)

# Blueprint route for targeted campaigns page
@bp.route('/campaigns/targeted_campaigns')
@login_required
def targeted_campaigns():

```



```

        # Check if logged-in user is registered as an admin and return 403
        (Forbidden) error message if not
        if not current_user.is_admin:
            abort(403)
        # Blueprint route for targeted campaigns page
        return render_template('admin/targeted_campaigns.html', title='Targeted
Training Campaigns')

# Blueprint route for departmental targeted voicemail campaigns page
@bp.route('/campaigns/target_campaigns/group_voicemail', methods=['GET',
'POST'])
@login_required
def group_voicemail():
    # Check if logged-in user is registered as an admin and return 403
    (Forbidden) error message if not
    if not current_user.is_admin:
        abort(403)
    # Instantiate form for Department selection
    form = DepartmentalGroups()
    # Generate options for dropdown Department list and set to form field
    departments = db.session.query(Organisation.department).distinct().all()
    form.department.choices = [(dept.department, dept.department) for dept in
departments]
    # Create new Campaign object and add to campaign table if request to page
submitted includes a completed form
    if form.validate_on_submit():
        department = form.department.data
        campaign = Campaign(name=f'Targeted {department} Voicemail
{datetime.now().strftime("%d%m%Y")}')
        db.session.add(campaign)
        db.session.commit()
        # Get list of targeted users in db
        users = db.session.query(User).where(User.department ==
department).all()
        for user in users:
            # Send targeted voicemail campaign email to all users in list
            send_voicemail(user, campaign)
            # Add entry to campaign result table for user
            result = CampaignResult(campaign_id=campaign.id, user_id=user.id,
username=user.username)
            db.session.add(result)
            db.session.commit()
        # Display info message on completion
        flash(f'Targeted voicemail campaign has been sent to all users in
{department}!')

```

```

        # Redirect to main admin console page on completion
        return redirect(url_for('admin.console'))

    # Render targeted voicemail campaign page if request contains no form data
    return render_template('admin/group_voicemail.html', title='Voicemail
Phishing Campaign for Departmental Groups', form=form)

# Blueprint route for risk score targeted voicemail campaigns page
@bp.route('/campaigns/target_campaigns/risk_voicemail', methods=['GET',
'POST'])
@login_required
def risk_voicemail():
    # Check if logged-in user is registered as an admin and return 403
    (Forbidden) error message if not
    if not current_user.is_admin:
        abort(403)

    # Instantiate form for Risk Score selection
    form=RiskGroups()

    # Generate options for dropdown risk score operator list and set to form
    field
    form.operator.choices = [('>', '>'), ('>=', '>='), ('<', '<'), ('<=',
'<='), ('==', '==')]

    # Create new Campaign object and add to campaign table if request to page
    submitted includes a completed form
    if form.validate_on_submit():
        # Map dropdown list choices to arithmetic expressions
        op_map = {'>': operator.gt, '>=': operator.ge, '<': operator.lt, '<=':
operator.le, '==': operator.eq}
        op_func = op_map[form.operator.data]
        # Set score as per form data
        score = form.score.data
        # Create new Campaign object and add to campaign table
        campaign = Campaign(name=f'Risk Score {form.operator.data} {score}
Voicemail {datetime.now().strftime("%d%m%Y")}')
        db.session.add(campaign)
        db.session.commit()
        # Get list of targeted users in db
        users =
db.session.query(User).join(Profile).filter(op_func(Profile.risk,
score)).all()
        for user in users:
            # Send targeted voicemail campaign email to all users in list
            send_voicemail(user, campaign)
            # Add entry to campaign result table for user
            result = CampaignResult(campaign_id=campaign.id, user_id=user.id,
username=user.username)

```

```

        db.session.add(result)
    db.session.commit()
    # Display info message on completion
    flash(f'Targeted voicemail campaign has been sent to all users with a
risk score {form.operator.data} {score}!')
    # Redirect to main admin console page on completion
    return redirect(url_for('admin.console'))
    # Render targeted voicemail campaign page if request contains no form data
    return render_template('admin/risk_voicemail.html', title='Voicemail
Phishing Campaign for Groups by Risk Score', form=form)

# Blueprint route for displaying a list of executed campaigns
@bp.route('/executed_campaigns', methods=['GET', 'POST'])
@login_required
def executed_campaigns():
    # Check if logged-in user is registered as an admin and return 403
    (Forbidden) error message if not
    if not current_user.is_admin:
        abort(403)
    # Instantiate form for filtering campaign list
    form = FilterCampaigns()
    # Generate options for dropdown campaigns list and set to form field
    campaigns = db.session.query(Campaign.name).distinct().all()
    form.campaign.choices = [(campaign.name, campaign.name) for campaign in
campaigns]
    # Query db to obtain campaign results per campaign
    results_query = (
        db.session.query(CampaignResult, Campaign, User)
        .join(Campaign, CampaignResult.campaign_id == Campaign.id)
        .join(User, CampaignResult.user_id == User.id)
    )
    # Return list of campaign results if request contains filtering data
    if form.validate_on_submit():
        selected_campaign = form.campaign.data
        results_query = results_query.filter(Campaign.name ==
selected_campaign)
    # Return list of all campaign results if no filtering data present
    results = results_query.all()
    # Count number of emails sent across campaign(s) displayed
    total_sent = results_query.count()
    # Count number of user clicks across campaign(s) displayed
    clicked_count = results_query.filter(CampaignResult.clicked ==
True).count()
    # Calculate click rate across campaign(s) displayed
    click_rate = (clicked_count / total_sent * 100) if total_sent else 0

```

```

# Render executed campaigns page
return render_template(
    'admin/executed_campaigns.html',
    title='Executed Campaigns',
    form=form,
    results=results,
    total_sent=total_sent,
    clicked_count=clicked_count,
    click_rate=round(click_rate, 2)
)

# Blueprint route for sending training invitation functionality
@bp.route('/training_invitations', methods=['GET', 'POST'])
@login_required
def training_invitation():
    # Check if logged-in user is registered as an admin and return 403
    (Forbidden) error message if not
    if not current_user.is_admin:
        abort(403)
    # Instantiate form for campaign filtering
    form = FilterCampaigns()
    # Query db to display all campaigns
    subquery = (
        db.session.query(
            Campaign.name,
            func.max(Campaign.id).label("latest_id")
        )
        .group_by(Campaign.name)
        .subquery()
    )
    campaigns = (
        db.session.query(Campaign)
        .join(subquery, Campaign.id == subquery.c.latest_id)
        .all()
    )
    # Generate options for dropdown campaign filter list and set to form field
    form.campaign.choices = [(str(c.id), c.name) for c in campaigns]
    # Obtain campaign results for selected campaign if request contains filter
    data
    if form.validate_on_submit():
        selected_campaign_id = int(form.campaign.data)
        campaign_results = (
            db.session.query(CampaignResult)
            .filter(CampaignResult.campaign_id == selected_campaign_id)
            .filter(CampaignResult.clicked.is_(True))

```

```

        .all()
    )
    # Generate list of users from campaign results table for selected
campaign
    users_to_email = [result.user for result in campaign_results]
    # Send email to each user in list
    for user in users_to_email:
        send_training_invitation(user)
    # Display info message on completion
    flash(f"{len(users_to_email)} training invitations sent.")
    # Redirect to admin console page on completion
    return redirect(url_for('admin.console'))

    # Render admin send training invitations page if request contains no form
data
    return render_template('admin/training_invitation.html', title='Send
training invitations', form=form)

```

app/auth/__init__.py

```

# Import Blueprint library
from flask import Blueprint

# Initialise auth Blueprint
bp = Blueprint('auth', __name__)

# Import routes (required here, prevents circular import issues)
from app.auth import routes

```

app/auth/email.py

```

# Import required libraries
from flask import render_template, current_app

# Import functions from other Blueprints
from app.email import send_email

# Function for sending email to user with password reset link
def send_password_reset_email(user):
    # Set JWT for requesting user
    token = user.get_reset_password_token()
    # Set variables for send_email function
    send_email('[PhishAdapt] reset your password',
                sender=current_app.config['ADMINS'][0],
                recipients=[user.email],
                ## Add text email option if email server/client requires
                # text_body=render_template('email/reset_password.txt',
user=user, token=token),

```

```
html_body=render_template('email/reset_password.html',
user=user, token=token))
```

app/auth/forms.py

```
# Import required libraries
from flask_wtf import FlaskForm
from wtforms import StringField, PasswordField, BooleanField, SubmitField
from wtforms.validators import DataRequired, Email, EqualTo

# Form for use with Login blueprint
class LoginForm(FlaskForm):
    username = StringField('Username', validators=[DataRequired()])
    password = PasswordField('Password', validators=[DataRequired()])
    remember_me = BooleanField('Remember Me')
    submit = SubmitField('Sign In')

# Form for use for requesting a password reset
class ResetPasswordRequestForm(FlaskForm):
    email = StringField('Email', validators=[DataRequired(), Email()])
    submit = SubmitField('Request Password Reset')

# Form for use when resetting password
class ResetPasswordForm(FlaskForm):
    password = PasswordField('Password', validators=[DataRequired()])
    password2 = PasswordField('Repeat Password', validators=[DataRequired(),
    EqualTo('password')])
    submit = SubmitField('Reset Password')
```

app/auth/routes.py

```
# Import Blueprint class instance
from app.auth import bp

# Import required libraries
from flask import render_template, flash, redirect, url_for, request
from flask_login import current_user, login_user, logout_user
import sqlalchemy as sa
from urllib.parse import urlsplit # Required for login route redirect

# Import required forms
from app.auth.forms import LoginForm
from app.auth.forms import ResetPasswordRequestForm, ResetPasswordForm

# Import functions from other Blueprints
from app.auth.email import send_password_reset_email
```

```

# Import classes from other functions
from app import db
from app.models import User

# Blueprint route for login page
@bp.route('/login', methods=['GET', 'POST'])
def login():
    # Redirect user to profile page if already authenticated
    if current_user.is_authenticated:
        return redirect(url_for('main.user', username=current_user.username))
    # Instantiate instance of login form
    form = LoginForm()
    # Attempt to login user if request to page submitted includes a completed
    form
    if form.validate_on_submit():
        # Set user as provided in form
        user = db.session.scalar(
            sa.select(User).where(User.username == form.username.data))
        # Provide error message if username cannot be found or password does
        not match db entry and redirect to login page
        if user is None or not user.check_password(form.password.data):
            flash('Invalid username or password')
            return redirect(url_for('auth.login'))
        # Login user if all details correct as per db entry
        login_user(user, remember=form.remember_me.data)
        # Set page for redirect on successful login (user profile)
        next_page = request.args.get('next')
        if not next_page or urlsplit(next_page).netloc != '':
            next_page = url_for('main.user', username=user.username)
        # Redirect user
        return redirect(next_page)
    # Render login page if request contains no form data
    return render_template('auth/login.html', title='Sign In', form=form)

# Blueprint route for logout functionality
@bp.route('/logout', methods=['GET', 'POST'])
def logout():
    # Logout user
    logout_user()
    # Redirect to login page on logout
    return redirect(url_for('auth.login'))

# Blueprint route for page to request a password request
@bp.route('/reset_password_request', methods=['GET', 'POST'])
def reset_password_request():

```

```

        # Redirect user to profile page if already authenticated
    if current_user.is_authenticated:
        next_page = url_for('main.user', username=current_user.username)
    # Instantiate instance of ResetPasswordRequestForm
    form = ResetPasswordRequestForm()
    # Send email to user if request to page submitted includes a completed
form
    if form.validate_on_submit():
        # Set user as provided in form
        user = db.session.scalar(sa.select(User).where(User.email ==
form.email.data))
        if user:
            # Send email if user found in db
            send_password_reset_email(user)
            # Provide information message to user
            flash('Check your email for the instructions to reset your password')
            # Redirect to login page
            return redirect(url_for('auth.login'))
        # Render reset_password_request page if request contains no form data
        return render_template('auth/reset_password_request.html', title='Reset
Password', form=form)

# Blueprint route for reset password page
@bp.route('reset_password/<token>', methods=['GET', 'POST'])
def reset_password(token):
    if current_user.is_authenticated:
        # Redirect user to profile page if already authenticated
        next_page = url_for('main.user', username=current_user.username)
    # Set user as provided in decoded JWT token
    user = User.verify_reset_password_token(token)
    # Redirect to login page if JWT token does not contain a valid user
    if not user:
        return redirect(url_for('auth.login'))
    # Instantiate instance of ResetPasswordForm
    form = ResetPasswordForm()
    # Reset user password if request to page submitted includes a completed
form
    if form.validate_on_submit():
        user.set_password(form.password.data)
        db.session.commit()
        # Provide information message to user
        flash('Your password has been reset.')
        # Redirect to login page
        return redirect(url_for('auth.login'))
    # Render reset_password_request page if request contains no form data

```



```
return render_template('auth/reset_password.html', form=form)
```

app/errors/__init__.py

```
# Import Blueprint library
from flask import Blueprint

# Initialise errors Blueprint
bp = Blueprint('errors', __name__)

# Import routes (required here, prevents circular import issues)
from app.errors import handlers
```

app/errors/handlers.py

```
# Import Blueprint class instance
from app.errors import bp

# Import required libraries
from flask import render_template

# Import classes from other functions
from app import db

# Blueprint route for error 403 (forbidden) page
@bp.app_errorhandler(403)
def forbidden(error):
    # Roll back any pending db changes
    db.session.rollback()
    # Render 403 page
    return render_template('errors/403.html'), 403

# Blueprint route for error 404 (page not found) page
@bp.app_errorhandler(404)
def not_found_error(error):
    # Render 404 page
    return render_template('errors/404.html'), 404

# Blueprint route for error 500 (internal server error) page
@bp.app_errorhandler(500)
def internal_error(error):
    # Roll back any pending db changes
    db.session.rollback()
    # Render 500 page
    return render_template('errors/500.html'), 500
```

app/main/__init__.py

```
# Import Blueprint library
from flask import Blueprint

# Initialise main Blueprint
bp = Blueprint('main', __name__)

# Import routes (required here, prevents circular import issues)

from app.main import routes
```

app/main/forms.py

```
# Import required libraries
from flask_wtf import FlaskForm
from wtforms import StringField, SubmitField, BooleanField
from wtforms.validators import Length

# Form for user edit profile functionality
class EditProfileForm(FlaskForm):
    firstname = StringField('First name', validators=[Length(min=0, max=16)])
    lastname = StringField('Last name', validators=[Length(min=0, max=16)])
    submit = SubmitField('Submit')

# Form for user edit training preferences functionality
class EditTrainingPreferences(FlaskForm):
    instructor = BooleanField('Tick here to add Instructor-led training to
your training preferences: ')
    group = BooleanField('Tick here to add Group-based training to your
training preferences: ')
    game = BooleanField('Tick here to add Gamification to your training
preferences: ')
    elearn = BooleanField('Tick here to add e-Learning to your training
preferences: ')
    quiz = BooleanField('Tick here to add Quizzes to your training
preferences: ')
    demo = BooleanField('Tick here to add Demonstrations to your training
preferences: ')
    video = BooleanField('Tick here to add Videos to your training
preferences: ')
    text = BooleanField('Tick here to add Articles to your training
preferences: ')
    visual = BooleanField('Tick here to add Visual training to your training
preferences: ')
    coach = BooleanField('Tick here to add Coaching to your training
preferences: ')
```

```

        audio = BooleanField('Tick here to add Audio training to your training
preferences: ')
        submit = SubmitField('Submit')

```

app/main/routes.py

```

# Import Blueprint class instance
from app.main import bp

# Import required libraries
from flask import redirect, render_template, url_for, flash, request
from flask_login import login_required, current_user
import sqlalchemy as sa
import io # For use with user profile (displaying risk score gauge)
import base64 # For use with user profile (displaying risk score gauge)
import matplotlib # For use with user profile (displaying risk score gauge)
matplotlib.use('Agg') # Needed for matplotlib to function in headless
environment
import matplotlib.pyplot as plt # For use with user profile (displaying risk
score gauge)
from matplotlib.patches import Rectangle # For use with user profile
(displaying risk score gauge)

# Import classes from other functions
from app import db
from app.models import User, Profile

# Import required forms
from app.main.forms import EditProfileForm, EditTrainingPreferences

# Blueprint route for index/home page
@bp.route('/')
@bp.route('/index')
def index():
    # Redirect to login page
    return redirect(url_for('auth.login'))

# Blueprint route for user profile page
@bp.route('/user/<username>')
@login_required
def user(username):
    # Set user variable for use with retrieving profile details
    user = db.first_or_404(sa.select(User).where(User.username == username))
    # Select profile from db
    profile = db.session.scalar(sa.select(Profile).where(Profile.user_id ==
user.id))

```

```

    # Set risk to 0 if no risk in profile otherwise retrieve risk score from
profile
    risk = profile.risk if profile and profile.risk is not None else 0

    # Set plot size for displaying risk score gauge
    fig, ax = plt.subplots(figsize=(5, 1.2))
    ax.set_xlim(0, 100)
    ax.set_ylim(0, 1)
    ax.axis('off')

    # Set colour scheme and limits for gauge display
    ax.add_patch(Rectangle((0, 0), 40, 1, color = 'green'))
    ax.add_patch(Rectangle((40, 0), 30, 1, color = 'orange'))
    ax.add_patch(Rectangle((70, 0), 30, 1, color = 'red'))

    # Set gauge line properties
    ax.plot([risk, risk], [0, 1], color='black', linewidth=5)

    # Set display text for risk score gauge
    ax.text(50, 1.1, f'Risk score: {risk}', ha='center', fontsize=12,
fontweight='bold')

    # Convert matplotlib object to base64 for embedding into page
    buf = io.BytesIO()
    plt.tight_layout()
    plt.savefig(buf, format='png', transparent=True, dpi=100,
bbox_inches='tight')
    buf.seek(0)
    gauge_base64 = base64.b64encode(buf.read()).decode('utf-8')
    plt.close(fig)

    # Render user profile page
    return render_template('user.html', user=user, profile=profile,
title='User Profile', gauge_image=gauge_base64)

# Blueprint route for user edit profile page
@bp.route('/edit_profile', methods=['GET', 'POST'])
@login_required
def edit_profile():
    # Instantiate instance of edit profile form
    form = EditProfileForm()

    # Change profile details as per form input if request to page submitted
includes a completed form
    if form.validate_on_submit():
        current_user.firstname = form.firstname.data

```

```

        current_user.lastname = form.lastname.data
        db.session.commit()
        # Display information message to user
        flash('Your changes have been saved.')
        # Redirect to profile page on completion
        return redirect(url_for('main.edit_profile'))
    # Prefill form with existing information if request to page not made
    through submit button
    elif request.method == 'GET':
        form.firstname.data = current_user.firstname
        form.lastname.data = current_user.lastname
        # Render user edit profile page if request contains no form data
        return render_template('edit_profile.html', title='Edit Profile',
                               form=form, user=user)

# Blueprint route for user edit training preferences page
@bp.route('/edit_training_preferences', methods=['GET', 'POST'])
@login_required
def edit_training_preferences():
    # Select profile from db
    profile = Profile.query.filter_by(user_id=current_user.id).first()
    # Instantiate instance of edit training preferences form, prefill with
    existing profile preferences
    form = EditTrainingPreferences(obj=profile)
    # Create profile entry in db for user if one does not already exist
    if not profile:
        profile = Profile(user_id=current_user.id)
        db.session.add(profile)
        db.session.commit()

    # Set user training preferences as per form input if request to page
    submitted includes a completed form
    if form.validate_on_submit():
        profile.game = form.game.data
        profile.quiz = form.quiz.data
        profile.video = form.video.data
        profile.text = form.text.data

    ### Fields below to be added when new content added to platform
    # profile.instructor = form.instructor.data
    # profile.group = form.group.data
    # profile.elearn = form.elearn.data
    # profile.demo = form.demo.data
    # profile.visual = form.visual.data
    # profile.coach = form.coach.data
    # profile.audio = form.audio.data

```

```

        db.session.commit()

        # Display info message to user
        flash('Your training preferences have been saved!')
        # Redirect user back to user profile page on completion
        return redirect(url_for('main.user', username=current_user.username))

    # Render edit training preference page if request contains no form data
    return render_template('edit_training_preferences.html', title='Edit
Training Preferences', form=form, profile=profile)

# Blueprint route for personalised training page
@bp.route('/user/<username>/personalised_training')
@login_required
def personalised_training(username):
    # Set user variable for use with retrieving profile details
    user = db.first_or_404(sa.select(User).where(User.username == username))
    # Select profile from db
    profile = Profile.query.filter_by(user_id=user.id).first()
    # Render personalised training page
    return render_template('personalised_training.html', user=user,
title='Personalised training recommendations', profile=profile)

```

app/static/styles.css

```

body {
    background-color: #FFFDD0;
    font-family: Helvetica, Verdana, san-serif;
    color: #12355B;
}

.center-container {
    justify-content: center;
    text-align: center;
    flex-direction: column;
}

.center-container img {
    max-width: 30%;
    max-height: 30%;
}

.accessdenied {
    color: red;
}

```

```

#admin_table {
    border: 1px solid black;
    border-collapse: collapse;
    width: 100%;
}

#admin_table th,
#admin_table td {
    border: 1px solid black;
    padding: 8px;
}

#admin_table tr:nth-child(even) {
    background-color: #f2f2f2;
}

.reset-button {
    display: inline-block;
    padding: 6px 12px;
    margin-left: 10px;
    background-color: #ccc;
    color: #000;
    text-decoration: none;
    border-radius: 4px;
}

.reset-button:hover {
    background-color: #aaa;
}

#click-table td {
    text-align: center;
}

```

app/templates/_personalised_training.html

```

<p>Looking for all of our training material? <a href="{{
url_for('training.content') }}">Click here.</a></p>

<!-- Check if user profile variable set and passed to template -->
{% if profile %}
    <!-- Set temporary variable for returning content if no preferences set --
>
    {% set pref_count = 0 %}

```

```

    <!-- Set temporary variable to provide else conditional evaluation
statement -->
    {% if profile.game %}{% set pref_count = pref_count + 1 %}{% endif %}
    {% if profile.quiz %}{% set pref_count = pref_count + 1 %}{% endif %}
    {% if profile.video %}{% set pref_count = pref_count + 1 %}{% endif %}
    {% if profile.text %}{% set pref_count = pref_count + 1 %}{% endif %}

    {% if pref_count == 0 %}
        <p>It doesn't look like you've set any training preferences. You can
set them from your profile <a href="{ url_for('main.edit_profile')
}}">here.</a></p>
        <p>For now, why not get started with the videos below?</p>
        <p><a href="https://www.youtube.com/watch?v=dKoRUFiwlFM"
target="_blank">How to Identify and Avoid Phishing Scams</a></p>
        <p><a href="https://www.youtube.com/watch?v=nEb5umERTbM&t=32s"
target="_blank">Phishing: 3 Tips</a></p><br>
        {% else %}
            {% if profile.game %}
                <h2>Game-based training</h2>
                <h3>Gone Phishing</h3>
                <p>You will become "James Ferguson", an elderly war veteran who
must distinguish which emails in his inbox can be trusted and which are
phishing attempts. James can explore his environment to find clues that point
to the legitimacy of these emails.</p>
                <p>The goal is to clear the inbox by answering legitimate emails
and ignoring phishing emails, all within a two-minutes timeframe. Good
luck!</p>
                <a href="https://lynsay.itch.io/gone-phishing"
target="_blank">Gone Phishing</a><br>
            {% endif %}

            {% if profile.quiz %}
                <h2>Quizzes</h2>
                <h3>Phishing Quiz by Google</h3>
                <p>Can you spot when you're being phished? Think you can tell
what's real or fake?</p>
                <p><a href="https://phishingquiz.withgoogle.com/"
target="_blank">Google Phishing Quiz</a></p><br>
            {% endif %}

            {% if profile.video %}
                <h2>Videos</h2>
                <h3>Klik Solutions - How to Identify and Avoid Phishing Scams</h3>
                <p>In this video, we break down how phishing works, the red flags
to watch for, and the best ways to protect yourself. Learn how to spot fake

```


emails, avoid falling for scams, and what to do if you've clicked a phishing link.</p>

<p>Length: approx. 3 minutes</p>

<p>How to Identify and Avoid Phishing Scams</p>

<h3>Cofense - Phishing: 3 Tips</h3>

<p>A short video with three actionable tips for spotting phishing.</p>

<p>Length: approx 1 minute</p>

<p>Phishing: 3 Tips</p>

{% endif %}

{% if profile.text %}

<h2>Articles</h2>

<p>Phishing: Spot and report scam emails, texts, websites and calls (NCSC)</p>

<p>How to Spot a Phishing Email (Crowdstrike)</p>

{% endif %}

{% endif %}

<!-- Return standard text if no profile found -->

{% else %}

<p>We couldn't find a training profile for you. Please visit your profile to set your preferences.</p>

<p>For now, why not get started with the videos below?</p>

<p>How to Identify and Avoid Phishing Scams</p>

<p>Phishing: 3 Tips</p>

{% endif %}

app/templates/_ training_preferences.html

<!-- Display for taking user training preferences input -->

<form action="" method="post" novalidate>

{{ form.hidden_tag() }}

<h2>Instructor-led training</h2>

<p>Think of these as lectures - you show up (in-person or virtually) and a trainer imparts wisdom upon you and the other attendees.</p>

```

    <p>Instructor-led training takes place at set dates and times - choosing
this option will prompt an invite to book on to a selection of available
sessions.</p>
    <h3>Coming soon!</h3>
    <!-- Content to include when learning material expanded
    <p>
        <strong>{{ form.instructor.label }}</strong>
        {{ form.instructor() }}<br>
    </p>-->
    <h2>Group-based training</h2>
    <p>Similar to instructor-led training, these sessions encourage a more
interactive type of training, and are led by a trainer.</p>
    <p>Working as a group, you'll be guided through a set of exercises that
will help you to understand what the risks are when it comes to phishing, and
how to avoid them.</p>
    <p>Group-based training takes place at set dates and times - choosing this
option will prompt an invite to book on to a selection of available
sessions.</p>
    <h3>Coming soon!</h3>
    <!--Content to include when learning material expanded
    <p>
        <strong>{{ form.group.label }}</strong>
        {{ form.group() }}<br>
    </p>-->
    <h2>Gaming</h2>
    <p>Don't learn much from being talked at? Why not try learning through
gamification instead?</p>
    <p>Gamification can be a useful option if you'd like to engage in shorter
sessions that you can complete at your own convenience.</p>
    <p>You can even come back to the same game time and time again to see if
you can better your last score!</p>
    <p>
        <strong>{{ form.game.label }}</strong>
        {{ form.game() }}<br>
    </p>
    <h2>e-Learning</h2>
    <p>If you like the idea of a game but aren't a fan of the bright colours
and cheery soundtracks, perhaps you might prefer an e-learning course.</p>
    <p>Similar to a game, e-learning allows you to learn at your own pace,
completing as much or as little of the course as suits you over as many
sessions as is required to complete it.</p>
    <h3>Coming soon!</h3>
    <!--Content to include when learning material expanded
    <p>
        <strong>{{ form.elearn.label }}</strong>

```

```

        {{ form.elearn() }}<br>
    </p>-->
    <h2>Quizzing</h2>
    <p>Quizzes are great for testing the knowledge that you already have.</p>
    <p>These can be used in conjunction with other training methods or on
their own if you want to give yourself a real challenge!</p>
    <p>
        <strong>{{ form.quiz.label }}</strong>
        {{ form.quiz() }}<br>
    </p>
    <h2>Demos</h2>
    <p>If you'd rather see an showcase of "red flags" in real life examples of
phishing e-mails, demonstration material might be for you.</p>
    <h3>Coming soon!</h3>
    <!--Content to include when learning material expanded
    <p>
        <strong>{{ form.demo.label }}</strong>
        {{ form.demo() }}<br>
    </p>-->
    <h2>Videos</h2>
    <p>If you like the idea of a lecture but can't commit to a specific
session, our video library is a solid alternative.</p>
    <p>Choose this option and we'll send you a link with a list of video
resources created by the experts that can help you get the most out of your
phishing training.</p>
    <p>
        <strong>{{ form.video.label }}</strong>
        {{ form.video() }}<br>
    </p>
    <h2>Articles</h2>
    <p>Maybe you want to go old-fashioned with your learning and just read all
about it?</p>
    <p>Our training library is packed with articles of all lengths that can
help you get to grips with the practical actions you can take to improve your
cybersecurity skills.</p>
    <p>
        <strong>{{ form.text.label }}</strong>
        {{ form.text() }}<br>
    </p>
    <h2>Visual training</h2>
    <p>If you're into Manga or Anime, or maybe even a bit of the Beano, this
one might be for you! We have comics and short graphic novels that might
engage you.</p>
    <h3>Coming soon!</h3>
    <!--Content to include when learning material expanded

```

```

<p>
    <strong>{{ form.visual.label }}</strong>
    {{ form.visual() }}<br>
</p>-->
<h2>Coaching</h2>
<p>If you're looking for the personal touch but don't like the crowds so
much, how about a coaching session?</p>
<p>Coaching sessions are available on set dates and times - choosing this
option will prompt an invite to book on to a selection of available
sessions.</p>
<h3>Coming soon!</h3>
<!--Content to include when learning material expanded
<p>
    <strong>{{ form.coach.label }}</strong>
    {{ form.coach() }}<br>
</p>-->
<h2>Audio training</h2>
<p>Are you an audiophile? Perhaps you like to listen to a audiobook on a
walk? Our selection of podcasts might be just what you're looking for!</p>
<h3>Coming soon!</h3>
<!--Content to include when learning material expanded
<p>
    <strong>{{ form.audio.label }}</strong>
    {{ form.audio() }}<br>
</p><br>-->
<p>{{ form.submit() }}</p>
</form>

```

app/templates/base.html

```

<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="utf-8">
        <meta name="viewport" content="width=device-width, initial-scale=1">
        <!-- Set web page title if not provided in Blueprint function -->
        {% if title %}
        <title>{{ title }} - PhishAdapt</title>
        {% else %}
        <title>Welcome to PhishAdapt</title>
        {% endif %}
        <link rel="stylesheet" href="{{ url_for('static',
filename='styles.css') }}">
    </head>
    <body>
        <div>

```

```

        <!-- Display navigation menu -->
        PhishAdapt:
        <!-- Display link to admin functions if logged-in user is an
adminstrator -->
        {% if current_user.is_admin %}
        <a href="{{ url_for('admin.console') }}">Admin console</a>
        {% endif %}
        <!-- Display links for pages applicable to all users -->
        <a href="{{ url_for('main.user', username=current_user.username)
}}">Profile</a>
        <a href="{{ url_for('training.content') }}">Training</a>
        <a href="{{ url_for('auth.logout') }}">Logout</a>
    </div>
    <hr>
    <!-- Display any messages held in Flash buffer -->
    {% with messages = get_flashed_messages() %}
    {% if messages %}
    <ul>
        {%for message in messages %}
        <li>{{ message }}</li>
        {% endfor %}
    </ul>
    {% endif %}
    {% endwith %}
    <main>
        <!-- Display individual block content for page -->
        {% block content %}{% endblock %}
    </main>
</body>
</html>

```

app/templates/edit_profile.html

```

<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1>Edit profile</h1>
    <p>Please note - changes may only be made to the first and last name held
on your profile.</p>
    <p>For any other changes, please contact the administrator at
phishadaptadmin@phishadapt.msc</p>
    <!-- Display for accepting user input -->
    <form action="" method="post">
        {{ form.hidden_tag() }}
        <p>

```

```

        {{ form.firstname.label }}<br>
        {{ form.firstname(size=32) }}<br>
        {% for error in form.firstname.errors %}
        <span style="color: red;">[{{ error }}]</span>
        {% endfor %}
    </p>
    <p>
        {{ form.lastname.label }}<br>
        {{ form.lastname(cols=50, rows=4) }}<br>
        {% for error in form.lastname.errors %}
        <span style="color: red;">[{{ error }}]</span>
        {% endfor %}
    </p>
    <p>{{ form.submit() }}</p>
</form>
{% endblock %}

```

app/templates/edit_training_preferences.html

```

<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1>Training options</h1>
    <p>You can choose as many of the training options as you want, or just
    stick with one if you prefer. You can even access all of the training material
    from our <a href="{{ url_for('training.content') }}">Training Content page, at
    any time.</a></p>
    <p>Once you're happy with your choices, hit the <strong>"Submit"</strong>
    button below the options, and we'll take care of the rest! And if you change
    your mind, you can change your preferences at any time from your PhishAdapt
    profile page.</p>
    <!-- Inherit content from child template -->
    {% include '_training_preferences.html' %}
{% endblock %}

```

app/templates/personalised_training.html

```

<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1>Training recommendations for {{ user.username }}</h1>
    <!-- Inherit content from child template -->
    {% include '_personalised_training.html' %}
{% endblock %}

```

app/templates/user.html

```
<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
<!-- Check to make sure that profile page being accessed belongs to logged-in
user -->
{% if not user == current_user %}
    <h1>Access denied!</h1>
{% else %}
    <h1>Welcome {{ user.firstname }}</h1>
    <table>
        <tr valign="top">
            <td></td>
        </tr>
        <tr>
            <td><p>Username: {{ user.username }}</p></td>
        </tr>
        <tr>
            <td><p>Name: {{user.firstname }} {{ user.lastname }}</p></td>
        </tr>
        <tr>
            <td><p>Job title: {{user.jobtitle }}</p></td>
        </tr>
        <tr>
            <td><p>Team: {{user.team }}</p></td>
        </tr>
        <tr>
            <td><p>Department: {{user.department }}</p></td>
        </tr>
        <tr>
            <td><p>Email address: {{user.email }}</p></td>
        </tr>
    </table><br>
    <a href="{{ url_for('main.edit_profile', username=user.username) }}">Edit
profile</a><br>
    <h2>Training</h2>
    <p><a href="{{ url_for('main.personalised_training',
username=user.username) }}">Your personalised training recommendations</a></p>
    <a href="{{ url_for('main.edit_training_preferences',
username=user.username) }}">Edit training preferences</a><br>

{% endif %}
{% endblock %}
```

app/templates/admin/_baseline_send.html

```
<!-- Display form for running training campaign -->
<form action="" method="post">
    {{ form.hidden_tag() }}
    <p>{{ form.send() }}</p>
</form>
```

app/templates/admin/_baseline_warning.html

```
<p>This campaign will be sent to <strong>all</strong> PhishAdapt users -
please ensure this is correct as there is <strong>no</strong> undo!</p><br>
```

app/templates/admin/_preview.html

```
<h1>Preview</h1>

<p>Below is an example of what your phishing simulation email will look
like to users.</p>

<p>Please check it and confirm that all the details are as you would
expect them to be.</p>
```

app/templates/admin/baselines.html

```
<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1>Baseline campaigns</h1>
    <a href="{{ url_for('admin.baseline_voicemail') }}">Voicemail phishing</a>
{% endblock %}
```

app/templates/admin/baseline_voicemail.html

```
<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <!-- Inherit content from preview header child template -->
    {% include 'admin/_preview.html' %}
    <!-- Inherit content from send to all users warning child template -->
    {% include 'admin/_baseline_warning.html' %}
    <!-- Display form for running training campaign -->
    <br>
    <!-- Inherit content from child template with form to run campaign -->
    {% include 'admin/_baseline_send.html' %}
{% endblock %}
```


app/templates/admin/campaigns.html

```
<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1>Training campaigns</h1>
    <a href="{{ url_for('admin.baselines') }}">Baseline Training campaigns</a>
    <p>Use a baseline training campaign if you want to send a phishing
simulation to all PhishAdapt users.</p><br>
    <a href="{{ url_for('admin.targeted_campaigns') }}">Targeted Training
campaigns</a>
    <p>Use a targeted training campaign if you only want to send a phishing
simulation to some groups of PhishAdapt users.</p>
{% endblock %}
```

app/templates/admin/console.html

```
<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1>Admin console</h1>
    <!-- List with links to admin functions -->
    <ul>
        <li><a href="{{ url_for('admin.register') }}">Register new
user</a></li><br>
        <li><a href="{{ url_for('admin.reset_user_password') }}">Reset user
password</a></li><br>
        <li><a href="{{ url_for('admin.users') }}">Users</a></li><br>
        <li><a href="{{ url_for('admin.campaigns') }}">Run a training
campaign</a></li><br>
        <li><a href="{{ url_for('admin.executed_campaigns') }}">Executed
campaigns</a></li><br>
        <li><a href="{{ url_for('admin.training_invitation') }}">Send training
invitations</a></li><br>
    </ul>
    <h2>Edit user</h2>
    <!--Form for search for editing user functionality -->
    <form method="POST">
        {{ form.hidden_tag() }}
        <p>{{ form.username.label }} {{ form.username }} {{ form.search }}</p>
    </form>
    <a href="{{ url_for('admin.delete_user') }}">Delete user</a>
{% endblock %}
```

app/templates/admin/delete_user.html

```
<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1>Delete user</h1>
    <p><strong>Deleting users cannot be undone!</strong></p>
    <p><strong>Please make sure you need to delete the user and that you enter
the right username!</strong></p>
    <!-- Display form to take username input for user deletion -->
    <form method="POST">
        {{ form.hidden_tag() }}
        <p>
            {{ form.username.label }}
            {{ form.username(size=32) }}
        </p>
        <p>{{ form.delete }}</p>
    </form>
{% endblock %}
```

app/templates/admin/executed_campaigns.html

```
<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1>Executed campaigns</h1>
    <p>
        Filter campaigns by name:
        <!-- Display form for accepting filter values -->
        <form method="POST">
            {{ form.hidden_tag() }}
            {{ form.campaign }}
            {{ form.submit }}
        </form>
    </p>
    <!-- Display list of matching results if filters applied using submit
button -->
    {% if request.method == 'POST' %}
        <div class="filter-summary">
            <p><strong>Filtered by: {{ form.campaign.data }}</strong></p>
            <table id="admin_table">
                <tr>
                    <!-- Display table with campaign statistics -->
                    <th>Total emails sent</th>
                    <th># of users who clicked</th>

```

```

        <th>Click rate</th>
    </tr>
    <tr id="click-table">
        <td>{{ total_sent }}</td>
        <td>{{ clicked_count }}</td>
        <td>{{ click_rate }}%</td>
    </tr>
</table>

<!-- Display button for resetting filters -->
<p><a href="{{url_for('admin.executed_campaigns')}}"
class="reset-button">Reset filters</a></p>
</div>

{% endif %}

<!-- Display table with all users if request to page made with GET -->
<table id="admin_table">
    <tr>
        <th>Campaign name</th>
        <th>Username</th>
        <th>Clicked?</th>

    </tr>

    <!-- Column with yes/no display for user "clicked" result -->
    {% for result, campaign, user in results %}
    <tr>
        <td>{{ campaign.name }}</td>
        <td>{{ user.username }}</td>
        <td>{{ 'Yes' if result.clicked else 'No' }}</td>

    </tr>

    {% endfor %}
</table>

{% endblock %}

```

app/templates/admin/group_voicemail.html

```

<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <!-- Inherit content from child template -->
    {% include 'admin/_preview.html' %}

    <!-- Display preview of training campaign email -->
    <br>

    <!-- Display form for running training campaign -->
    <form action="" method="post">
        {{ form.hidden_tag() }}

        <p>Please select the Departmental Group you want to send this campaign
to: {{ form.department }}</p>

```

```

        <p>{{ form.send() }}</p>
    </form>
{% endblock %}

```

app/templates/admin/register.html

```

<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1 class=>Register new user</h1>
    <!-- Display form for inputting new user data -->
    <form action="" method="post">
        {{ form.hidden_tag() }}
        <p>
            {{ form.username.label }}
            {{ form.username(size=32) }}<br>
            {% for error in form.username.errors %}
            <span style="color:red;">[{{ error }}]</span>
            {% endfor %}
        </p>
        <p>
            {{ form.firstname.label }}
            {{ form.firstname(size=32) }}<br>
            {% for error in form.firstname.errors %}
            <span style="color:red;">[{{ error }}]</span>
            {% endfor %}
        </p>
        <p>
            {{ form.lastname.label }}
            {{ form.lastname(size=32) }}<br>
            {% for error in form.lastname.errors %}
            <span style="color:red;">[{{ error }}]</span>
            {% endfor %}
        </p>
        <p>
            {{ form.email.label }}
            {{ form.email(size=32) }}<br>
            {% for error in form.email.errors %}
            <span style="color:red;">[{{ error }}]</span>
            {% endfor %}
        </p>
        <p>
            {{ form.jobtitle.label }}
            {{ form.jobtitle(size=32) }}<br>
            {% for error in form.jobtitle.errors %}

```

```

        <span style="color:red;">[{{ error }}]</span>
    {% endfor %}
</p>
<p>
    {{ form.team.label }}
    {{ form.team }}<br>
    {% for error in form.team.errors %}
    <span style="color:red;">[{{ error }}]</span>
    {% endfor %}
</p>
<p>
    {{ form.department.label }}
    {{ form.department }}<br>
    {% for error in form.department.errors %}
    <span style="color:red;">[{{ error }}]</span>
    {% endfor %}
</p>
<p>
    {{ form.password.label }}
    {{ form.password(size=32) }}<br>
    {% for error in form.password.errors %}
    <span style="color:red;">[{{ error }}]</span>
    {% endfor %}
</p>
<p>
    {{ form.password2.label }}
    {{ form.password2(size=32) }}<br>
    {% for error in form.password2.errors %}
    <span style="color:red;">[{{ error }}]</span>
    {% endfor %}
</p>
<p>
    {{ form.isadmin.label }}
    {{ form.isadmin(size=32) }}<br>
    {% for error in form.isadmin.errors %}
    <span style="color:red;">[{{ error }}]</span>
    {% endfor %}
</p>
<p>{{ form.submit() }}</p>
</form>
{% endblock %}

```

app/templates/admin/reset_user_password.html

```
<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1 class=>Register new user</h1>
    <!-- Display form for inputting new user data -->
    <form action="" method="post">
        {{ form.hidden_tag() }}
        <p>
            {{ form.username.label }}
            {{ form.username(size=32) }}<br>
            {% for error in form.username.errors %}
            <span style="color:red;">[{{ error }}]</span>
            {% endfor %}
        </p>
        <p>
            {{ form.firstname.label }}
            {{ form.firstname(size=32) }}<br>
            {% for error in form.firstname.errors %}
            <span style="color:red;">[{{ error }}]</span>
            {% endfor %}
        </p>
        <p>
            {{ form.lastname.label }}
            {{ form.lastname(size=32) }}<br>
            {% for error in form.lastname.errors %}
            <span style="color:red;">[{{ error }}]</span>
            {% endfor %}
        </p>
        <p>
            {{ form.email.label }}
            {{ form.email(size=32) }}<br>
            {% for error in form.email.errors %}
            <span style="color:red;">[{{ error }}]</span>
            {% endfor %}
        </p>
        <p>
            {{ form.jobtitle.label }}
            {{ form.jobtitle(size=32) }}<br>
            {% for error in form.jobtitle.errors %}
            <span style="color:red;">[{{ error }}]</span>
            {% endfor %}
        </p>
        <p>
```

```

        {{ form.team.label }}
        {{ form.team }}<br>
        {% for error in form.team.errors %}
        <span style="color:red;">{{{ error }}}</span>
        {% endfor %}
    </p>
    <p>
        {{ form.department.label }}
        {{ form.department }}<br>
        {% for error in form.department.errors %}
        <span style="color:red;">{{{ error }}}</span>
        {% endfor %}
    </p>
    <p>
        {{ form.password.label }}
        {{ form.password(size=32) }}<br>
        {% for error in form.password.errors %}
        <span style="color:red;">{{{ error }}}</span>
        {% endfor %}
    </p>
    <p>
        {{ form.password2.label }}
        {{ form.password2(size=32) }}<br>
        {% for error in form.password2.errors %}
        <span style="color:red;">{{{ error }}}</span>
        {% endfor %}
    </p>
    <p>
        {{ form.isadmin.label }}
        {{ form.isadmin(size=32) }}<br>
        {% for error in form.isadmin.errors %}
        <span style="color:red;">{{{ error }}}</span>
        {% endfor %}
    </p>
    <p>{{ form.submit() }}</p>
</form>
{% endblock %}

```

app/templates/admin/risk_voicemail.html

```

<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <!-- Inherit content from child template -->
    {% include 'admin/_preview.html' %}

```

```

<!-- Display preview of training campaign email -->
<br>
<!-- Display form for running training campaign -->
<form action="" method="post">
    {{ form.hidden_tag() }}
    <p>Please specify what level of risk you want to send this campaign
to: {{ form.operator }} {{ form.score }}</p>
    <p>{{ form.send() }}</p>
</form>
{% endblock %}

```

app/templates/admin/target_user.html

```

<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1>User: {{ form.username.data }}</h1>
    <!-- Display form for inputting new user data -->
    <form method="POST">
        {{ form.hidden_tag() }}
        <p>
            {{ form.username.label }}<br>
            {{ form.username(size=32) }}<br>
            {% for error in form.username.errors %}
            <span style="color: red;">[{{ error }}]</span>
            {% endfor %}
        </p>
        <p>
            {{ form.firstname.label }}<br>
            {{ form.firstname(size=32) }}<br>
            {% for error in form.firstname.errors %}
            <span style="color: red;">[{{ error }}]</span>
            {% endfor %}
        </p>
        <p>
            {{ form.lastname.label }}<br>
            {{ form.lastname(size=32) }}<br>
            {% for error in form.lastname.errors %}
            <span style="color: red;">[{{ error }}]</span>
            {% endfor %}
        </p>
        <p>
            {{ form.email.label }}<br>
            {{ form.email(size=32) }}<br>
            {% for error in form.email.errors %}

```



```

        <span style="color: red;">[{{ error }}]</span>
    {% endfor %}
</p>
<p>
    {{ form.jobtitle.label }}<br>
    {{ form.jobtitle(size=32) }}<br>
    {% for error in form.jobtitle.errors %}
    <span style="color: red;">[{{ error }}]</span>
    {% endfor %}
</p>
<p>
    {{ form.current_team.label }}<br>
    {{ form.current_team(size=32) }}<br>
    {% for error in form.current_team.errors %}
    <span style="color: red;">[{{ error }}]</span>
    {% endfor %}
</p>
<p>
    {{ form.new_team.label }}<br>
    {{ form.new_team }}<br>
    {% for error in form.new_team.errors %}
    <span style="color: red;">[{{ error }}]</span>
    {% endfor %}
<p>
    {{ form.current_department.label }}<br>
    {{ form.current_department(size=32) }}<br>
    {% for error in form.current_department.errors %}
    <span style="color: red;">[{{ error }}]</span>
    {% endfor %}
</p>
<p>
    {{ form.new_department.label }}<br>
    {{ form.new_department }}<br>
    {% for error in form.new_department.errors %}
    <span style="color: red;">[{{ error }}]</span>
    {% endfor %}
</p>
<p>
    {{ form.isadmin.label }}<br>
    {{ form.isadmin(size=32) }}<br>
    {% for error in form.isadmin.errors %}
    <span style="color: red;">[{{ error }}]</span>
    {% endfor %}
</p>
<p>{{ form.update }}</p>

```

```
        </form>
    {% endblock %}
```

app/templates/admin/targeted_campaigns.html

```
<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1>Targeted campaigns</h1>
    <h2>By Departmental groups</h2>
    <a href="{% url_for('admin.group_voicemail') %}">Voicemail phishing</a>
    <h2>By Risk Score</h2>
    <a href="{% url_for('admin.risk_voicemail') %}">Voicemail phishing</a>
{% endblock %}
```

app/templates/admin/training_invitation.html

```
<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1>Send training invitations</h1>
    <p>
        Select campaign to send training invitations for:
        <!-- Display form for sending training invitations -->
        <form method="POST">
            {{ form.hidden_tag() }}
            {{ form.campaign }}
            {{ form.send }}
        </form>
    </p>
{% endblock %}
```

app/templates/admin/users.html

```
<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1>Users</h1>
    <p>
        Filter users by:
        <!-- Display form for accepting filter values -->
        <form method="POST">
            {{ form.hidden_tag() }}
            {{ form.department.label }} {{ form.department }}
        </form>
    </p>
{% endblock %}
```

```

        {{ form.score_operator.label }} {{ form.score_operator }} {{
form.score }}
        {{ form.training_preference.label }} {{ form.training_preference
}}

        {{ form.submit }}
    </form>
</p>
<!-- Display list of matching results if filters applied using submit
button -->
    {% if request.method == 'POST' %}
        <div class="filter-summary">
            <p><strong>Filtered by:</strong></p>
            <ul>
                <li>Department: {{ form.department.data }}</li>
                <li>Risk Score: {{ form.score_operator.data }} {{
form.score.data or "" }}</li>
                <li>Training Preference: {{ form.training_preference.data
}}</li>
            </ul>
            <!-- Display button for resetting filters -->
            <p><a href="{{url_for('admin.users') }}" class="reset-
button">Reset filters</a></p>
        </div>
    {% endif %}
<!-- Display table with all users if request to page made with GET -->
<table id="admin_table">
    <tr>
        <th>Username</th>
        <th>First name</th>
        <th>Last name</th>
        <th>Team</th>
        <th>Department</th>
        <th>Risk score</th>
        <th>Training preferences</th>
    </tr>
    {% for user in users %}
        <tr>
            <td>{{ user.username }}</td>
            <td>{{ user.firstname }}</td>
            <td>{{ user.lastname }}</td>
            <td>{{ user.team }}</td>
            <td>{{ user.department }}</td>
            <td>
                {% if user.profile %}
                    {{ user.profile.risk }}

```

```

        {% else %}
            N/A
        {% endif %}
    </td>
    <td>
        <!-- Column with user training preferences -->
        {% if user.profile %}
            <ul>
                {% if user.profile.instructor %}<li>Instructor-
led</li>{% endif %}
                {% if user.profile.group %}<li>Group-based</li>{%
endif %}
                {% if user.profile.game %}<li>Games</li>{% endif %}
                {% if user.profile.elearn %}<li>e-Learning</li>{%
endif %}
                {% if user.profile.quiz %}<li>Quizzes</li>{% endif %}
                {% if user.profile.demo %}<li>Demonstration</li>{%
endif %}
                {% if user.profile.video %}<li>Videos</li>{% endif %}
                {% if user.profile.text %}<li>Articles</li>{% endif %}
                {% if user.profile.visual %}<li>Visual-based</li>{%
endif %}
                {% if user.profile.coach %}<li>Coaching</li>{% endif
%}
                {% if user.profile.audio %}<li>Audio-based</li>{%
endif %}
            </ul>
        {% else %}
            N/A
        {% endif %}
    </td>

</tr>
{% endfor %}
</table>
{% endblock %}

```

app/templates/auth/login.html

```

<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="utf-8">
        <meta name="viewport" content="width=device-width, initial-scale=1">
        <!-- Set web page title if not provided in Blueprint function -->
        {% if title %}

```

```

<title>{{ title }} - PhishAdapt</title>
{% else %}
<title>Welcome to PhishAdapt</title>
{% endif %}
<link rel="stylesheet" href="{{ url_for('static',
filename='styles.css') }}">
</head>
<body>
    <hr>
    <!-- Display any messages held in Flash buffer -->
    {% with messages = get_flashed_messages() %}
    {% if messages %}
    <ul>
        {%for message in messages %}
        <li>{{ message }}</li>
        {% endfor %}
    </ul>
    {% endif %}
    {% endwith %}
    <div class="center-container">
        <!-- Display PhishAdapt logo -->
        <br>
        <!-- Display LoginForm -->
        <h1 class=>Please log in:</h1>
        <form action="" method="post" novalidate>
            {{ form.hidden_tag() }}
            <p>
                {{ form.username.label }}<br>
                {{ form.username(size=32) }}<br>
                {% for error in form.username.errors %}
                <span style="color:red;">[{{ error }}]</span>
                {% endfor %}
            </p>
            <p>
                {{ form.password.label }}<br>
                {{ form.password(size=32) }}<br>
                {% for error in form.password.errors %}
                <span style="color:red;">[{{ error }}]</span>
                {% endfor %}
            </p>
            <p>{{ form.remember_me() }} {{ form.remember_me.label }}</p>
            <p>{{ form.submit() }}</p>
        </form>
    <p>

```

```

        Forgot Your Password?
        <a href="{{ url_for('auth.reset_password_request') }}">Click
to Reset It</a>
    </p>
</div>
</body>
</html>

```

app/templates/auth/reset_password_request.html

```

<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="utf-8">
        <meta name="viewport" content="width=device-width, initial-scale=1">
        <!-- Set web page title if not provided in Blueprint function -->
        {% if title %}
        <title>{{ title }} - PhishAdapt</title>
        {% else %}
        <title>Welcome to PhishAdapt</title>
        {% endif %}
        <link rel="stylesheet" href="{{ url_for('static',
filename='styles.css') }}">
    </head>
    <body>
        <hr>
        <!-- Display any messages held in Flash buffer -->
        {% with messages = get_flashed_messages() %}
        {% if messages %}
        <ul>
            {%for message in messages %}
            <li>{{ message }}</li>
            {% endfor %}
        </ul>
        {% endif %}
        {% endwith %}
        <div class="center-container">
            <!-- Display PhishAdapt logo -->
            <br>
            <h1>Request Password Reset</h1>
            <form action="" method="post">
                <!-- Display ResetPasswordRequestForm -->
                {{ form.hidden_tag() }}
                <p>
                    {{ form.email.label }}<br>

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        {{ form.email(size=64) }}<br>
        {% for error in form.email.errors %}
        <span style="color: red;">[{{ error }}]</span>
        {% endfor %}
    </p>
    <p>{{ form.submit() }}</p>
    <a href="{{ url_for('main.index') }}">Home</a>
</form>
</div>
</body>
</html>

```

app/templates/auth/reset_password.html

```

<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="utf-8">
        <meta name="viewport" content="width=device-width, initial-scale=1">
        <!-- Set web page title if not provided in Blueprint function -->
        {% if title %}
        <title>{{ title }} - PhishAdapt</title>
        {% else %}
        <title>Welcome to PhishAdapt</title>
        {% endif %}
        <link rel="stylesheet" href="{{ url_for('static',
filename='styles.css') }}">
    </head>
    <body>
        <hr>
        <!-- Display any messages held in Flash buffer -->
        {% with messages = get_flashed_messages() %}
        {% if messages %}
        <ul>
            {%for message in messages %}
            <li>{{ message }}</li>
            {% endfor %}
        </ul>
        {% endif %}
        {% endwith %}
        <div class="center-container">
            <!-- Display PhishAdapt logo -->
            <br>
            <h1>Reset Your Password</h1>
            <!-- Display ResetPasswordForm -->

```

```

<form action="" method="post">
    {{ form.hidden_tag() }}
    <p>
        {{ form.password.label }}<br>
        {{ form.password(size=32) }}<br>
        {% for error in form.password.errors %}
        <span style="color: red;">[{{ error }}]</span>
        {% endfor %}
    </p>
    <p>
        {{ form.password2.label }}<br>
        {{ form.password2(size=32) }}<br>
        {% for error in form.password2.errors %}
        <span style="color: red;">[{{ error }}]</span>
        {% endfor %}
    </p>
    <p>{{ form.submit() }}</p>
</form>
</div>
</body>
</html>

```

app/templates/email/baseline_voicemail.html

```

<!DOCTYPE html>
<html>
    <body>
        <p>Dear {{user.firstname }},</p>
        <p><strong>New Message from 311-152-3414</strong></p><br>
        <p>Attention: <a href="{{ url_for('training.clicked', token=token,
        _external=True) }}">__ *****@*****.t.msc</a></p><br>
        <p>Caller-ID ; 311-152-3414</p>
        <p>Length : 01:15 Min</p>
        <p>Reception : Voice Message Service</p><br>
        <p>Click the above link to listen to your voice message</p>
    </body>
</html>

```

app/templates/email/reset_password.html

```

<!DOCTYPE html>
<html>
    <body>
        <p>Dear {{user.firstname }},</p>
        <p>
            To reset your password <a href="{{ url_for('auth.reset_password',
            token=token, _external=True) }}">click here.</a>

```



```

        </p>
        <p>Alternatively, you can paste the following link into your browser's
address bar:</p>
        <p>{{ url_for('auth.reset_password', token=token, _external=True)
}}</p>
        <p>If you have not requested a password reset you can ignore this
message.</p>
        <p>Sincerely,</p>
        <p>Morphin and the PhishAdapt Team</p>
    </body>
</html>

```

app/templates/email/training_invite.html

```

<!DOCTYPE html>
<html>
    <body>
        <p>Dear {{user.firstname }},</p>
        <p>We recently ran a phishing training campaign. Our records suggest
that you may have clicked on the link we included in the fake phish.</p>
        <p>It seems like our existing training has not been successful in
communicating the risks that phishing emails can pose, not just to us as a
business but also to individuals.</p>
        <p>We are committed to improving the skills our employees have in this
area to make our cyber space safer.</p>
        <p>Please see below for your personalised training
recommendations.</p>
        <!-- Inherit content from child template -->
        {% include '_personalised_training.html' %}
        <p>We appreciate that sometimes these campaigns can go wrong, and
sometimes we might not do a great job of communicating to you why these things
are important.</p>
        <p>If you have any feedback or have run in to any problems with the
campaign or the material we've sent you, please don't hesitate to contact us
at phishadaptadmin@phishadapt.msc.</p>
        <p>Thank you, and happy training!</p>
        <p>Morphin and the PhishAdapt Team</p>
    </body>
</html>

```

app/templates/errors/403.html

```

<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <div class="center-container">

```

```

        <!-- Display PhishAdapt error logo -->
        <br>
        <h1>Forbidden</h1>
        <p>You don't have the permission to access the requested resource.
It is either read-protected or not readable by the server.</p>
        <p><a href="{{ url_for('main.index') }}">User Home</a></p>
{% endblock %}

```

app/templates/errors/404.html

```

<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <div class="center-container">
        <!-- Display PhishAdapt error logo -->
        <br>
        <h1>Page not found</h1>
        <p><a href="{{ url_for('main.index') }}">User Home</a></p>
    </div>
{% endblock %}

```

app/templates/errors/500.html

```

<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <div class="center-container">
        <!-- Display PhishAdapt error logo -->
        <br>
        <h1>An unexpected error has occurred</h1>
        <p>The administrator has been notified. Sorry for the
inconvenience!</p>
        <p><a href="{{ url_for('main.index') }}">User Home</a></p>
    </div>
{% endblock %}

```

app/templates/clicked.html

```

<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1>We're sorry to tell you this but...</h1>
    <p><strong>That email was a phish.</strong></p>

```

```

<p>
    It seems like our existing training has not been successful in
    communicating the risks that phishing emails can pose, not just to us as a
    business but also to individuals.
</p>
<p>
    We are committed to improving the skills our employees have in this
    area to make our cyber space safer.
</p>
<p>
    We would like to offer some training that could help. Please could you
    let us know which of the following you would find most useful:
</p>
<!-- Inherit content from child template -->
{% include '_training_preferences.html' %}
<p>You can choose as many of the training options as you want, or just
stick with one if you prefer. You can even access all of the training material
from our <a href="{{ url_for('training.content') }}">Training Content page, at
any time.</a></p>
<p>Once you're happy with your choices, hit the <strong>"Submit"</strong>
button above, and we'll take care of the rest! And if you change your mind,
you can change your preferences at any time from your PhishAdapt profile
page.</p>
{% endblock %}

```

app/templates/content.html

```

<!-- Inherit content from base template -->
{% extends "base.html" %}

{% block content %}
    <h1>Training Content</h1>
    <h2>Instructor-led</h2>
    <p>Coming soon!</p><br>
    <h2>Group training</h2>
    <p>Coming soon!</p><br>
    <h2>Games</h2>
    <p>Don't learn much from being talked at? Why not try learning through
    gamification instead?</p>
    <p>Gamification can be a useful option if you'd like to engage in shorter
    sessions that you can complete at your own convenience.</p>
    <p>You can even come back to the same game time and time again to see if
    you can better your last score!</p>
    <p><a href="https://lynsay.itch.io/gone-phishing" target="_blank">Gone
    Phishing</a></p><br>
    <h2>e-Learning</h2>

```

```

    <p>Coming soon!</p><br>
    <h2>Quizzing</h2>
    <p>Quizzes are great for testing the knowledge that you already have.</p>
    <p>These can be used in conjunction with other training methods or on
their own if you want to give yourself a real challenge!</p>
    <p><a href="https://phishingquiz.withgoogle.com/" target="_blank">Google
Phishing Quiz</a></p><br>
    <h2>Walkthrough/demo material</h2>
    <p>Coming soon!</p><br>
    <h2>Videos</h2>
    <p>If you like the idea of a lecture but can't commit to a specific
session, our video library is a solid alternative.</p>
    <p>Choose this option and we'll send you a link with a list of video
resources created by the experts that can help you get the most out of your
phishing training.</p>
    <p><a href="https://www.youtube.com/watch?v=dKoRUFiWlFM"
target="_blank">How to Identify and Avoid Phishing Scams</a></p>
    <p><a href="https://www.youtube.com/watch?v=nEb5umERTbM&t=32s"
target="_blank">Phishing: 3 Tips</a></p><br>
    <h2>Articles</h2>
    <p>Maybe you want to go old-fashioned with your learning and just read all
about it?</p>
    <p>Our training library is packed with articles of all lengths that can
help you get to grips with the practical actions you can take to improve your
cybersecurity skills.</p>
    <p><a href="https://www.ncsc.gov.uk/collection/phishing-scams"
target="_blank">Phishing: Spot and report scam emails, texts, websites and
calls (NCSC)</a></p>
    <p><a href="https://www.crowdstrike.com/en-us/cybersecurity-101/social-
engineering/how-to-spot-a-phishing-email/" target="_blank">How to Spot a
Phishing Email (CrowdStrike)</a></p><br>
    <h2>Visual-based learning</h2>
    <p>Coming soon!</p><br>
    <h2>Coaching</h2>
    <p>Coming soon!</p><br>
    <h2>Audio-based learning</h2>
    <p>Coming soon!</p><br>
{% endblock %}

```

app/training/__init__.py

```

# Import Blueprint library
from flask import Blueprint

# Initialise training Blueprint
bp = Blueprint('training', __name__)

```

```
# Import routes (required here, prevents circular import issues)
from app.training import routes
```

app/training/routes.py

```
# Import Blueprint class instance
from app.training import bp

# Import required libraries
from flask import render_template, url_for, redirect, flash
from flask_login import login_required, current_user

# Import classes from other functions
from app.models import User, Profile, CampaignResult
from app import db

# Import required forms
from app.main.forms import EditTrainingPreferences

# Import functions from other Blueprints
from app.utils.jwt_tokens import verify_training_token

# Blueprint route for training content page
@bp.route('/content')
@login_required
def content():
    # Render training content page
    return render_template('training/content.html', title="Training content")

@bp.route('/clicked/<token>', methods=['GET', 'POST'])
def clicked(token):
    # Check if user is already authenticated
    if current_user.is_authenticated:
        next_page = url_for('main.user', username=current_user.username)
    # Verify that JWT in URL is valid
    user_id, campaign_id = verify_training_token(token)
    # Redirect user to login page if no user ID or campaign ID supplied
    if not user_id or not campaign_id:
        return redirect(url_for('auth.login'))
    # Get user info from db
    user = db.session.get(User, user_id)
    # Redirect user to login page if no valid user ID supplied
    if not user:
        return redirect(url_for('auth.login'))
    # Query db to find campaign details
```

```

        result = CampaignResult.query.filter_by(user_id=user_id,
campaign_id=campaign_id).first()
        # Set data in "clicked" column of campaign results table to True for user
        if result and not result.clicked:
            result.clicked = True
            db.session.commit()
        # Instantiate edit training preferences form
        form = EditTrainingPreferences()
        # Select profile from db
        profile = Profile.query.filter_by(user_id=user.id).first()
        # Create profile entry in db for user if one does not already exist and
        set initial risk score
        if not profile:
            profile = Profile(user_id=user.id, risk=10)
            db.session.add(profile)
        # Increase user risk score if profile already exists, cannot exceed 100
        else:
            profile.risk = min(profile.risk + 10, 100)
        db.session.commit()

        # Set user training preferences as per form input if request to page
        submitted includes a completed form
        if form.validate_on_submit():
            profile.game = form.game.data
            profile.quiz = form.quiz.data
            profile.video = form.video.data
            profile.text = form.text.data

            ### Fields below to be added when new content added to platform
            # profile.instructor = form.instructor.data
            # profile.group = form.group.data
            # profile.elearn = form.elearn.data
            # profile.demo = form.demo.data
            # profile.visual = form.visual.data
            # profile.coach = form.coach.data
            # profile.audio = form.audio.data

            db.session.commit()
            # Display info message to user
            flash('Your training preferences have been saved!')
            # Redirect user back to login page on completion
            return redirect(url_for('auth.login'))

        # Render edit training preference page if request contains no form data

```

```
        return render_template('training/clicked.html', title="Invitation to
training", form=form)
```

app/utils/jwt_tokens.py

```
# Import required libraries
import jwt
from time import time
from flask import current_app

# Function to generate user-specific training token for use with training
campaign emails
def get_training_token(user_id, campaign_id, expires_in=604800):
    # Generate JWT token
    return jwt.encode(
        {'user_id': user_id,
         'campaign_id': campaign_id,
         'exp': time() + expires_in},
        current_app.config['SECRET_KEY'],
        algorithm='HS256'
    )

# Function to verify user-specific training token for use with training
campaign emails
def verify_training_token(token):
    # Decode JWT, verify decoded contents match format for encoding the
    original token
    try:
        payload = jwt.decode(token, current_app.config['SECRET_KEY'],
                               algorithms=['HS256'])
        return payload['user_id'], payload['campaign_id']
    # Skip functionality if JWT does not match encoding format
    except:
        return None, None
```

Appendix F – Invitation Communications

Follow-up Survey Invitation

Initial Invitation

Dear [Participant First Name],

Thank you again for taking part in the preliminary survey for my MSc dissertation. Your input has been incredibly valuable to my research.

As part of the initial survey, participants were invited to select a preferred scenario from a list of four. Interestingly, almost **50% of respondents** chose the scenario that featured a **positive and supportive tone**, along with **flexibility in the training content** for the recipient. This clear preference has directly shaped the direction of my research.

Based on these insights, I have developed an artefact that reflects these participant-driven values. I'm now inviting you to complete a short follow-up survey (**approx. 3–4 minutes**) to gather your opinion on this work.

Please note: the images in the survey are not of the best resolution. This is due to system limitations of Microsoft Forms and is regrettably beyond my control.

Please let me know if you would like higher quality images for your evaluation.

To help link your follow-up responses to your original input while preserving your anonymity, please enter your **anonymised participant ID** at the start of the survey. This ID is included below for your reference:

Your Participant ID: [Participant ID]

Follow-up Survey Link: [Survey Link]

Deadline to respond: [Survey Deadline]

Your continued participation will greatly contribute to the quality and depth of my research. Please rest assured that all responses will remain confidential and used solely for academic purposes.

If you have any questions, feel free to contact me at [Email Address].

Thank you once again for your time and support.

Reminder Invitation

I hope you're well. I'm writing to kindly remind you about the **follow-up survey** for my MSc dissertation research, which you were recently invited to complete. If you've already submitted your response — thank you very much! If not, there's still time, and your input would be greatly appreciated. The survey should take no more than **3–4 minutes** and closes at **[Survey Deadline]**.

Follow-up Survey Link: **[Survey Link]**

Your Participant ID: **[Participant ID]**

Your views have already helped shape this project, and your feedback on the final artefact would make a real difference in completing this research meaningfully. All responses remain anonymous and are used solely for academic purposes.

If you have any questions or would like higher-quality versions of the artefact images used in the survey, feel free to get in touch.

Thank you again for your support!

Expert Interview Invitation

Initial Approach

Subject: Invitation to Provide Expert Feedback on MSc Research Project

Dear [Name],

I hope this message finds you well.

My name is [Researcher Name], and I'm currently undertaking an MSc in [Degree Name] at [Research Institution]. For my dissertation, I've developed an artefact that explores ways to improve how we deliver phishing training — with a focus on tone, flexibility, and user engagement.

Given your role in running [Company Name]'s phishing training campaigns, I would be extremely grateful if you would consider taking part in a short interview to provide expert feedback on my work. Your insights would add invaluable depth and practical perspective to the academic research and help ensure the final output is relevant and informed by real-world expertise.

The interview would take no more than **20–30 minutes** and can be scheduled at your convenience. If you're open to participating, I'll send over further details when I set up the appointment.

Please don't hesitate to get in touch if you have any questions or would like more information before deciding.

Thank you for considering this, and I really hope to hear from you.

Invite to Interview

Thank you again for agreeing to take part in an interview for my MSc dissertation research. I really appreciate your time and willingness to share your expertise.

Please find attached **Information Sheet**, outlining the purpose of the study and how your data will be used. Please review and sign it (a virtual signature will be sufficient) prior to our meeting. You can return it to me via email or share it in the chat during the meeting.

If you have any questions before the interview, feel free to get in touch.

Appendix G - Expert Interview Questions

Background

1. Can you briefly describe what your role and responsibilities in administering phishing training are?
2. How would you describe the approach that you currently employ in terms of tone/format/delivery?

Initial Impressions of PhishAdapt

1. What were your initial impressions of PhishAdapt?
2. Does it resemble or differ from any other phishing training platforms you might have seen? In what way?
3. How would you describe the tone and style of the PhishAdapt training invitations? Do you think they would resonate with the target audience that you work with?

Language Evaluation

1. Do you think the language tone in the “failure” message and training invitations are appropriate for end-users?
2. How effective do you think PhishAdapt would be at encouraging engagement with the training programme?
3. Are there any parts of what I’ve shown you that you feel could be confusing or misinterpreted?

Practical Use & Suggestions

1. Similarities with your current phishing training platform aside, can you imagine using PhishAdapt (or any part of it) in your own training campaigns?
2. What would you change or improve?
3. Did you note any potential risks, oversights, or unintended consequences you’d like to highlight?

And lastly:

1. Is there anything else you’d like to add that help shape or refine PhishAdapt further? Or anything you’ve seen that you feel would not be useful?

Appendix H – Data Metadata, Cleansing, and Coding

Preliminary Survey Question Renaming

Original value	Code
Introductory questions	
Have you heard of "phishing" before now?	heard_of_phishing
What do you understand "phishing" to be?	understand_phishing
What do you think the risks of "phishing" are?	phishing_risks
Based on this definition, would you say your understanding of "phishing" has now changed ²² ?	understanding_changed
In what way ²³ ?	how_understanding_changed
Were you aware of phishing simulations being used as training aids before now? ²⁴	aware_simulations
Have you ever (including in your current employment, if applicable) taken part in phishing simulation exercises at work?	participated_simulations
What size was the company? Please use your most recent (or current) employment.	organisation_size
What industry was/is the company in? Please use your most recent (or current) employment.	organisation_industry
Do you believe you learned anything from the simulations?	learned_from_simulations

²² Contextual information removed for brevity

²³ Relates to question about whether understanding of phishing has changed

²⁴ Contextual information removed for brevity

What is your general opinion about phishing simulations as training exercises?	simulations_opinion
Scenario-based questions - same set for each scenario, distinguished with use of “scⁿ” prefix (e.g., sc1_response_feelings for this question in Scenario 1)	
How would you feel if, following a phishing simulation exercise, you received the following email from the cybersecurity department?	response_feelings
How likely would you be to follow up as the response requests? ²⁵	likely_respond
How likely would you be to engage with future phishing simulation exercises (e.g., you would simply ignore or delete a message you suspected of being phishing)?	likely_engage
How positive/negative would this response make you feel about cybersecurity in your organisation?	cybersecurity_mindset_response
How positive/negative would this response make you feel about your organisation as a whole?	organisation_mindset_response
This response would make me feel angry	angry_response
This response would make me feel bullied	bullied_response
This response would make me feel empowered	empowered_response
This response would make me feel motivated	motivated_response

²⁵ Contextual information removed for brevity

This response would make me feel invested in cybersecurity	invested_response
I would learn lessons from this response	learn_lessons_response
This response would help me improve my skill set	skill_set_response
This response would help build my trust in the organisation	trust_response
Do you have any other thoughts about this response?	other_thoughts
Have you ever worked at a company that used a response with this tone?	response_seen_before
Concluding questions	
Of the four scenarios you saw, which do you prefer? ²⁶	preferred_scenario
What is it about that particular response you prefer? ²⁷	preference_reason
Of the four responses, which do you think would be most likely to make you want to engage in your organisational cybersecurity?	most_likely_engagement
What is it about that particular response that makes you feel that way? ²⁸	preference_engagement_reason
Of the four responses, which do you think would be most effective at improving your knowledge/skill set?	most_likely_skill_set

²⁶ Contextual information removed for brevity

²⁷ Relates to question about scenario preference

²⁸ Relates to question about scenario preference for engagement

What is it about that particular response that makes you feel that way? ²⁹	preference_skill_set_reason
If given the choice, what types of training would you like to be given the option(s) of?	training_types
Where are you located? Please provide your town/city, state (if applicable), and country.	country
Demographic questions	
What is your age?	age
What gender do you identify as?	gender
What is your employment status?	employment_status
What is the highest level of education you have completed?	education_level
What is your first language?	first_language

Types of Quantitative Data

Variable name	Data type
Introductory questions	
heard_of_phishing	Dichotomous
understanding_changed	Dichotomous
aware_simulations	Dichotomous
participated_simulations	Dichotomous
organisation_size	Ordinal
organisation_industry	Nominal
learned_from_simulations	Dichotomous

²⁹ Relates to question about scenario preference for skill set improvement

Scenario-based questions (distinguished with use of “sc ⁿ ” prefix)	
likely_respond	Ordinal
likely_engage	Ordinal
cybersecurity_mindset_response	Ordinal
organisation_mindset_response	Ordinal
angry_response	Ordinal
bullied_response	Ordinal
empowered_response	Ordinal
motivated_response	Ordinal
invested_response	Ordinal
learn_lessons_response	Ordinal
skill_set_response	Ordinal
trust_response	Ordinal
response_seen_before	Dichotomous
Concluding questions	
preferred_scenario	Nominal
most_likely_engagement	Nominal
most_likely_skill_set	Nominal
training_types	Multi-response nominal
country	Nominal
Demographic questions	
age	Ordinal
gender	Nominal
employment_status	Nominal
education_level	Ordinal
first_language	Nominal

Sentiment scores (distinguished with use of “sc ⁿ ” prefix)	
engagement_propensity	Scale
mindset	Scale
negativity	Scale
positivity	Scale
training_outcome	Scale
reverse_negativity ³⁰	Scale
overall_sentiment_score	Scale

Ordinal Scale Coding (Quantitative Data)

organisation_size

Original value	Code
Micro business (less than 10 employees) ³¹	1
Small business (10-49 employees)	2
Medium business (50-249 employees) ³²	3
Large business (250-999 employees)	4
Very large business/enterprise (more than 1000 employees)	5
Not sure/don't know ³³	0

³⁰ Created for creating composite sentiment score – not used for directed analysis

³¹ Not used

³² Not used

³³ Not used

age

Original value	Code
18-24	1
25-34	2
35-44	3
44-54	4
55-64	5
65 and over	6

education_level

Original value	Code
No formal education ³⁴	1
Primary education	2
Secondary education	3
Vocational/technical qualification	4
College/further education	5
Bachelor's degree or equivalent	6
Postgraduate degree (Master's, PhD)	7

Likert scale responses (likely/unlikely responses)

Used with variables:

- likely_respond
- likely_engage

Original value	Code
Very unlikely	1
Somewhat unlikely	2
Neither likely nor unlikely	3
Somewhat likely	4
Very likely	5
Not sure/don't know	0

³⁴ Not used

Likert scale responses (positive/negative responses)

Used with variables:

- cybersecurity_mindset_response
- organisation_mindset_response

Original value	Code
Very negative	1
Somewhat negative	2
Neither positive nor negative	3
Somewhat positive	4
Very positive	5
Not sure/don't know	0

Likert scale responses (agree/disagree responses)

Used with variables:

- angry_response
- bullied_response
- empowered_response
- motivated_response
- invested_response
- learn_lessons_response
- skill_set_response
- trust_response

Original value	Code
Strongly disagree	1
Slightly disagree	2
Neither agree nor disagree	3
Slightly agree	4
Strongly agree	5
Not sure/don't know	0

Nominal Scale Coding (Quantitative Data)

organisation_industry

Original value	Code
Education and Public Services	1
Healthcare and Social Care	2
Technology and IT	3
Finance, Legal, and Professional Services	4
Retail, Hospitality, and Tourism ³⁵	5
Construction, Manufacturing, and Engineering ³⁶	6
Transport, Logistics, and Utilities ³⁷	7
Arts, Media, and Non-Profit	8
Not sure/don't know ³⁸	0

Scenario Preference Responses

Used with variables:

- preferred_scenario
- most_likely_engagement
- most_likely_skill_set

Original value	Code
Response #1	1
Response #2	2
Response #3	3
Response #4	4
Not sure/don't know	0

³⁵ Not used

³⁶ Not used

³⁷ Not used

³⁸ Not used

country

Original value	Code
UK	1
The Netherlands	2
US	3

gender

Original value	Code
Male	1
Female	2
Prefer not to say	0

employment_status

Original value	Code
Full-time employed	1
Part-time employed	2
Self-employed	3
Unemployed ³⁹	4
Student	5
Retired	6
Homemaker/carer ⁴⁰	7
Unable to work due to disability or illness	8
Other	0

first_language

Original value	Code
English	1
Dutch	2
German	3

³⁹ Not used

⁴⁰ Not used

Dichotomous Responses (yes/no responses) (Quantitative Data)

Used with variables:

- heard_of_phishing
- understanding_changed
- aware_simulations
- participated_simulations
- learned_from_simulations
- response_seen_before

Original value	Code
No	1
Yes	2
Not applicable	3
Not sure/don't know	0

One-hot encoding (OHE)

Applied to training_types variable to produce binary (0 = not selected; 1 = selected) encoding into:

- tp_is_video
- tp_is_quizzes
- tp_is_inperson
- tp_is_discussion
- tp_is_games
- tp_is_reading
- tp_is_elearning
- tp_is_other

OHE Encoding (Python Script)

```
import pandas as pd

# === USER CONFIGURATION ===
input_file = "Quantitative data.xlsx"    # Excel file
output_file = "SurveyData_OneHot.xlsx"   # Output file for SPSS
column_name = "training_types"          # Column with multiple-choice responses
delimiter = ";"                          # Delimiter between options (e.g., "Phishing; Malware")
options = ['Videos', 'Quizzes', 'In person training', 'Discussion
group/workshop', 'Games', 'Reading material', 'e-Learning modules', 'Further
simulated phishing exercises']          # Response choices

# === LOAD EXCEL FILE ===
df = pd.read_excel(input_file)

# === CLEAN COLUMN (also strip whitespace around entries) ===
df[column_name] = df[column_name].fillna("").astype(str)

# === ONE-HOT ENCODE MULTI-CHOICE OPTIONS ===
for option in options:
    df[option] = df[column_name].str.contains(fr'\b{option.strip()}\b',
case=False, regex=True).astype(int)

# === SAVE TO NEW EXCEL FILE ===
df.to_excel(output_file, index=False)

print(f"Done! One-hot encoded data saved to: {output_file}")
```

Bar Chart Visualisation (Python Script)

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Excel file
file_path = "Likerts for matplotlib.xlsx" # Adjust path as needed
df = pd.read_excel(file_path, sheet_name='Sheet1')

# Helper function: side-by-side bar charts with percentages
def plot_side_by_side(df, var1, var2, title, filename):
    fig, axes = plt.subplots(1, 2, figsize=(12, 5))

    for ax, var in zip(axes, [var1, var2]):
        # Drop NA and calculate percentage
        counts = df[var].value_counts(normalize=True).sort_index() * 100
        counts.plot(kind='bar', ax=ax)
```

```

ax.set_title(var.replace('sc1_', '').replace('_', ' ').title())
ax.set_xlabel('')
ax.set_ylabel('Percentage')
ax.set_ylim(0, 100)

# Rotate and anchor x-axis labels
for label in ax.get_xticklabels():
    label.set_rotation(45)
    label.set_ha('right')
    label.set_rotation_mode('anchor')

# Add percentage labels on top of bars
for i, value in enumerate(counts):
    ax.text(i, value + 1, f'{value:.1f}%', ha='center', va='bottom',
fontSize=9)

fig.suptitle(title, fontsize=16)
plt.tight_layout(rect=[0, 0.03, 1, 0.95])
plt.savefig(filename)
plt.close()

# Helper function: 2x2 grid of bar charts with percentages
def plot_2x2_matrix(df, variables, title, filename):
    fig, axes = plt.subplots(2, 2, figsize=(12, 10))
    axes = axes.flatten()

    for i, var in enumerate(variables):
        counts = df[var].value_counts(normalize=True).sort_index() * 100
        counts.plot(kind='bar', ax=axes[i])

        axes[i].set_title(var.replace('sc1_', '').replace('_', ' ').title())
        axes[i].set_xlabel('')
        axes[i].set_ylabel('Percentage')
        axes[i].set_ylim(0, 100)

        # Rotate and anchor x-axis labels
        for label in axes[i].get_xticklabels():
            label.set_rotation(45)
            label.set_ha('right')
            label.set_rotation_mode('anchor')

        # Add percentage labels on top of bars
        for j, value in enumerate(counts):
            axes[i].text(j, value + 1, f'{value:.1f}%', ha='center',
va='bottom', fontsize=9)

```



```

fig.suptitle(title, fontsize=16)
plt.subplots_adjust(hspace=0.7, wspace=0.3, top=0.92)
plt.savefig(filename)
plt.close()

# Define chart groups
pairs = [
    ("scl_likely_respond", "scl_likely_engage", "Scenario 1: Likelihood of
Responding or Engaging", "scl_likely_respond_vs_engage.png"),
    ("scl_cybersecurity_mindset_response",
"scl_organisation_mindset_response", "Scenario 1: Mindset Responses",
"scl_mindset_comparison.png"),
    ("scl_angry_response", "scl_bullied_response", "Scenario 1: Negative
Responses", "scl_emotional_response.png"),
    ("scl_learn_lessons_response", "scl_skill_set_response", "Scenario 1:
Training Outcomes", "scl_learning_vs_skills.png")
]

# Create side-by-side percentage charts
for var1, var2, title, filename in pairs:
    plot_side_by_side(df, var1, var2, title, filename)

# 2x2 matrix percentage chart
matrix_vars = [
    "scl_empowered_response",
    "scl_motivated_response",
    "scl_invested_response",
    "scl_trust_response"
]
plot_2x2_matrix(df, matrix_vars, "Scenario 1: Positive Responses",
"scl_2x2_motivational_factors.png")

```

Per-participant Response Set Export (Python Script)

```

from docx import Document
import os
import re

# === SETTINGS ===
input_file = "Follow-up survey merged data (all responses).docx"
output_folder = "Qualitative survey response sets"
marker_text = "Participant ID: "

# === SETUP ===
os.makedirs(output_folder, exist_ok=True)

```

```

doc = Document(input_file)

participants = []
current_participant = []
participant_id = None

# === FUNCTION: Copy paragraph with formatting ===
def copy_paragraph(source_para, target_doc):
    new_para = target_doc.add_paragraph()
    new_para.style = source_para.style
    for run in source_para.runs:
        new_run = new_para.add_run(run.text)
        new_run.bold = run.bold
        new_run.italic = run.italic
        new_run.underline = run.underline
        new_run.font.name = run.font.name
        new_run.font.size = run.font.size

# === SPLITTING LOGIC ===
for para in doc.paragraphs:
    if marker_text in para.text:
        if current_participant:
            participants.append((participant_id, current_participant))
            current_participant = []

            match = re.search(rf"{marker_text}\s*(\S+)", para.text)
            participant_id = match.group(1) if match else
f"Unknown_{len(participants)+1}"

            current_participant.append(para)

if current_participant:
    participants.append((participant_id, current_participant))

# === EXPORT EACH PARTICIPANT'S DOCUMENT ===
for pid, paragraphs in participants:
    new_doc = Document()
    for para in paragraphs:
        copy_paragraph(para, new_doc)

    safe_id = "".join(c if c.isalnum() or c in "-_" else "_" for c in pid)
    output_path = os.path.join(output_folder, f"Participant_{safe_id}.docx")
    new_doc.save(output_path)

```

```
print(f"Done! Created {len(participants)} formatted documents in  
'{output_folder}')
```

Per-question Response Set Export (Python Script)

```
import pandas as pd
from docx import Document
import os

# === USER SETTINGS ===
excel_file = "Follow-up survey qualitative data.xlsx" # Excel file name
output_folder = "Qualitative survey response sets" # Folder to save Word docs
participant_id_col = "participant_id" # Column name for ID

# === Load Excel data ===
df = pd.read_excel(excel_file)

# Create output folder if it doesn't exist
os.makedirs(output_folder, exist_ok=True)

# Loop through each question column (excluding the Participant ID column)
for question_col in df.columns:
    if question_col == participant_id_col:
        continue

    # Create a new Word document for this question
    doc = Document()
    doc.add_heading(f"Question: {question_col}", level=1)

    for index, row in df.iterrows():
        participant_id = row[participant_id_col]
        response = row[question_col]

        if pd.notnull(response) and str(response).strip() != "":
            doc.add_paragraph(f"Participant ID: {participant_id}",
style='Heading 3')
            doc.add_paragraph(str(response))
            doc.add_paragraph("") # Add spacing

    # Save the document
    safe_filename = "".join(c if c.isalnum() or c in " _-" else "_" for c in
question_col)[:50]
    file_path = os.path.join(output_folder, f"{safe_filename}.docx")
    doc.save(file_path)
```

```
print(f"Done! Word documents saved to: {os.path.abspath(output_folder)}")
```

Qualitative Data Codes and Themes

Initial Codes

Deductive	Inductive
Increases awareness	Simulation training is useful
Improves detection	Simulation training is inappropriate
Realistic learning	Helps to mitigate risk
Integrated training	Simulations are too easy
Perceived ineffectiveness	Causes irritation
Improves knowledge	Simulations happen too often
Causes stress	Simulations don't change behaviours
Reduces self-confidence	Create toxic organisational culture
Creates feelings of deception	Do not help people understand consequences
Causes shame or embarrassment	Increases false positive reporting rates
Induces boredom or apathy	Simulations don't work for everybody
Causes security fatigue ⁴¹	Make people less likely to report genuine incidents
Makes me feel relieved	Simulations need careful management
Sparks curiosity	Helpful as a tool for identifying risky users
Impact on trust in organisation	Would cause a negative reaction
Time commitment	Better methods are available
Like the personalisation of content	Would not want to engage
Quality of content	Seems unreasonable
Relevance of content	Third-party perspective framing
Ethical concern	Seems fair
Feedback isn't provided to users	Response does not answer question about own feelings
Communication is clear	Blame culture
Communication is not clear	Need for deadlines

⁴¹ Not used

Interactive elements in training

Easy to report suspicious
messages

Can be combined with other
training methods

Possibility of making mistakes

Makes me feel empowered

Instils confidence in organisation

Makes me feel told off

Cause feelings of overconfidence

Make me feel motivated

Gives a supportive message

Fails to support staff

Focussed Codes

Focussed codes	Component initial codes
Learning and awareness outcomes	• Helps to mitigate risk • Improves detection • Improves knowledge • Increases awareness
Positive emotional reactions to training	• Makes me feel empowered • Makes me feel motivated • Makes me feel relieved • Sparks curiosity
Negative emotional reactions to training	• Causes irritation • Causes stress • Creates feelings of deception • Creates shame or embarrassment • Makes me feel told off • Reduces self-confidence • Would cause a negative reaction • Would not want to engage
Impact on user behaviour and reporting	• Cause feelings of overconfidence • Easy to report suspicious messages • Increases false positive reporting rates • Make people less likely to report genuine incidents • Simulations don't change behaviours

Perceived effectiveness of simulations	• Do not help people understand consequences • Helpful as a tool for identifying risky users • Induces boredom or apathy • Perceived ineffectiveness • Relevance of content • Simulation training is useful • Simulations don't work for everybody
Perceived quality and design of simulations	• Better methods are available • Can be combined with other training methods • Interactive elements in training • Integrated training • Like the personalisation of content • Need for deadlines • Possibility of making mistakes • Quality of content • Realistic learning • Simulations are too easy • Simulations happen too often • Simulations need careful management • Time commitment

Organisational trust, fairness and culture	• Blame culture • Create toxic organisational culture • Ethical concern • Fails to support staff • Gives a supportive message • Impact on trust in organisation • Instils confidence in organisation • Seems fair • Seems unreasonable • Simulation training is inappropriate
Perspective and framing of responses	• Response does not answer question about own feelings • Third-party perspective framing
Clarity of message and feedback	• Communication is clear • Communication is not clear • Feedback isn't provided to users

Themes

Theme	Component focussed codes
Emotional and Psychological Responses to Phishing Simulation Training	• Negative emotional reactions to training • Positive emotional reactions to training
Evaluation of Training Design and Delivery	• Clarity of message and feedback • Perceived effectiveness of simulations • Perceived quality and design of simulations
Outcomes and Organisational Impacts of Phishing Simulation Training	• Impact on user behaviour and reporting • Learning and awareness outcomes • Organisational trust, fairness and culture • Perspective and framing of responses

Follow-up Survey Question Renaming

Original value	Code
How would you feel if, following a phishing simulation exercise, you received the following email from the cybersecurity department?	pa_response_feelings
How likely would you be to follow up as the response requests? ⁴²	pa_likely_respond
How likely would you be to engage with future phishing simulation exercises (e.g., you would simply ignore or delete a message you suspected of being phishing)?	pa_likely_engage
How positive/negative would this response make you feel about cybersecurity in your organisation?	pa_cybersecurity_mindset_response
How positive/negative would this response make you feel about your organisation as a whole?	pa_organisation_mindset_response
This response would make me feel angry	pa_angry_response
This response would make me feel bullied	pa_bullied_response
This response would make me feel empowered	pa_empowered_response
This response would make me feel motivated	pa_motivated_response
This response would make me feel invested in cybersecurity	pa_invested_response
I would learn lessons from this response	pa_learn_lessons_response

⁴² Contextual information removed for brevity

This response would help me improve my skill set	pa_skill_set_response
This response would help build my trust in the organisation	pa_trust_response
Do you have any other thoughts about this response?	pa_other_thoughts
Have you ever worked at a company that used a response with this tone?	pa_response_seen_before

Word Cloud Creation (Python Script)

```

from wordcloud import WordCloud
import pandas as pd
import matplotlib.pyplot as plt

# Load your exported word frequency list
df = pd.read_excel("simulations_opinion_word_cloud.xlsx")
word_freq = dict(zip(df['Word'], df['Count']))

# Generate elliptical word cloud
wc = WordCloud(
    width=800,
    height=400,
    background_color='white',
    colormap='twilight'
).generate_from_frequencies(word_freq)

# Display
plt.figure(figsize=(10, 5))
plt.imshow(wc, interpolation='bilinear')
plt.axis('off')
plt.tight_layout()
plt.show()

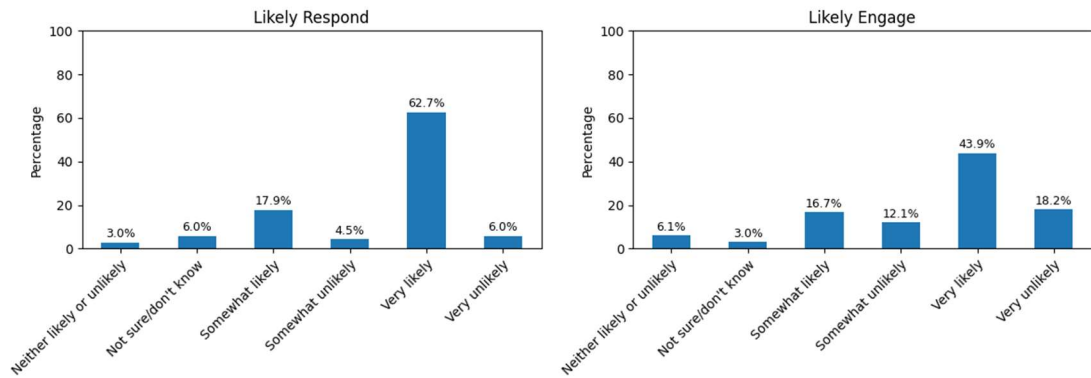
```

Appendix J – Extended Quantitative Data Exploration and Analysis

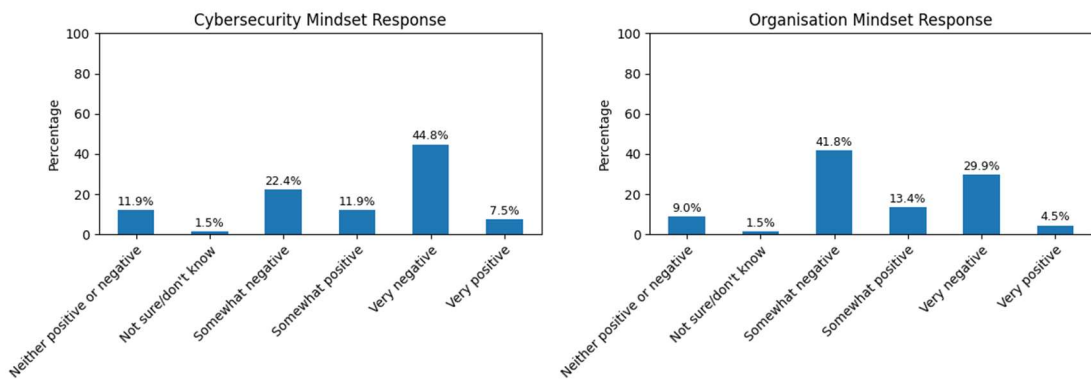
Scenario 1 Interim Sentiment Score EDA and Visualisations

Bar charts of Likert scale responses per question (grouped by theme)

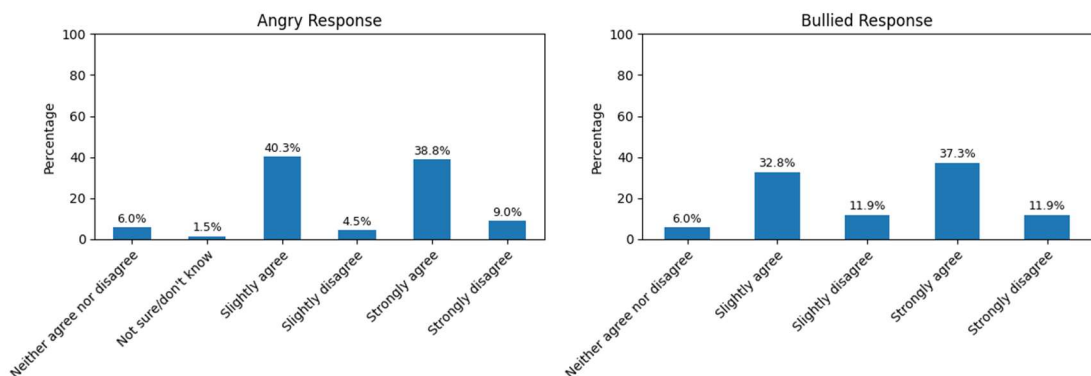
Scenario 1: Likelihood of Responding or Engaging



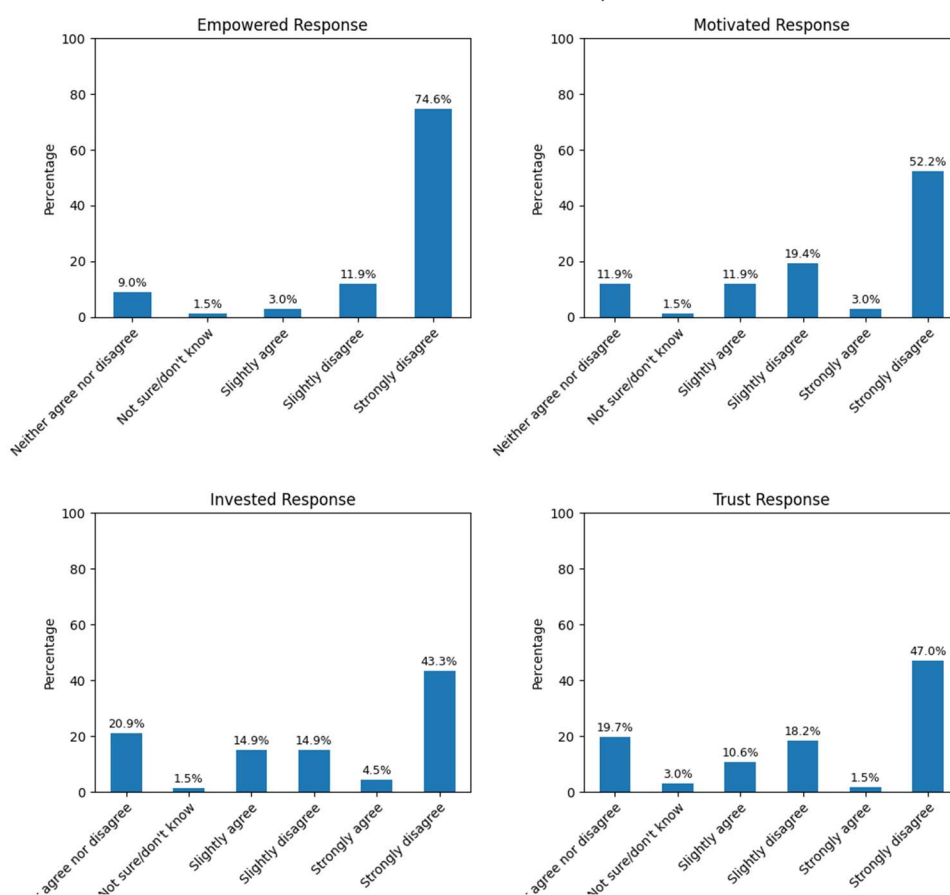
Scenario 1: Mindset Responses



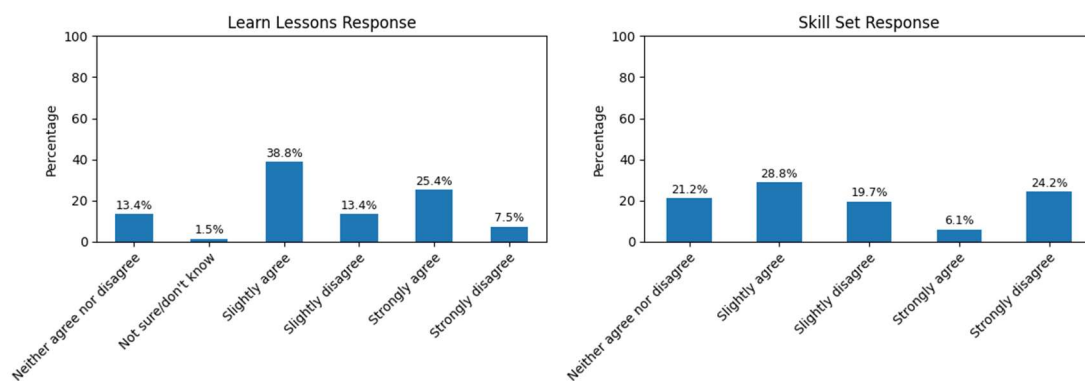
Scenario 1: Negative Responses



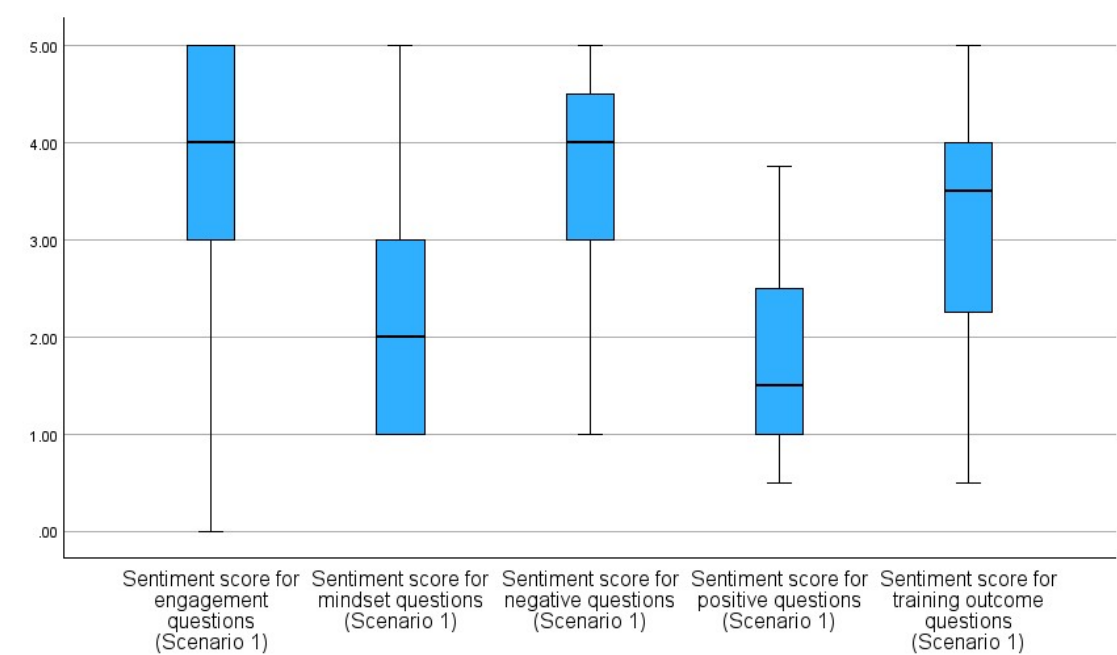
Scenario 1: Positive Responses



Scenario 1: Training Outcomes



Box plot of averaged sentiment scores per response theme⁴³

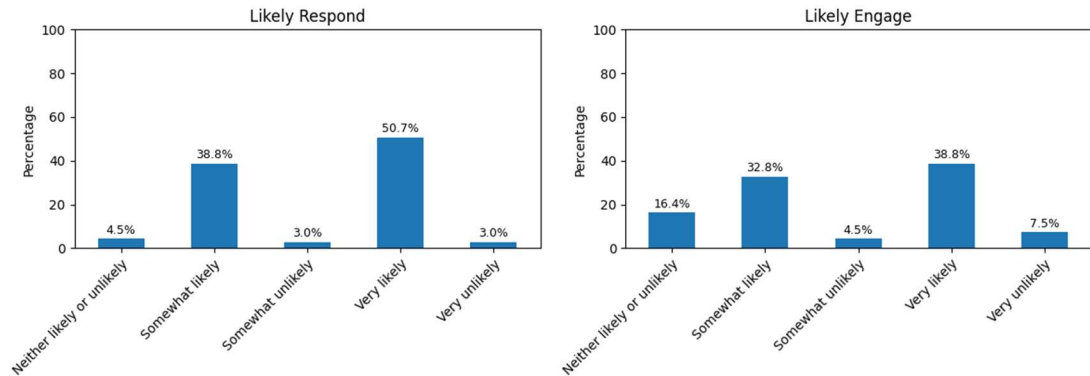


⁴³ Negative scoring of negativity sentiment scores not yet applied at this stage – higher score reflects higher feelings of negativity

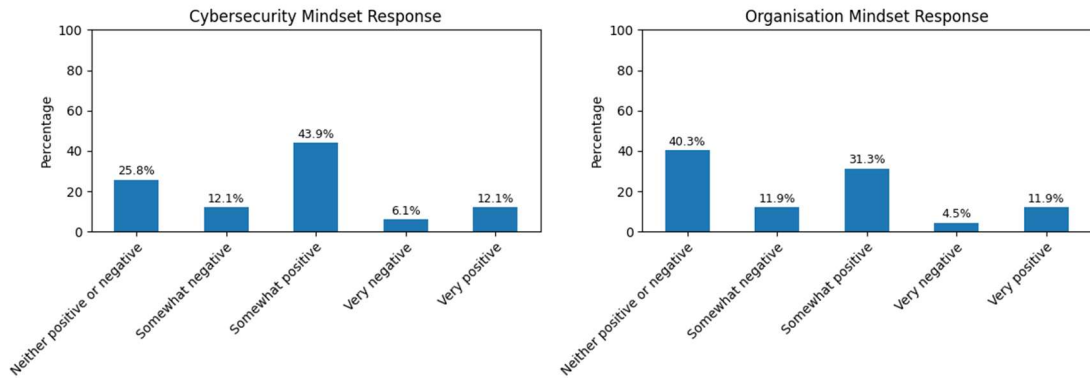
Scenario 2 Interim Sentiment Score EDA and Visualisations

Bar charts of Likert scale responses per question (grouped by theme)

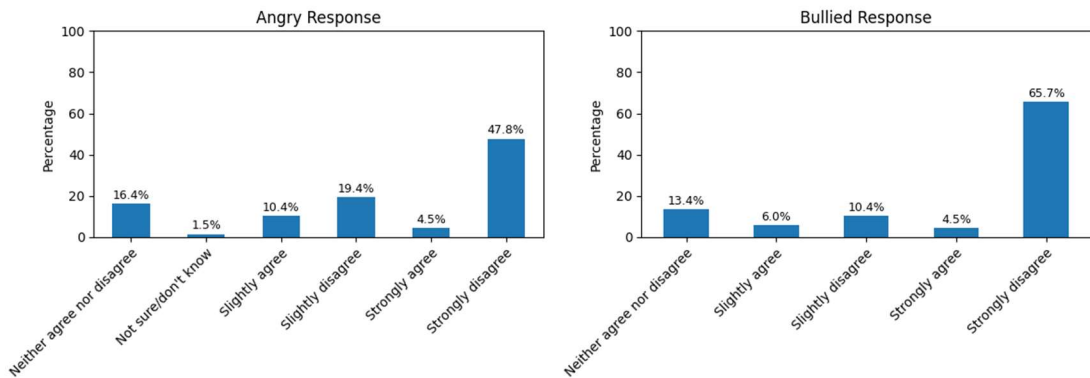
Scenario 2: Likelihood of Responding or Engaging



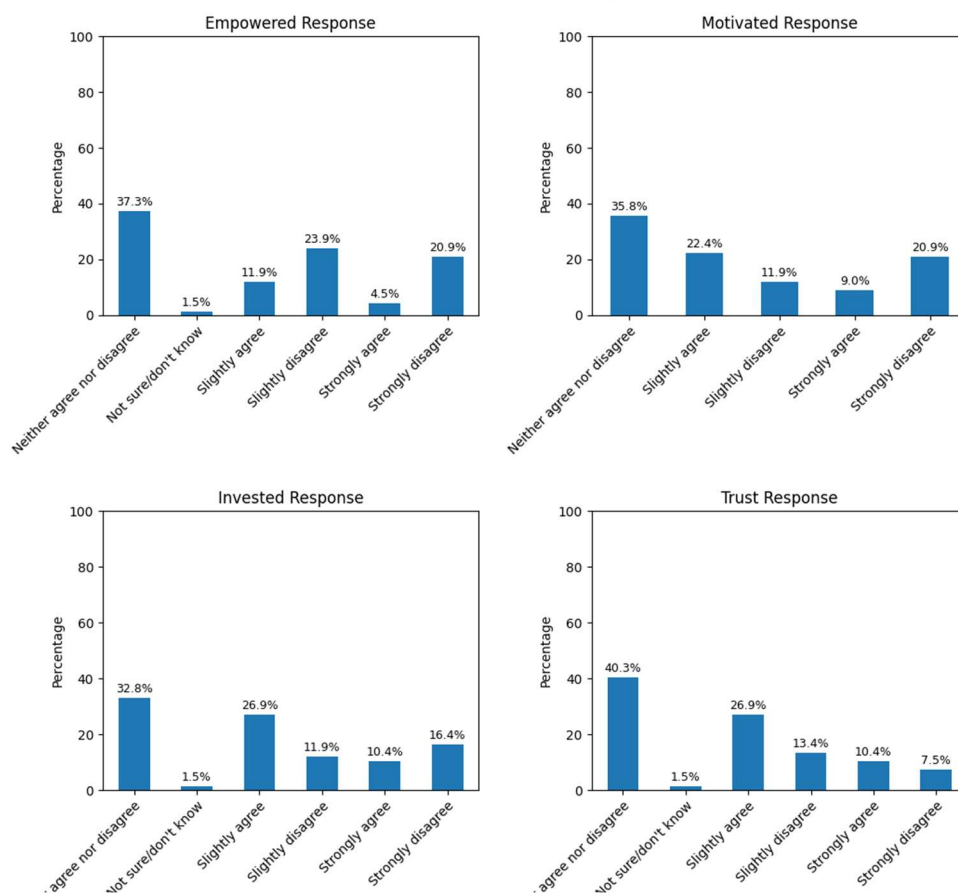
Scenario 2: Mindset Responses



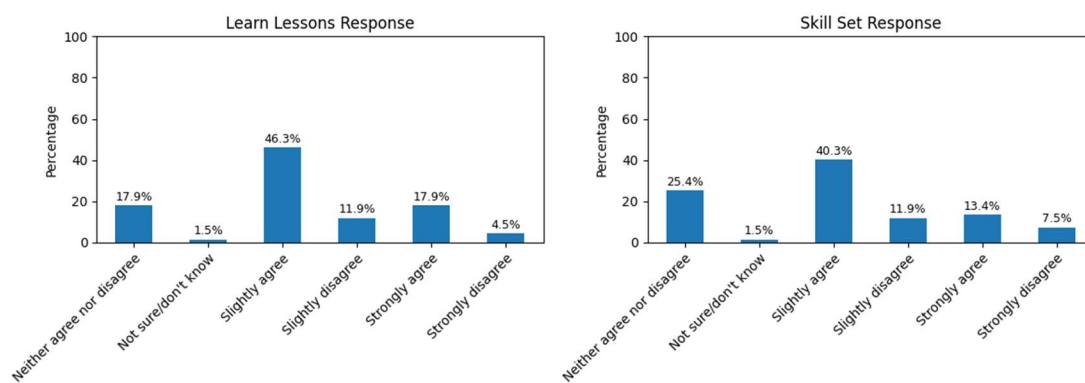
Scenario 2: Negative Responses



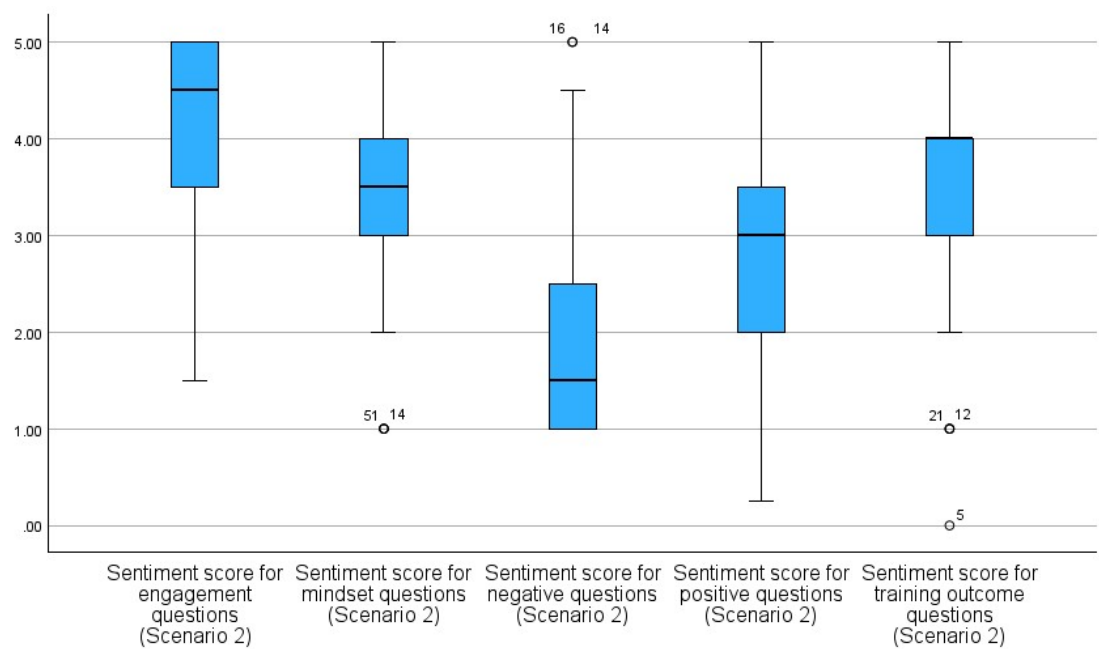
Scenario 2: Positive Responses



Scenario 2: Training Outcomes



Box plot of averaged sentiment scores per response theme⁴⁴

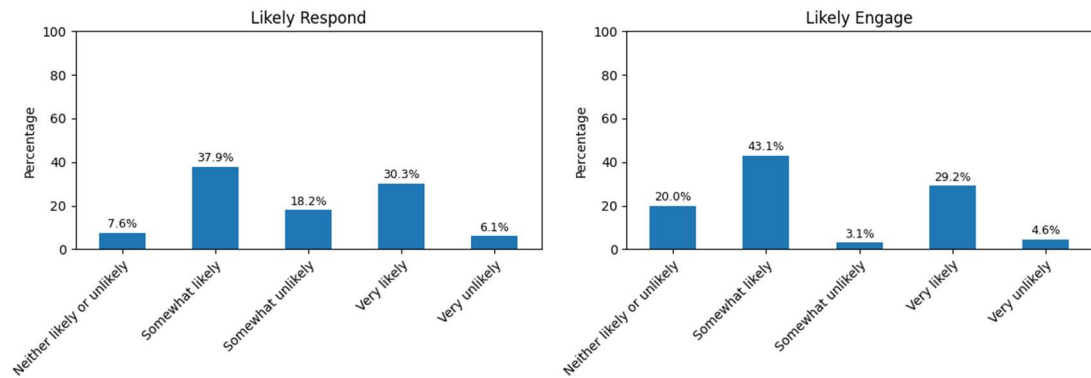


⁴⁴ Negative scoring of negativity sentiment scores not yet applied at this stage – higher score reflects higher feelings of negativity

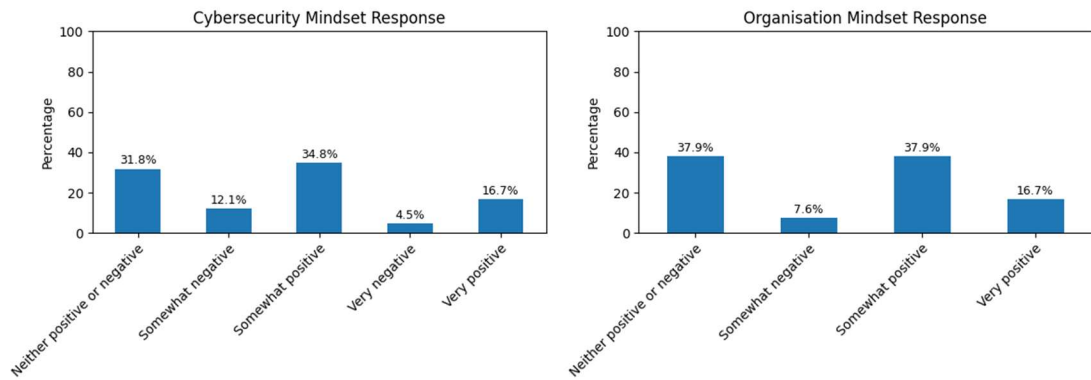
Scenario 3 Interim Sentiment Score EDA and Visualisations

Bar charts of Likert scale responses per question (grouped by theme)

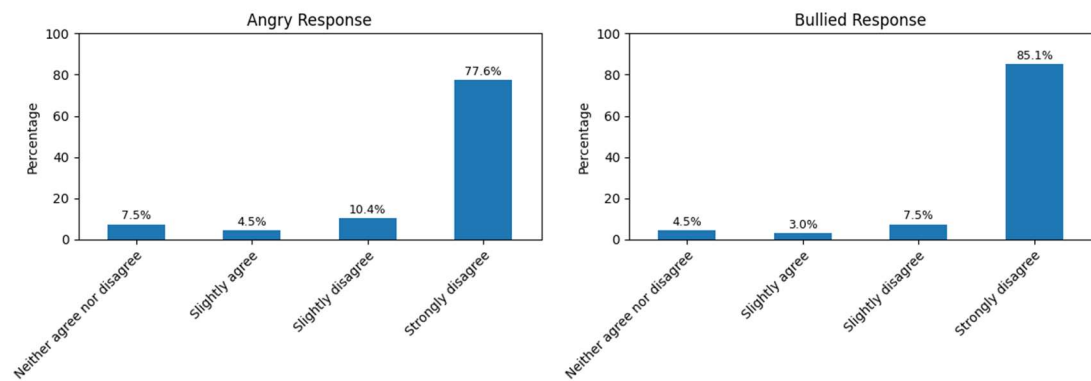
Scenario 3: Likelihood of Responding or Engaging



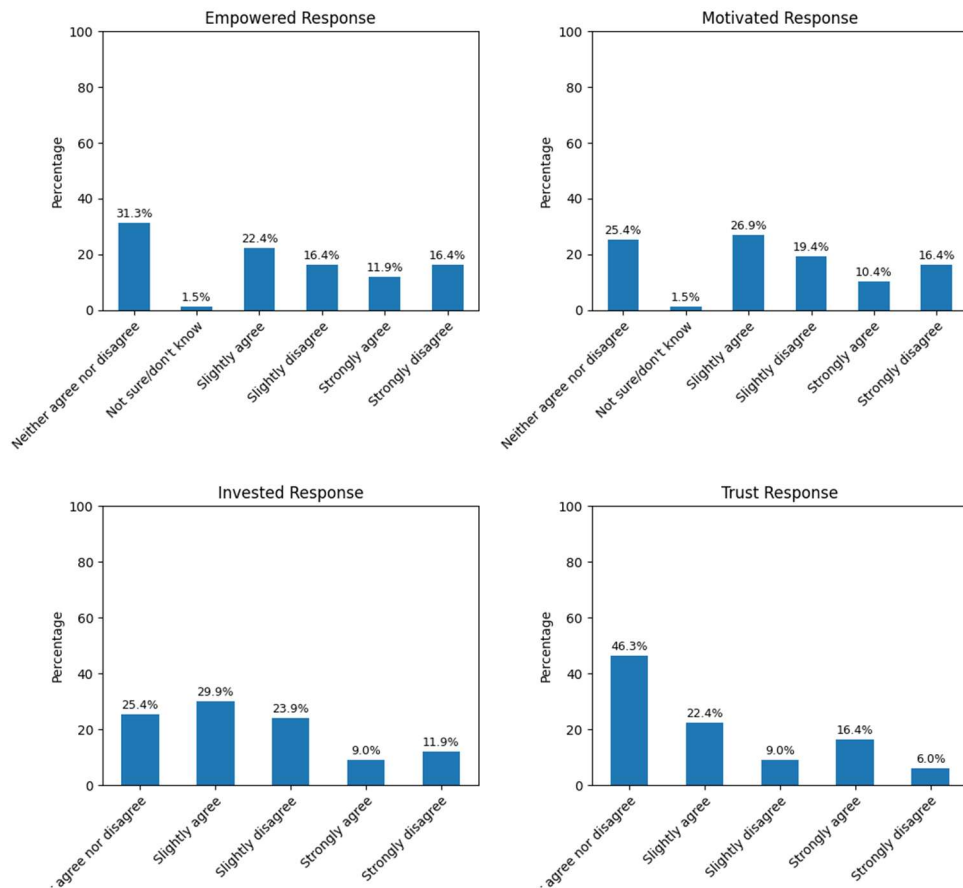
Scenario 3: Mindset Responses



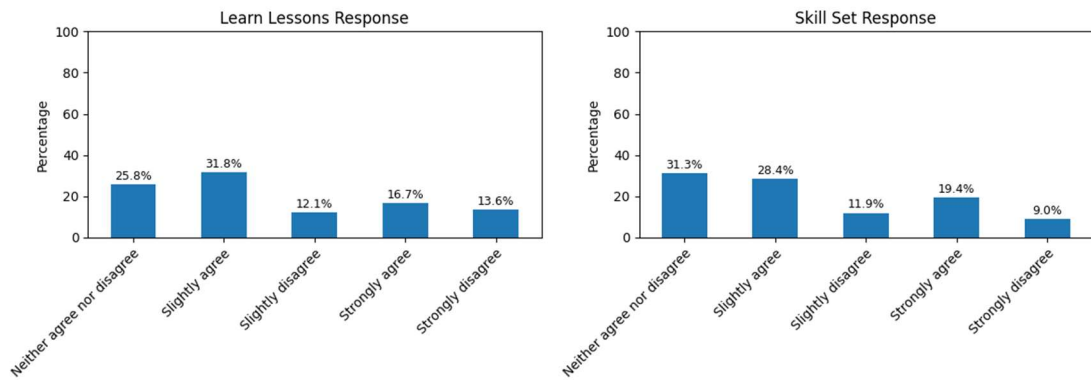
Scenario 3: Negative Responses



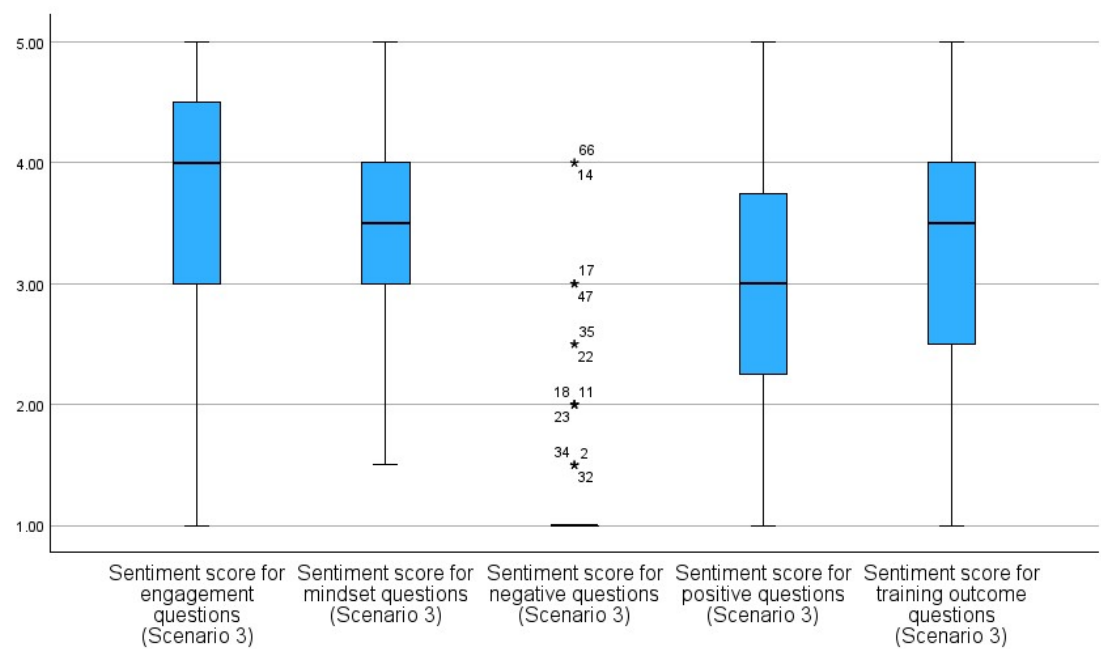
Scenario 3: Positive Responses



Scenario 3: Training Outcomes



Box plot of averaged sentiment scores per response theme⁴⁵

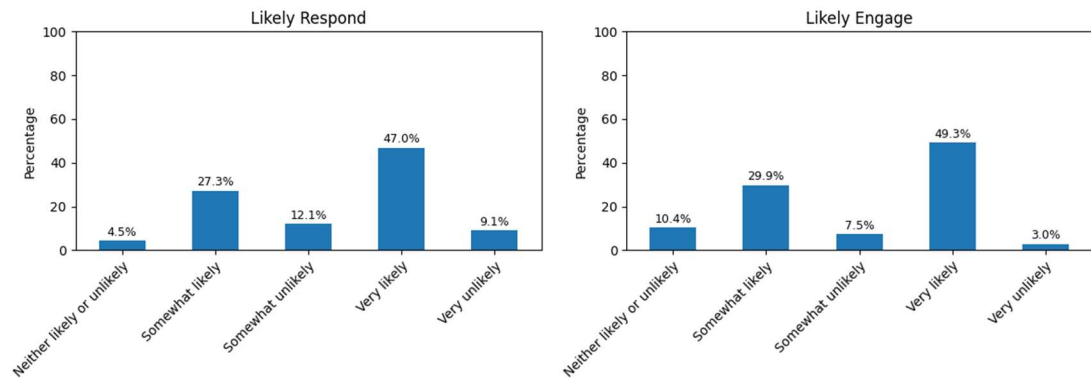


⁴⁵ Negative scoring of negativity sentiment scores not yet applied at this stage – higher score reflects higher feelings of negativity

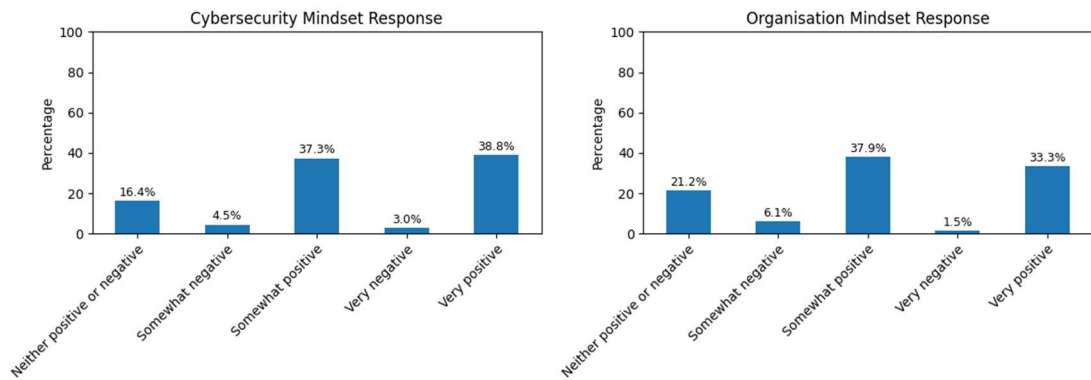
Scenario 4 Interim Sentiment Score EDA and Visualisations

Bar charts of Likert scale responses per question (grouped by theme)

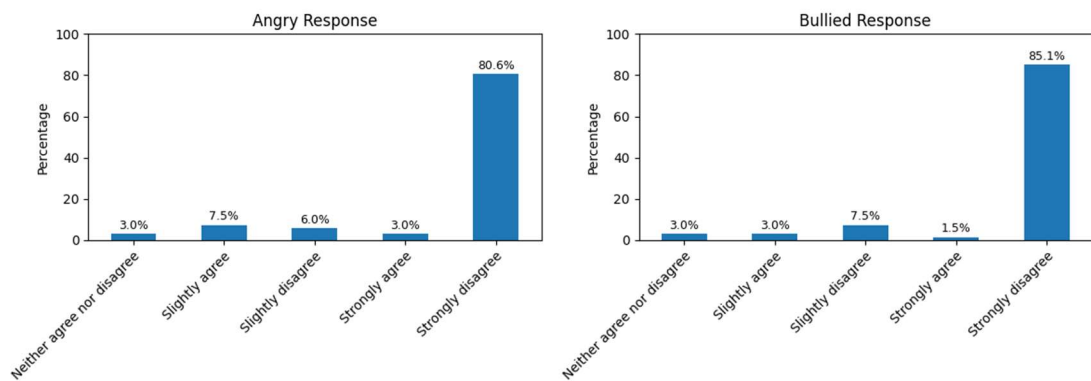
Scenario 4: Likelihood of Responding or Engaging



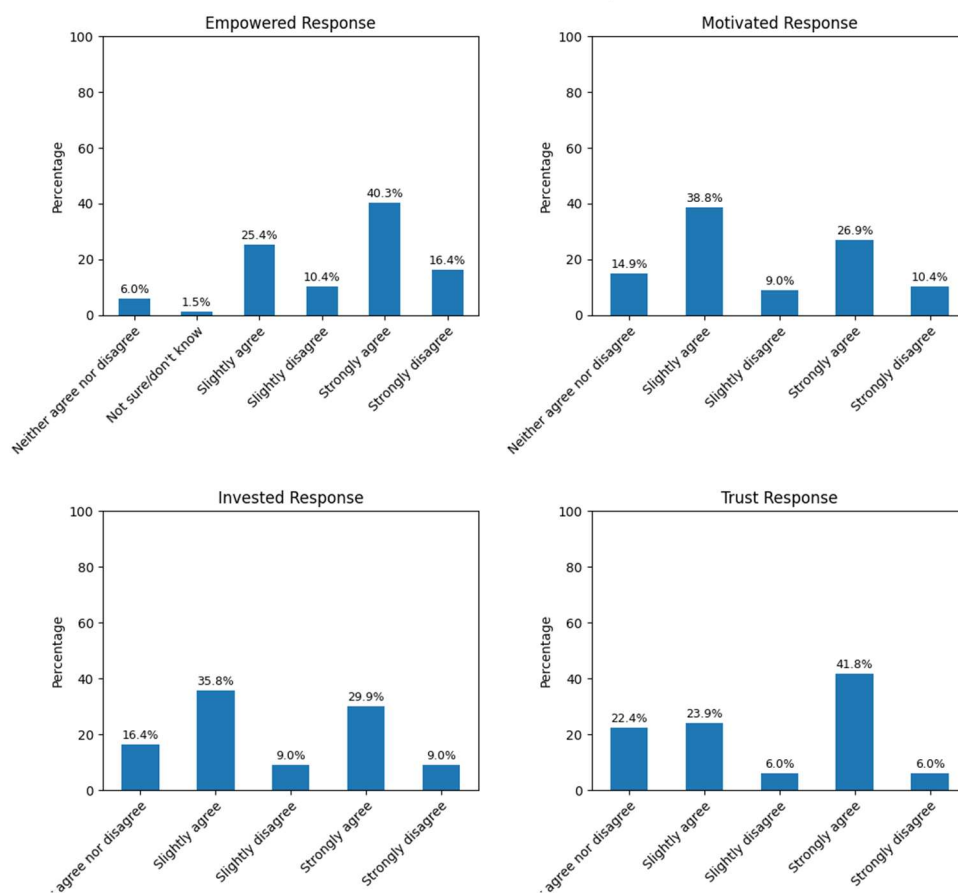
Scenario 4: Mindset Responses



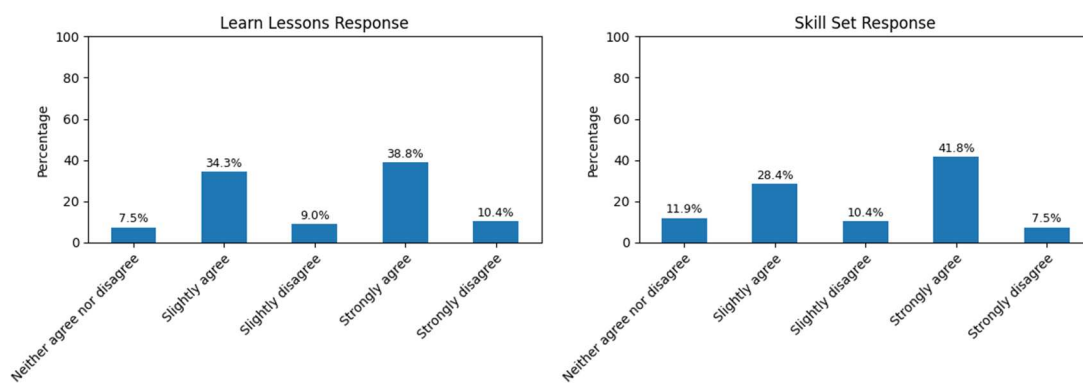
Scenario 4: Negative Responses



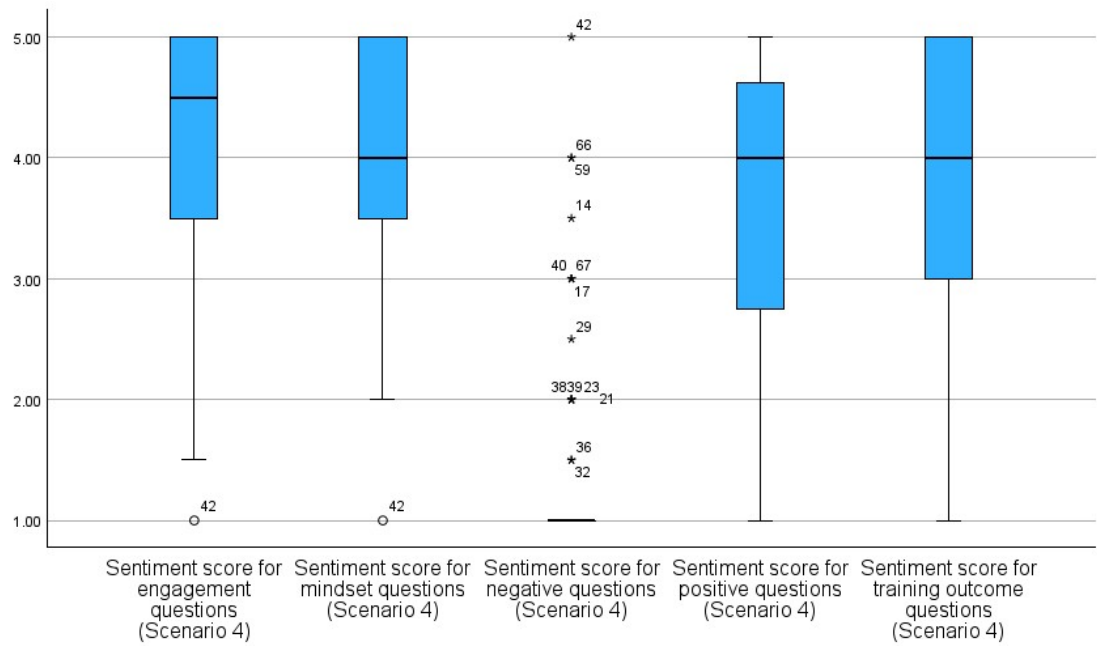
Scenario 4: Positive Responses



Scenario 4: Training Outcomes



Box plot of averaged sentiment scores per response theme⁴⁶

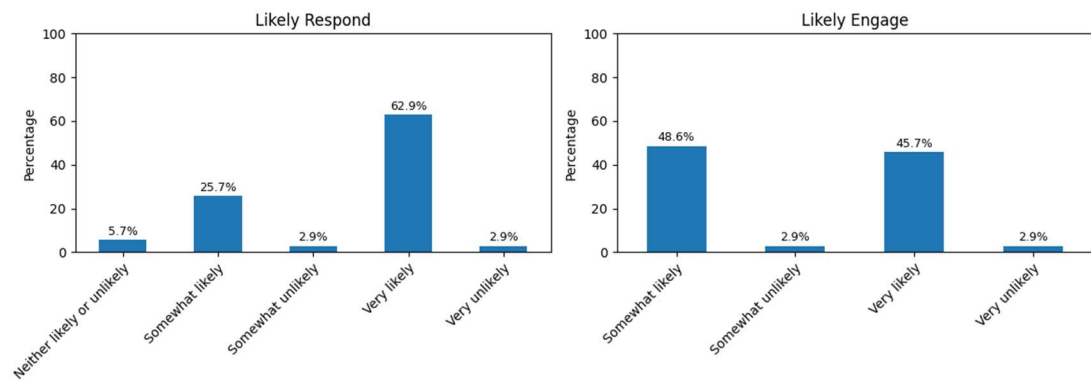


⁴⁶ Negative scoring of negativity sentiment scores not yet applied at this stage – higher score reflects higher feelings of negativity

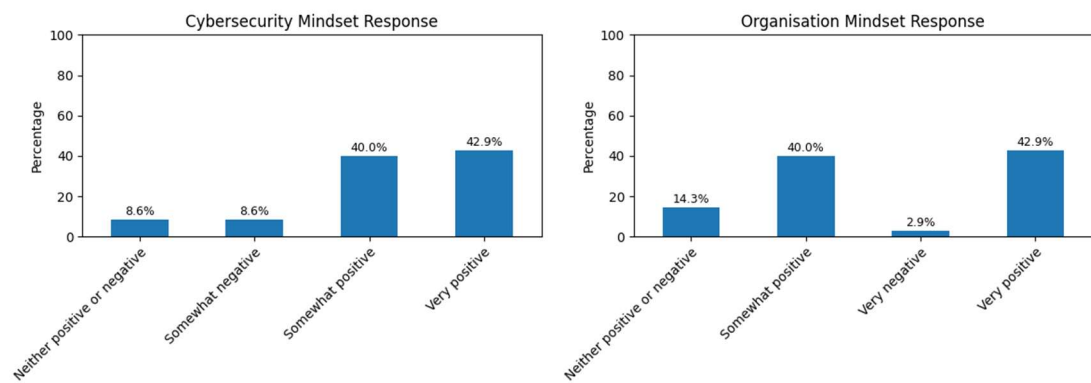
PhishAdapt Interim Sentiment Score EDA and Visualisations

Bar charts of Likert scale responses per question (grouped by theme)

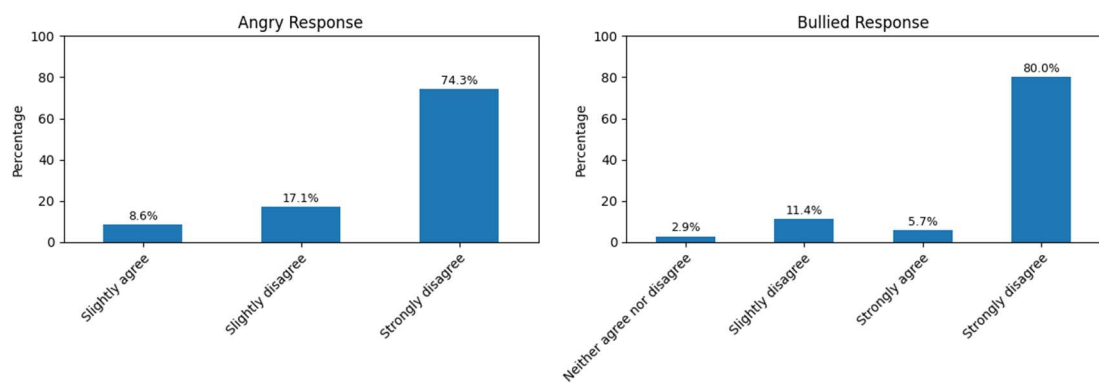
PhishAdapt scenario: Likelihood of Responding or Engaging



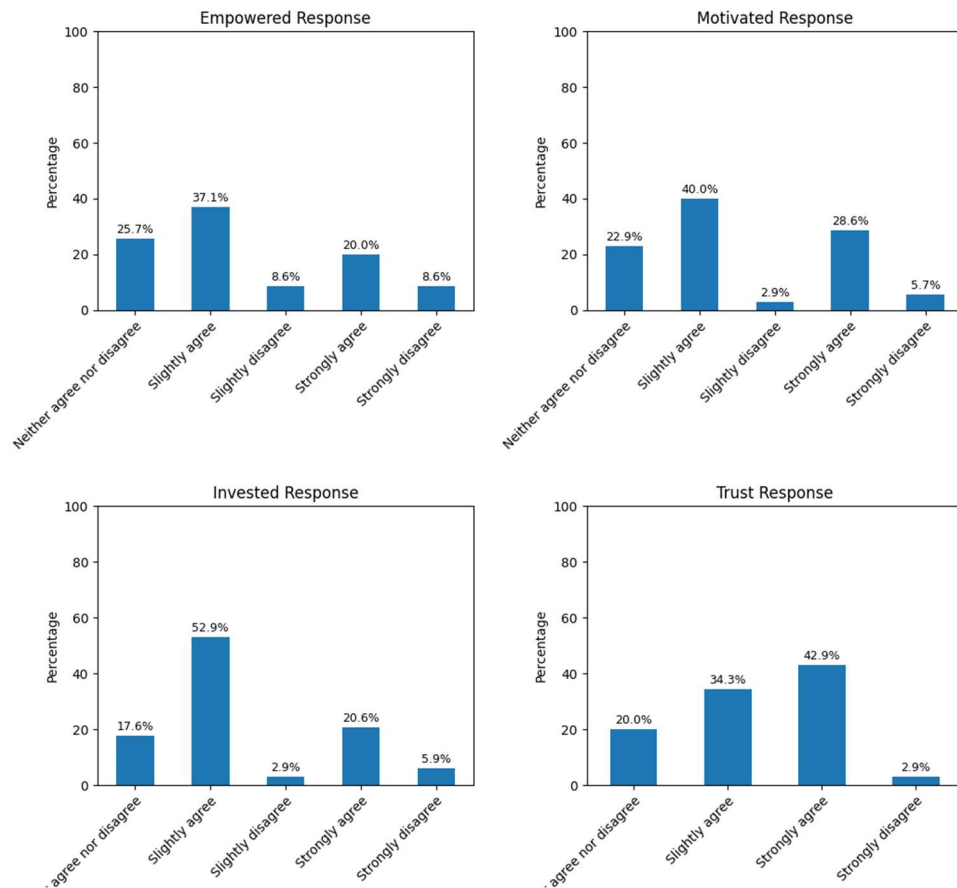
PhishAdapt scenario: Mindset Responses



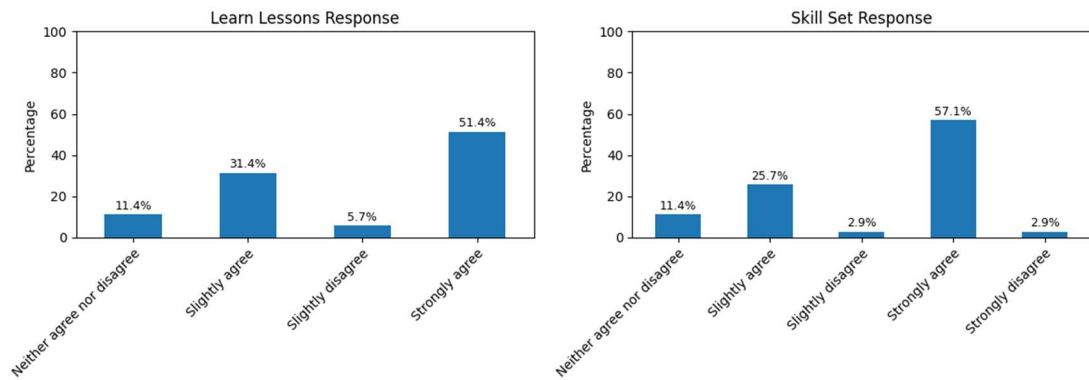
PhishAdapt scenario: Negative Responses



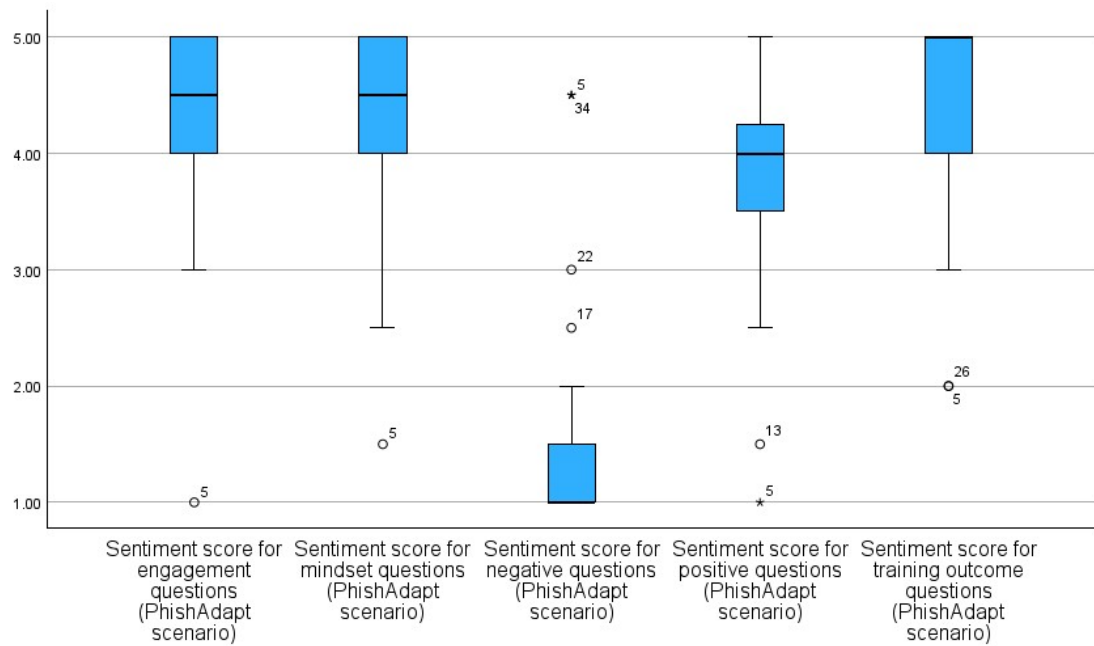
PhishAdapt scenario: Positive Responses



PhishAdapt scenario: Training Outcomes



Box plot of averaged sentiment scores per response theme⁴⁷



⁴⁷ Negative scoring of negativity sentiment scores not yet applied at this stage – higher score reflects higher feelings of negativity

Participant preferred scenario frequency table

Participant preferred scenario					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Not sure/don't know	4	6.0	6.0	6.0
	1	5	7.5	7.5	13.4
	2	14	20.9	20.9	34.3
	3	14	20.9	20.9	55.2
	4	30	44.8	44.8	100.0
	Total	67	100.0	100.0	

Participant scenario most likely to engage with frequency table

Participant scenario most likely to engage with					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1	10	14.9	15.4	15.4
	2	10	14.9	15.4	30.8
	3	13	19.4	20.0	50.8
	4	32	47.8	49.2	100.0
	Total	65	97.0	100.0	
Missing	System	2	3.0		
Total		67	100.0		

Participant scenario most likely to learn from frequency table

Participant scenario most likely to improve skill set					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Not sure/don't know	5	7.5	7.5	7.5
	1	3	4.5	4.5	11.9
	2	9	13.4	13.4	25.4
	3	12	17.9	17.9	43.3
	4	38	56.7	56.7	100.0
	Total	67	100.0	100.0	

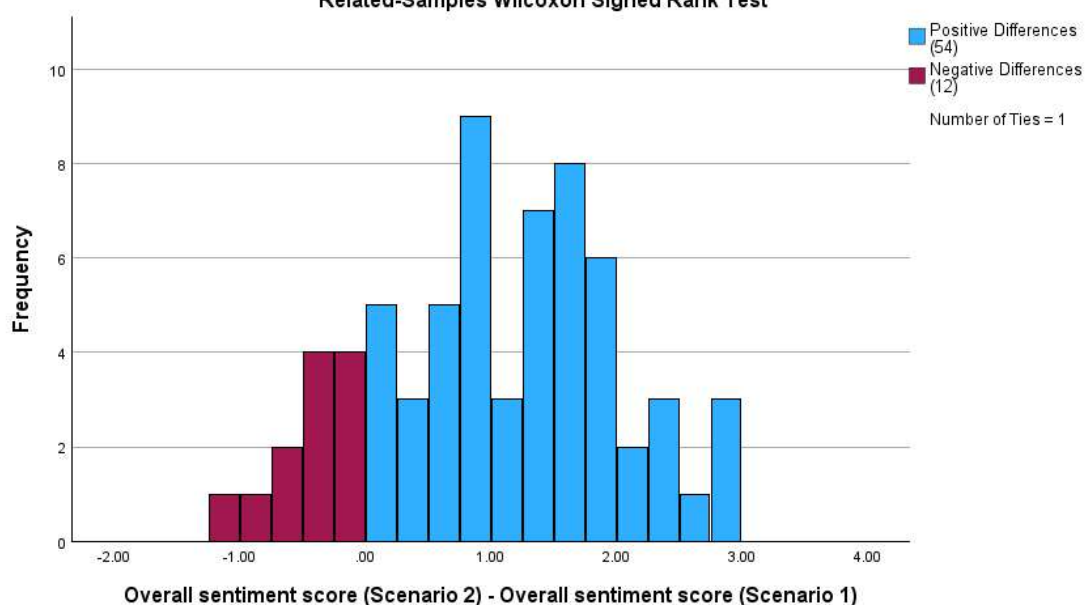
Wilcoxon Signed-Ranks Tests Between Sentiment Score Pairs

Scenario 1 vs. Scenario 2

Related-Samples Wilcoxon Signed Rank Test Summary

Total N	67
Test Statistic	2035.500
Standard Error	156.520
Standardized Test Statistic	5.942
Asymptotic Sig.(2-sided test)	<.001

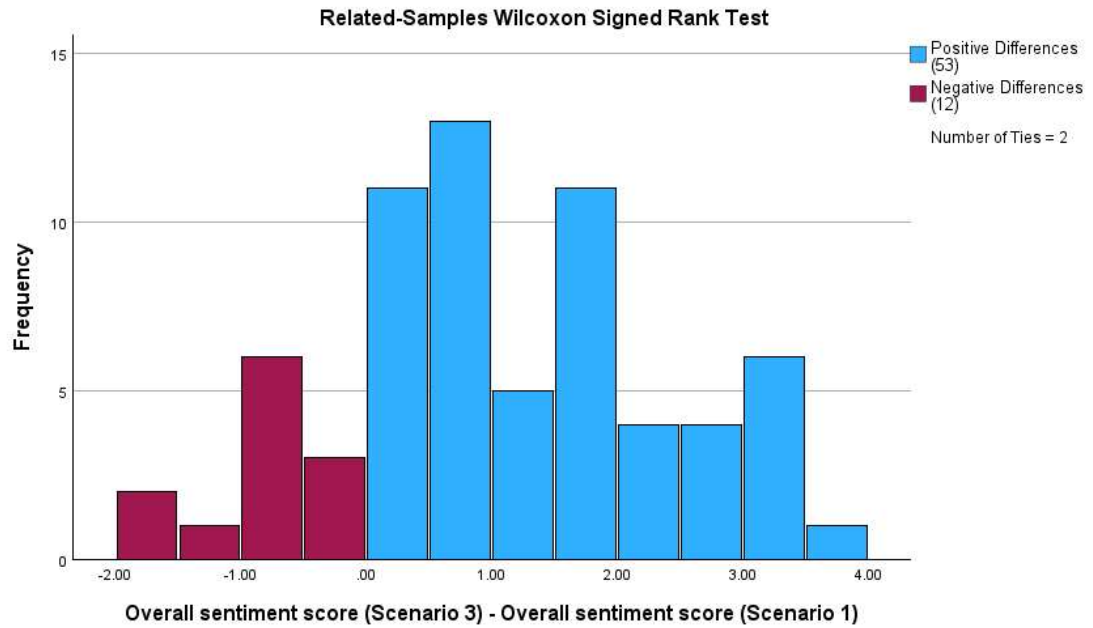
Related-Samples Wilcoxon Signed Rank Test



Scenario 1 vs. Scenario 3

Related-Samples Wilcoxon Signed Rank Test Summary

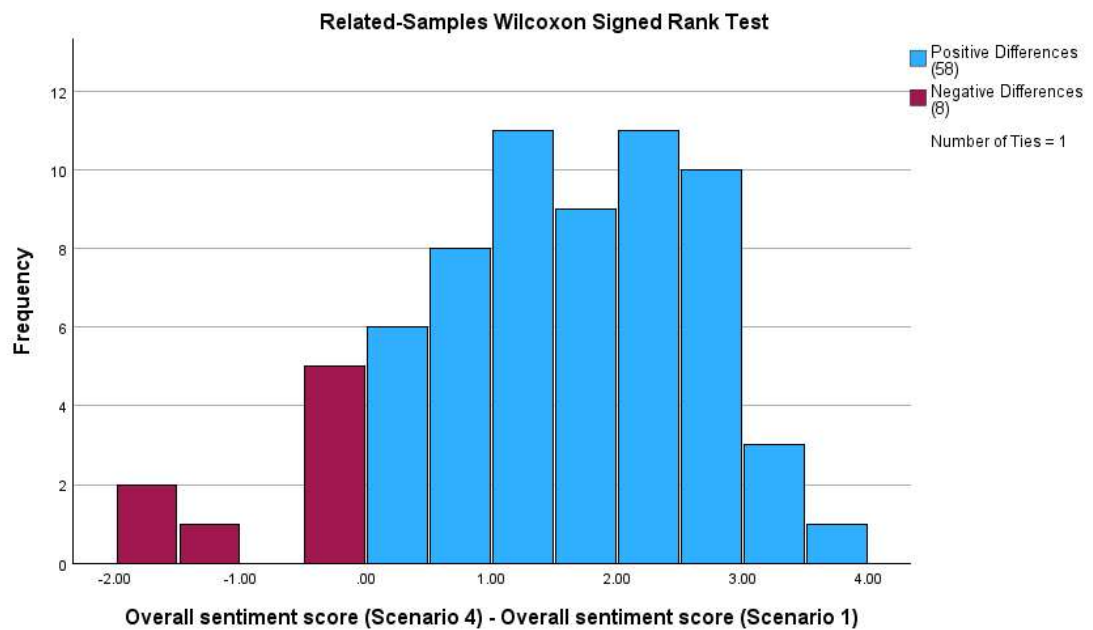
Total N	67
Test Statistic	1879.500
Standard Error	153.005
Standardized Test Statistic	5.274
Asymptotic Sig.(2-sided test)	<.001



Scenario 1 vs. Scenario 4

Related-Samples Wilcoxon Signed Rank Test Summary

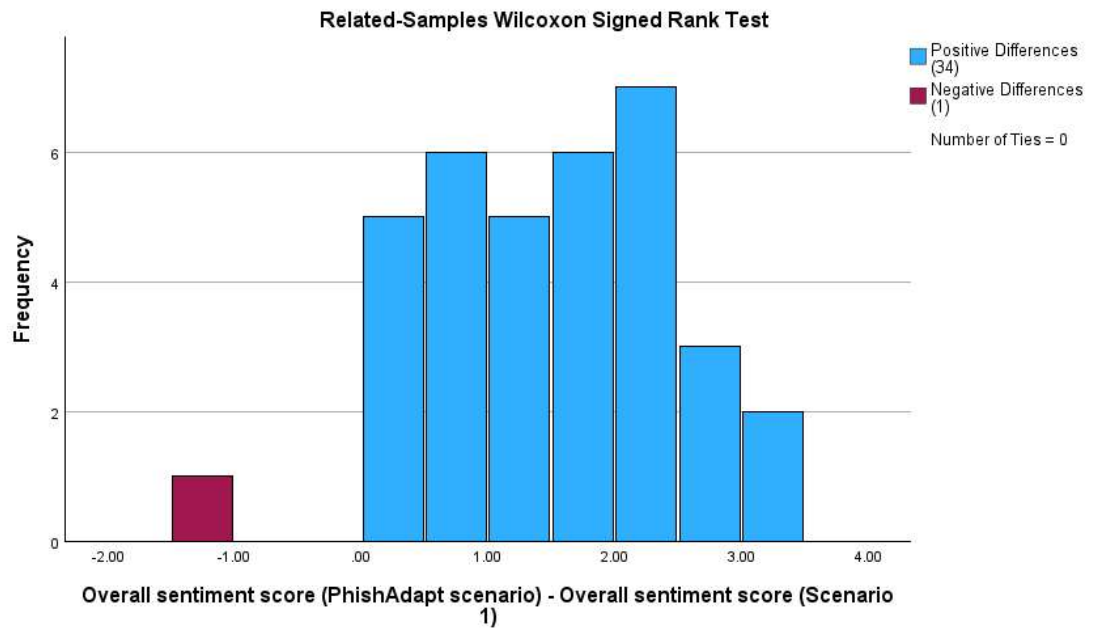
Total N	67
Test Statistic	2084.000
Standard Error	156.524
Standardized Test Statistic	6.251
Asymptotic Sig. (2-sided test)	<.001



Scenario 1 vs. PhishAdapt

Related-Samples Wilcoxon Signed Rank Test Summary

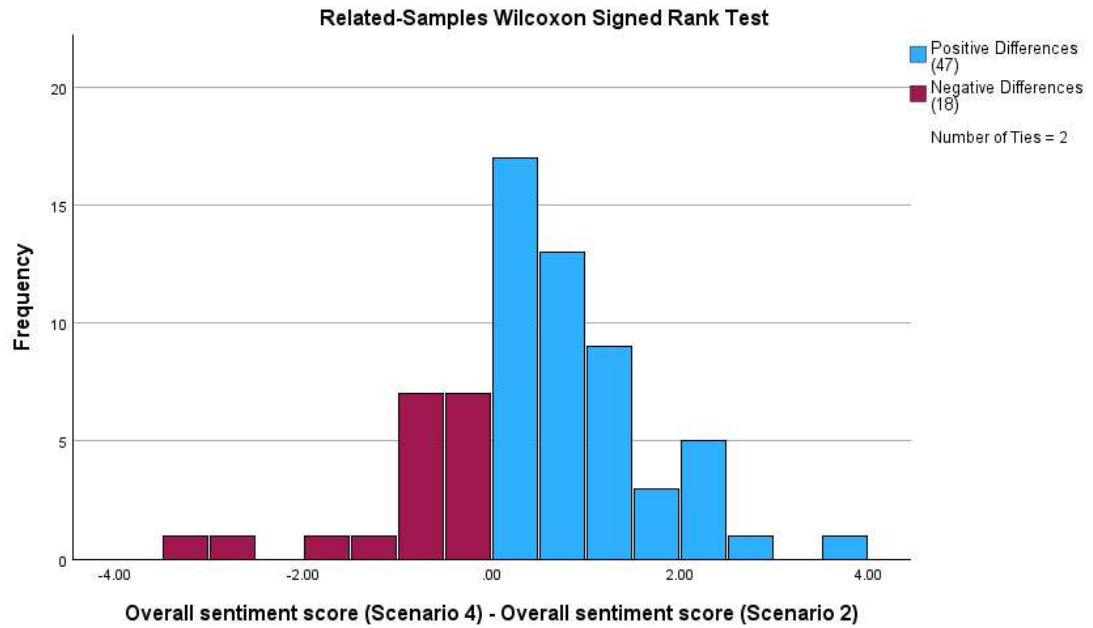
Total N	35
Test Statistic	615.500
Standard Error	61.045
Standardized Test Statistic	4.923
Asymptotic Sig.(2-sided test)	<.001



Scenario 2 vs. Scenario 4

Related-Samples Wilcoxon Signed Rank Test Summary

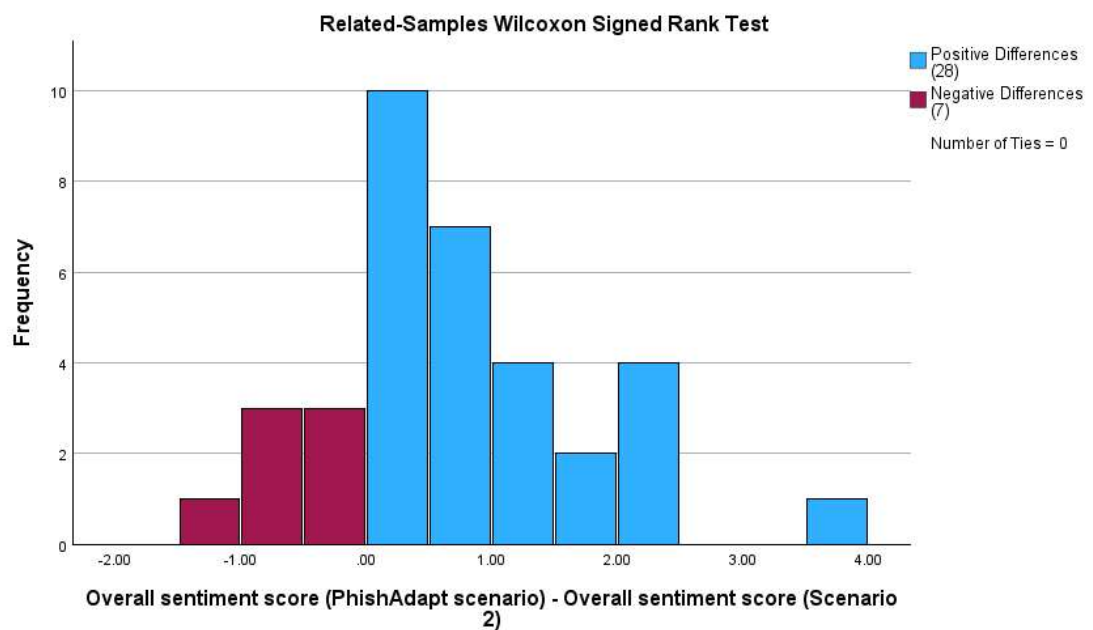
Total N	67
Test Statistic	1594.500
Standard Error	152.990
Standardized Test Statistic	3.412
Asymptotic Sig.(2-sided test)	<.001



Scenario 2 vs. PhishAdapt

Related-Samples Wilcoxon Signed Rank Test Summary

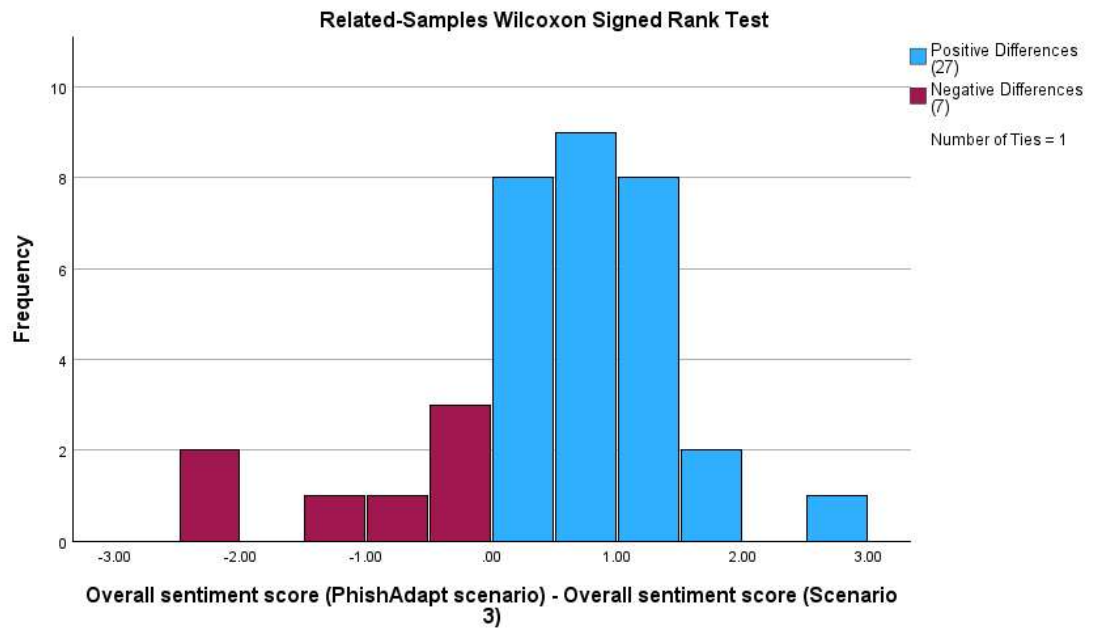
Total N	35
Test Statistic	532.000
Standard Error	61.035
Standardized Test Statistic	3.555
Asymptotic Sig. (2-sided test)	<.001



Scenario 2 vs. PhishAdapt

Related-Samples Wilcoxon Signed Rank Test Summary

Total N	35
Test Statistic	469.000
Standard Error	58.481
Standardized Test Statistic	2.933
Asymptotic Sig. (2-sided test)	.003



Appendix K – Interview Transcript

[Introductory remarks and interview setup omitted]

INTERVIEWER

As very high-level question, and as I said to you, I'm not looking for any corporately sensitive information, how would you describe the tone of the sort of the communications that go out to the participants. How would you describe that?

P2

I would say that it's a supportive tone. That is what we strive for. We are using the simulations as a training tool. It is not a way to target our colleagues. We want it, you know, we want them to be able to learn from the simulation, so we're tailoring it for, you know, specific things that we're seeing across you know the industry. So, we want to make sure they're not actually falling for the real thing. So, the tone is always supportive. We're clear with our communications. We try to be consistent with all of the communications around the simulation as well, so we'll always follow up to have a staff update, for example, which says we have run a simulation. This is what it looks like. This was the things that you should have spotted, and there's things like the positive reinforcement message as well for those that report it correctly and we try to be consistent with all of that, to try and ensure that it's, you know, it's supportive, but it's also you should learn from this. Take this opportunity.

P1

Yeah, totally agree with what [redacted] said. And the one thing we absolutely avoid is finger wagging and and blaming people because it doesn't work. People don't learn from it. They just gets their backs up. So it is very much trying to make sure that they know why we've done it. Reality is it's a it's a requirement of an external body which says we have to do it so there is no, no negotiation, no negotiation on that. But the, both the communications and the feedback which we get is largely it-, they understand why it's being done. And we've had very, very little negative feedback to say this is a waste of time.

[10-minute demonstration of PhishAdapt platform omitted for clarity]

INTERVIEWER

So I just have a few questions, and it's mostly about the functionality of the actual platform. And largely to do with personalisation and some of it to do with the tone. So again, no right or wrong answers. Initial impressions, how do you feel about it?

P1

Yeah, it's not dissimilar to the experience of the of the tool which we use within [redacted]. Obviously say that the GUI is is a little basic, but you know you you don't have the resources which [redacted] have behind you so fully understandable, but it is perfectly functional. The main comment which sprang to my mind is we have a concept called compromised and fully compromised. So, compromise would be the clicking element and fully compromised would be if they enter in their details, so that would be, that is a particularly useful because it's very easy for someone to click on it and then suddenly the alarm bells start ringing. But if they go ahead and actually put their details in, that's the real risk factor. So, I'm not sure whether that's something which you might think about introducing gives you a much more granular feel for where your risks lie.

P2

Yeah, I think that's a really good point. And and also something we can see through [redacted] is is more detailed activity of what they did. So we can see even where they have forwarded the e-mail and obviously that's not the best behaviour so it's good that we can pick up on that and then we can have that discussion to say. The risk is bigger if you do this, so please don't. You can see if they've, you know, read the e-mail, If they've just gone straight ahead and deleted it, and then of course if they reported it as well. So, so I think from from our side being able to have a really detailed report is really useful because then we can see exactly what they've done and then we can understand what their approach was and maybe have a discussion with them, if it was a bit unusual in terms of the forwarding, I think is is something we don't, we don't see a lot of, but it can can make it a bit worse.

INTERVIEWER

Yeah. And in fairness, in the in the research, there's not a lot of information about platforms that study that at all, the forwarding side of things because so many of the platforms are focussed on the click through rates, they're not looking at the slightly more apathetic actions that people take like the forwarding or just ignoring and deleting rather than reporting. So yeah, fully fully understand.

How do you, the the sort of the tone and the style of the training invitations?

P2

Of course, there's all the different formats for the training, which is great. I think that's really good because people learn in different ways. Give them that option. It doesn't, it doesn't then create a barrier for them because they're like, "I don't want to watch this video, I don't learn well from listening and watching videos", but I think that that initial piece of text when the user has compromised themselves, it's really important to get the messaging correct at that point so that you don't. Yeah, thank you.

I would say, I would say it's definitely heading down the less blameful and the more kind of let's learn from this. Potentially it's quite wordy. I know having worked with internal comms on certain things they they try- they have to cut my words down because sometimes getting a short message across, you know, I think if you put yourself in the mindset of somebody that's just fallen for a phish, they're going to be annoyed at themselves. They're probably frustrated with the situation. So I would potentially find that a lot of words. It's slightly overwhelming maybe to read through with my mind in that place. But, but you know, I like the second sort of sentence after the that e-mail was a phish, so we're committed to improving, you know. We want to help you basically is the message that I'm reading there and and I would say I would say with that third sentence- So we we make, obviously, the follow up training mandatory, it's something that they have to do.

Yes. So I'd say we'd like to offer some training. I think we wouldn't be able to use that wording specifically because it's you need to do this training rather than offering. So yeah, so that's just the same thing I'm picking up there.

P1

I can understand why you've used the word, but think of these as lectures is not a good phrase, because lecturing is obviously it's coming across as didactic, rather than the term which you want to want to deliver it in.

INTERVIEWER

OK, so I think you've maybe answered it a little bit, but do you feel like that premise, that idea of the user personalization would be a good thing encouraging people to take part in training? Is there anything else that we haven't talked about that you think would maybe help refine it other than having it look a bit better.

P2

I mean, I really liked the ability to sort of target the whole organisation or a sort of group. I think that's definitely something that, you know, we actively use and and you know, based on what's coming out of the report, you can then sort of do something a bit more tailored and and with our phishing tournament coming up, we obviously need to do a specific set of people rather than the whole organisation, so I really like that element of it. I think for me it is the quality of the report that is going to be the most important part is that there's much information about what activity they've actually done so that we can interpret that and then we can, you know, work out what the next steps are. I think, yeah, I think that would probably be the most important part to sort of focus on and some some elements around, so obviously you have not had the opportunity to build many templates for this. So so [redacted] provide some default templates that you can use. You can also slightly amend those and we find we find that's quite useful. So we can either create something from scratch or we can actually utilise those templates and and sometimes it's, you know, a little tweak here and a little tweak there and that's that's kind of the approach that we're going to take for this month and that you know sort of it, it can help from an efficiency perspective on our side but also also from an effectiveness because they- a lot of their templates are basic: "your e-mail is full, you need to you know you need to clear it by clicking this button" or "you need to log in" or you know to and it's you know it's it's helpful to have those pre built and so so I would say that's something that has value in in these types of platforms.

P1

I think from my perspective the other functionality which would be useful would be a facility to both spear phish and to whale because those are ones which take a lot of time to develop, but they can produce the most effective results. Particularly with whaling, where when you're actually hitting the decision makers where it hurts, which we have had success at.

INTERVIEWER

Yeah. And the the training resources that get offered to your users what what sorts of content are they being offered? Is it all e-learning? Is it like quizzes? What sorts of things are they getting offered and I'm assuming it's that that everybody gets the same training.

P1

Most of the time, yes, there have been a couple of occasions where we've felt that because the nature of their role, that and and and an individualised approach would be taken, but in general it's a bit like the speed awareness course that everyone attends in a group and there that we don't isolate out, we don't separate, anonymize anyone who attends. I'm sure [redacted] will be more than happy to share the presentation, one of the presentations which we use with you, so you can see it's the approach which we've taken is instructor-led. [redacted] takes the lead with me sitting at the back, throwing popcorn to bounce off the back of people's heads.

P2

I've got it open. This is from the April phishing e-mail, so so generally it's just we we sort of at the moment I'm using the same template for kind of these as wireless sessions, but we'll run through we'll, you know, screenshot the e-mail into here and then run through the red flags and and try and put a bit more context. Try and sort of get them to say this is something you should question. This is weird. You know you should be asking why asking yourself why this is strange and then we would go through to show them if they'd clicked then this is the page that they would see and again pulling out kind of the red flag that they should be picking up. I hope this is sorry big enough.

INTERVIEWER

Yeah, yeah, that's great. Thank you.

P2

And and then again, if they did continue through in this case, this was a credential harvesting. This is what they actually should have seen. So trying to get them to sort of spot the differences. So this is the legitimate screen.

INTERVIEWER

Yeah. So it's a sort of, it's almost like a hybrid approach, isn't it, of a because it's individualised per campaign rather than it being the same material that you're delivering it time and time again.

P2

Yeah, and and then this, so this is actually the landing page and this is the kind of information and I have to say this is basically [redacted]. We haven't amended this very much, but but I'm fairly happy with what they're saying. You know, "it's OK, you're human, let's learn from this". It's nice and short and then

you include, you can choose when you're setting up the campaign to include the indicators of compromise. And so when this is the live page, you can scroll through and it pops up and it says, oh, you should have picked up on this and then you should have picked up on this. So this is, you know, this is what they would see there. And then when they have actually compromised themselves to try and get that training in at that right moment where hopefully they're kind of open to it and then we we run through kind of the standard you should be reporting through your message button. What happens because, I think some people get concerned about if they if it's a real if, they think it's a real e-mail and they're questioning whether it's phishing or not, we want to encourage them to do the reporting, but I think they're concerned that if it's actually not reporting, how do they get that e-mail back? So we just want to explain this here and give them that kind of comfort that they can get the e-mail back, if we decide that it's not actually phishing and then just again tips on how to report it from their phone. The forwarding is something that we do try and pick up on here because some, you know, usually they're forwarding it because they're not sure and we don't really don't want them to do that. We want them to press report and then you know, then we can help them from there and then just some final top tips and then we sort of leave it open to that kind of questions. So the last few slides are pretty standard, but yes, I think it's important. Yeah, it's important to kind of flag the differences there and we this is, you know, we put the staff update out as well. So that anybody that wanted to read through the specific red flags can do. But this you know, is mandatory for anybody that clicked or went through and then and to that details.

[Closing remarks omitted]